

Materials Characterization Lab Notebook

Measurement & Analysis Record

This Notebook Belongs To

Lab / Organization

How to Use This Notebook

1. Record every measurement run immediately, in ink.
2. Capture the full setup: instrument, technique, parameters.
3. Log raw data file names, storage location and any deviations.
4. Have a witness sign off on critical or non-standard runs.

Materials Characterization Record

No. SAMPLE — DEMO

Date 2025-04-05	Operator M. Wang	Sample ID SLG-2025-018
Technique X-ray Diffraction (XRD)	Instrument Bruker D8 Discover	
Sample Preparation As-deposited thin film on Si(100), no further preparation.		
Measurement Parameters Parameter Value Note Source: Cu Kα λ=1.5406Å 2θ range: 10-80° Step: 0.02° Scan speed: 2°/min		
Environment (vacuum / ambient / temp) Ambient, 22°C, 45% RH		
Raw Data File & Location Z:/Lab/XRD/2025-04/SLG-2025-018.raw		
Observations Strong (111) peak at 2θ=38.2°, FWHM=0.35°. Weak (200) peak detected.		
Analysis / Results Grain size (Scherrer): 24 nm. Lattice parameter: 4.08Å (PDF#04-0783 match).	Conclusion Polycrystalline film, preferential (111) orientation.	
Follow-up / Next Step Cross-section TEM for grain morphology.		

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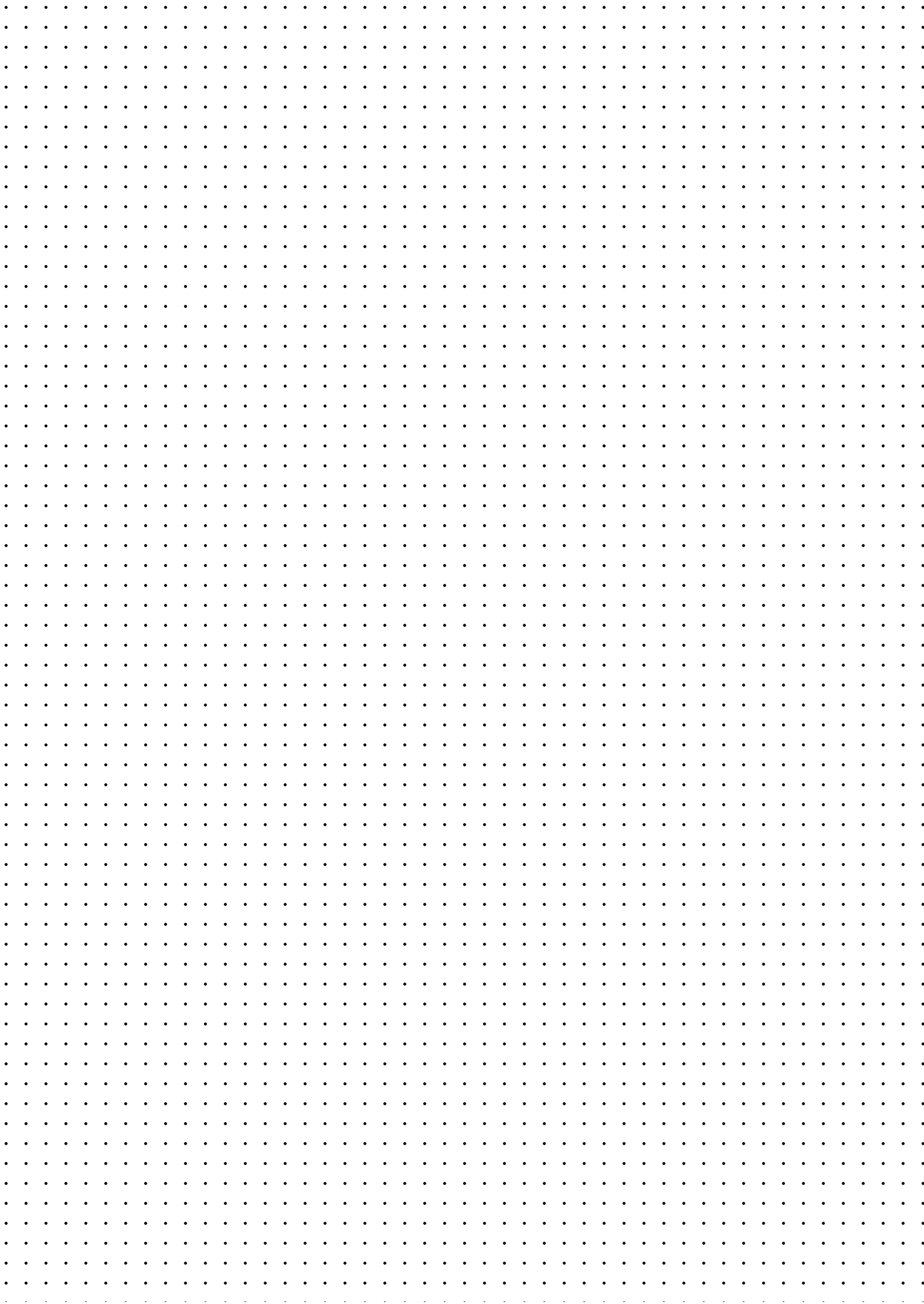
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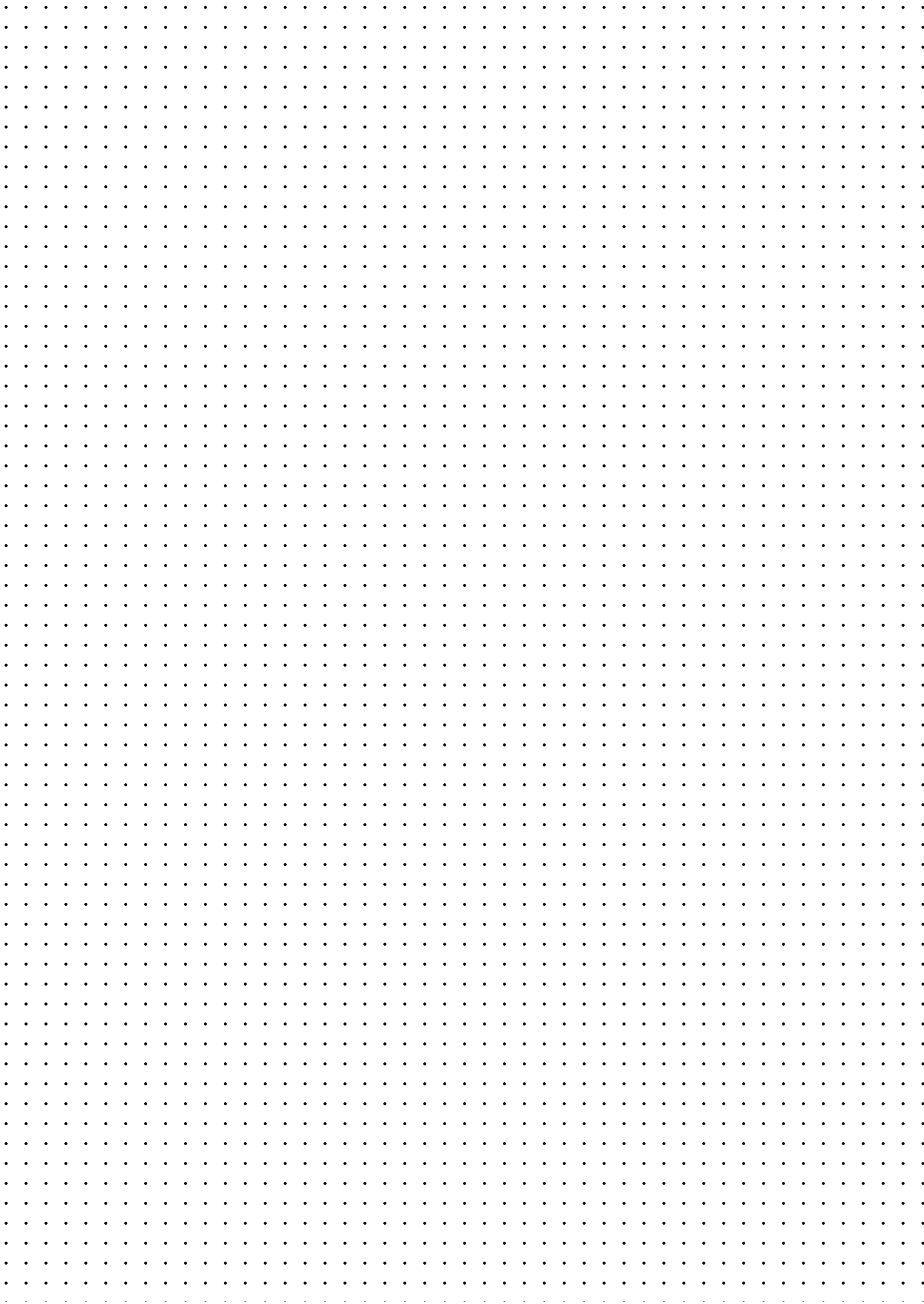
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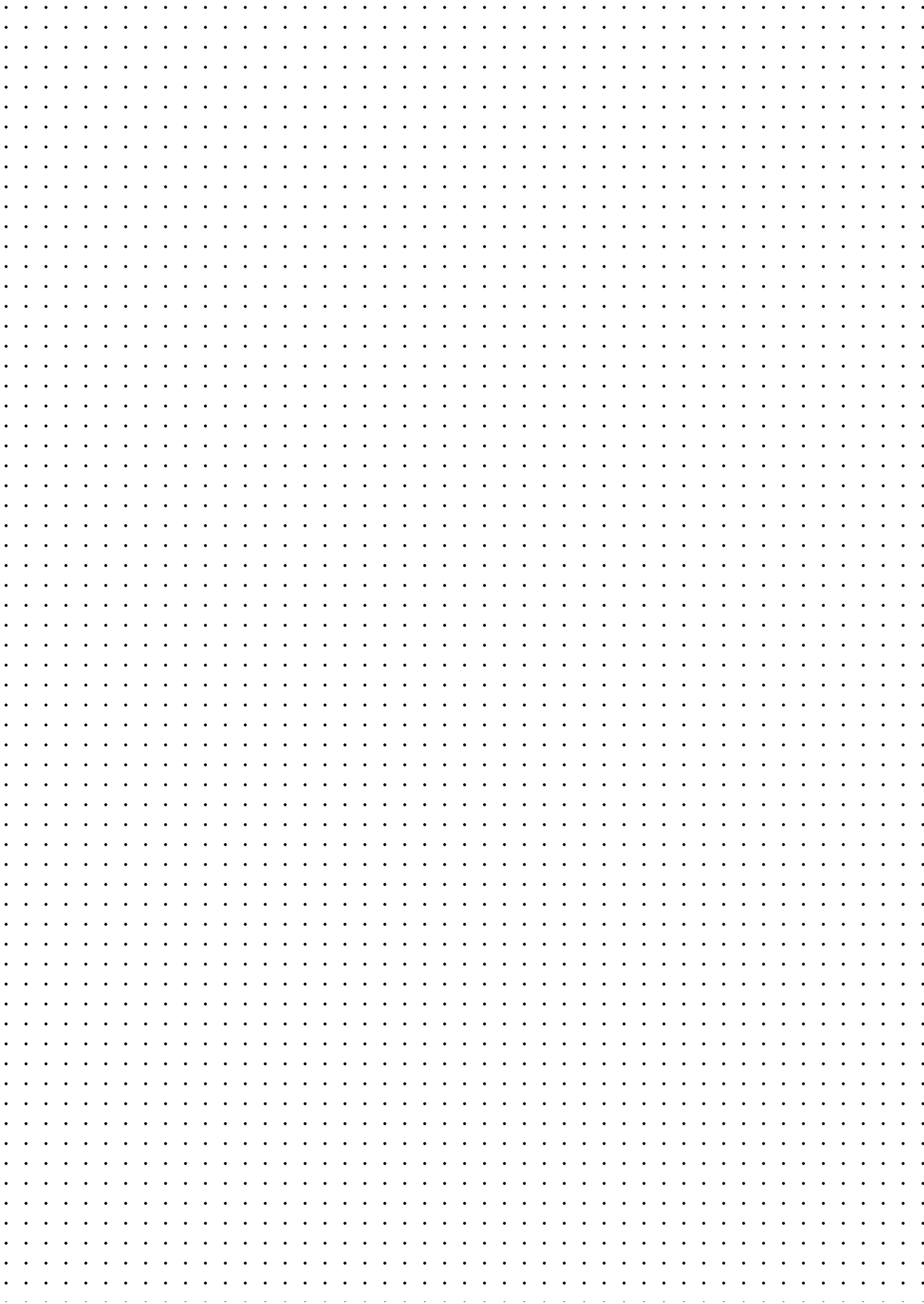
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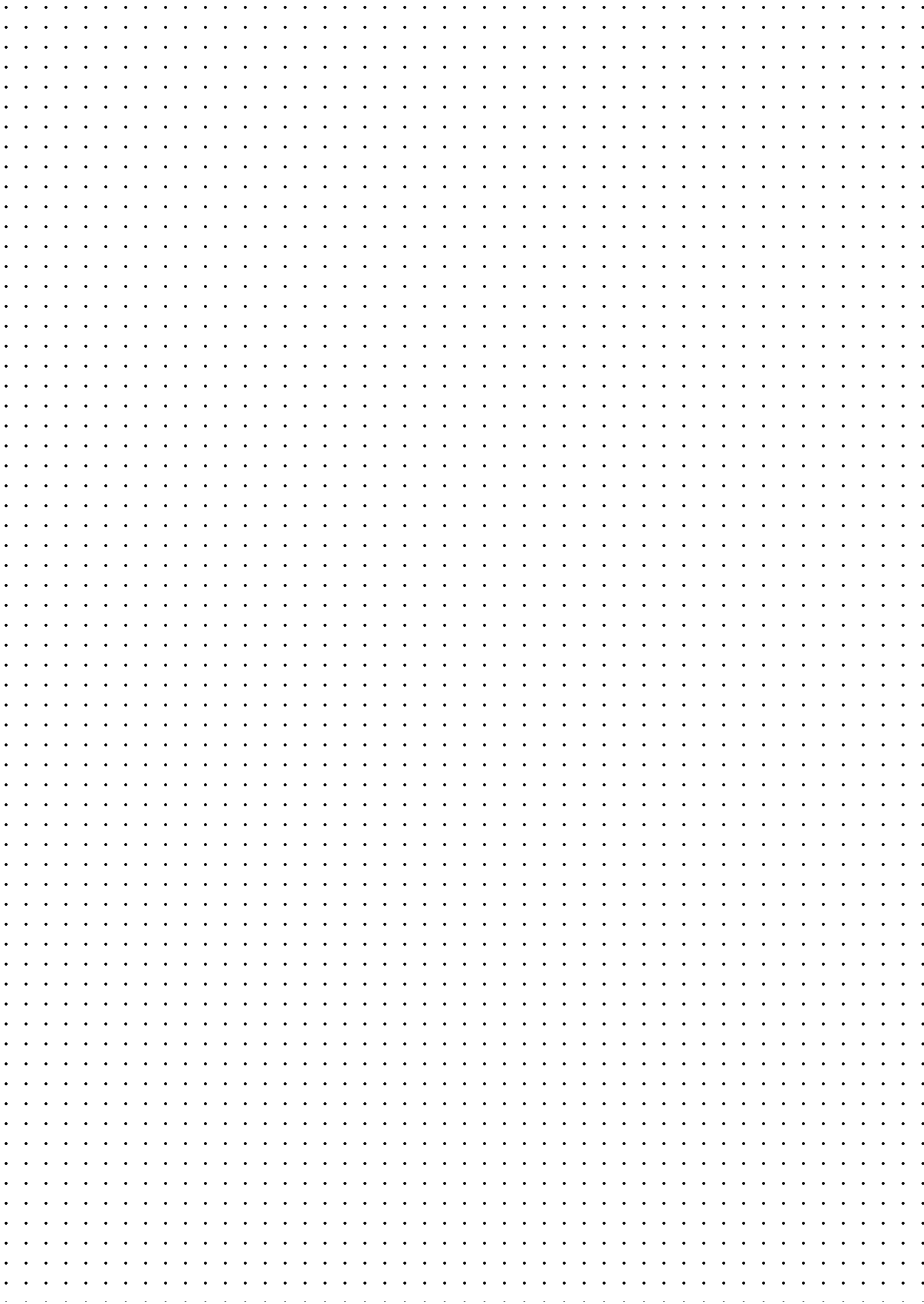
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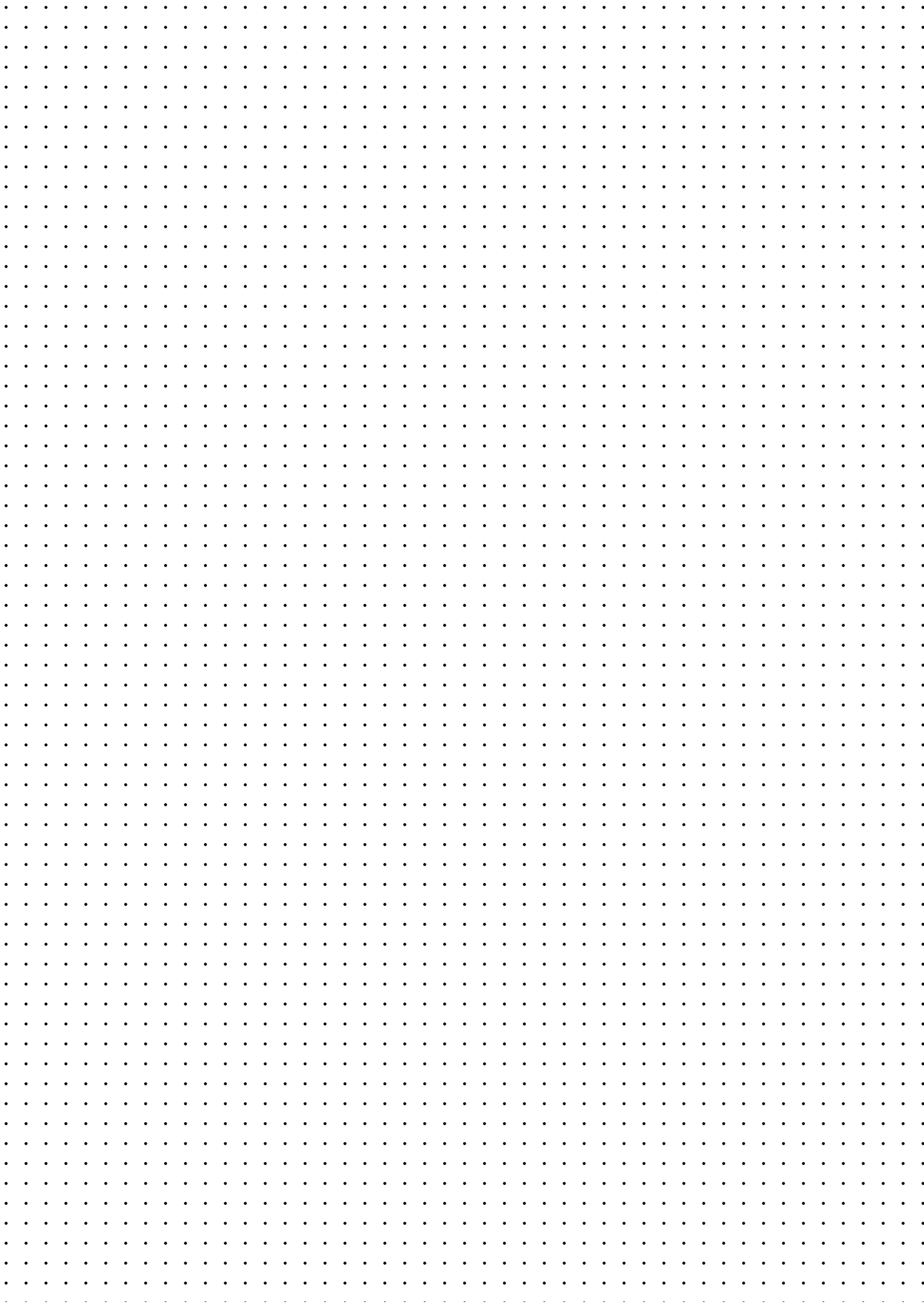
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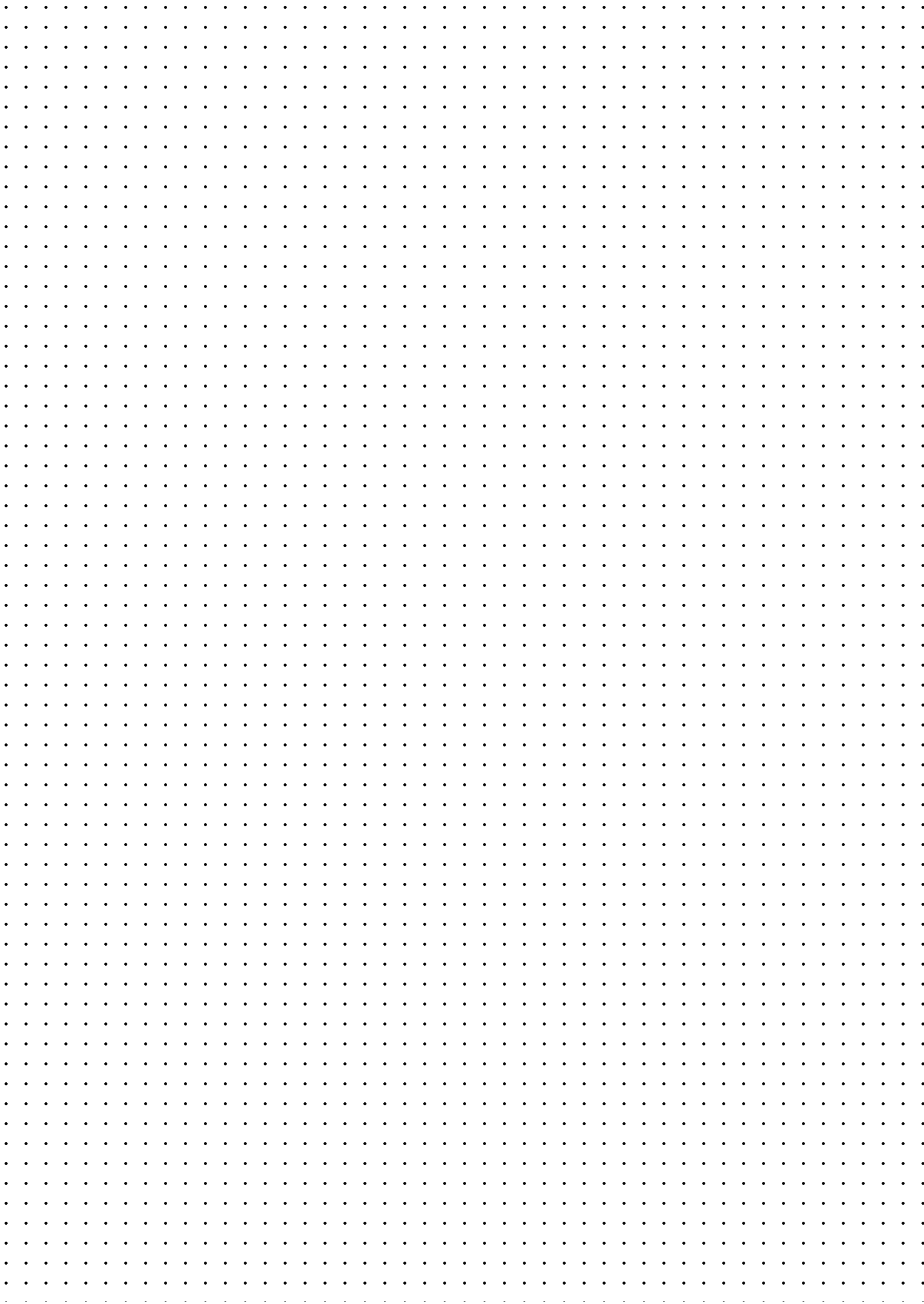
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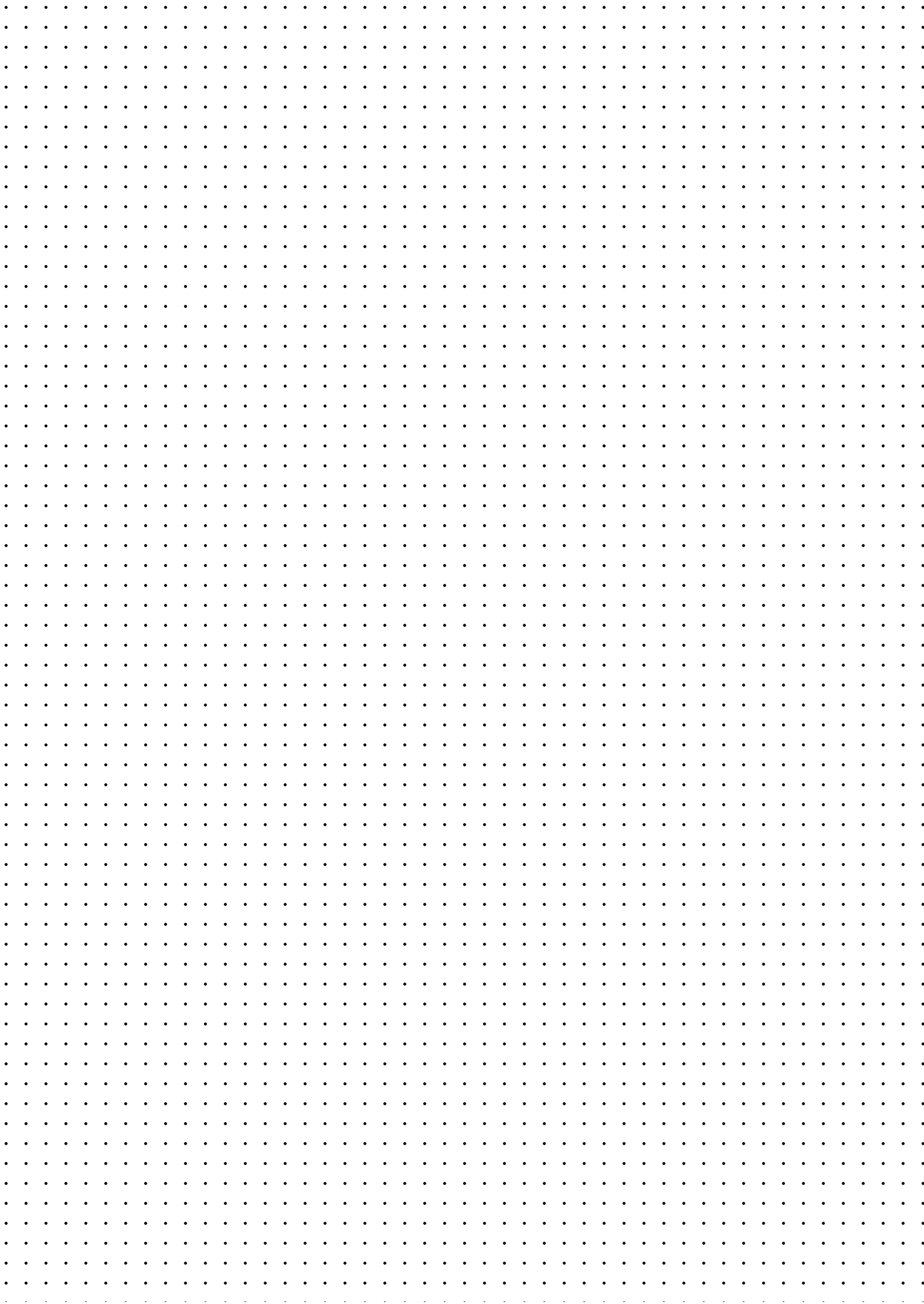
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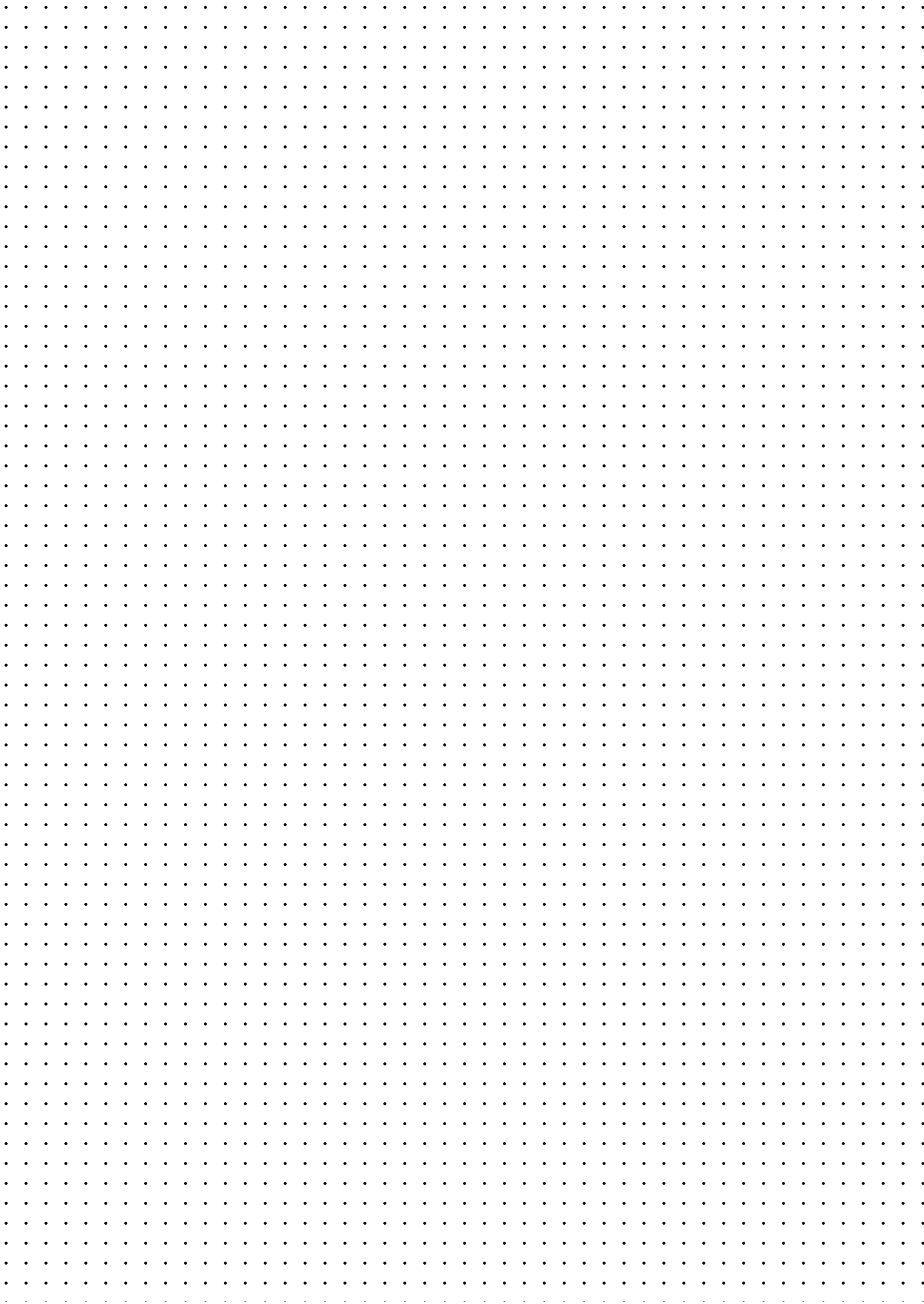
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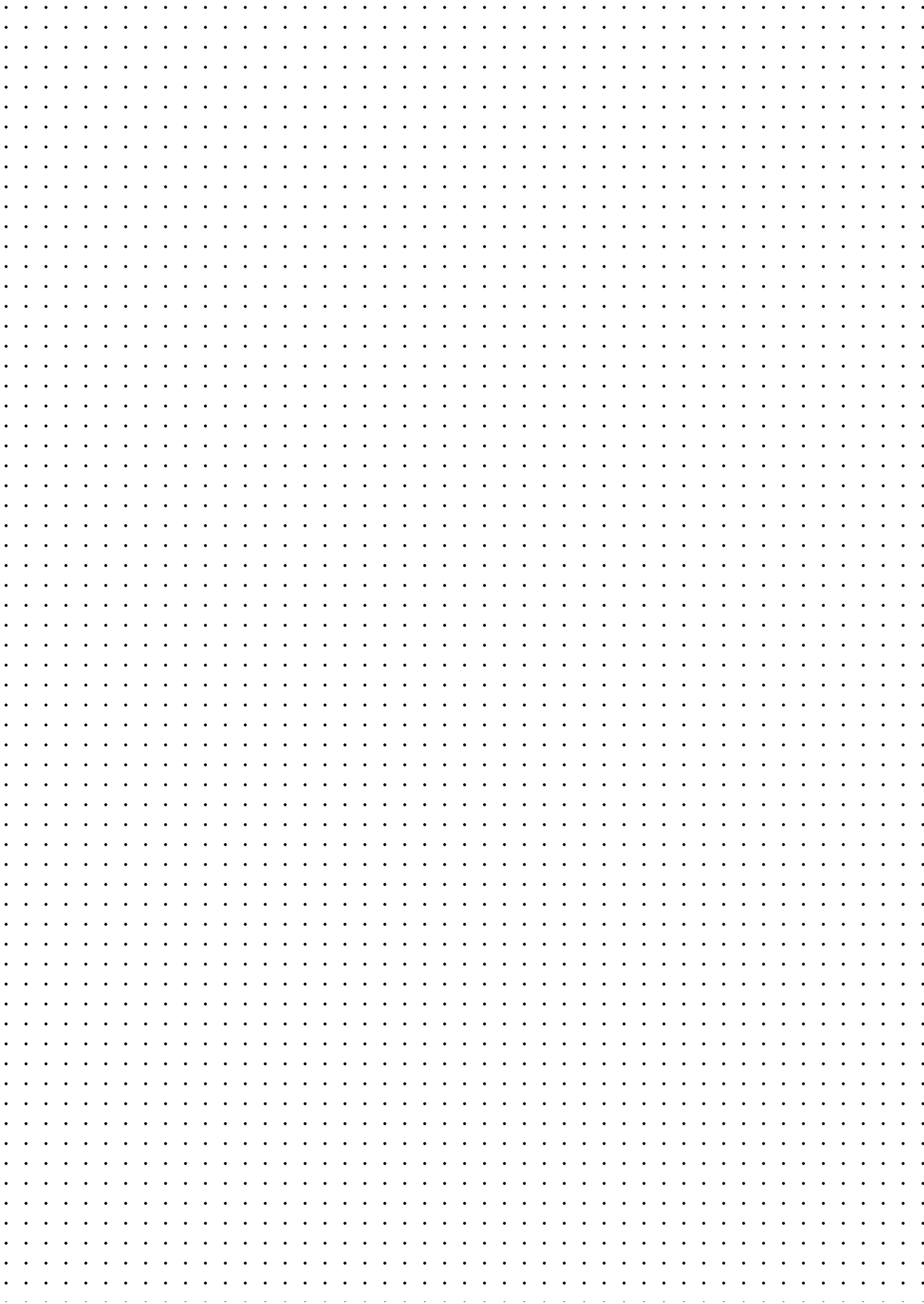
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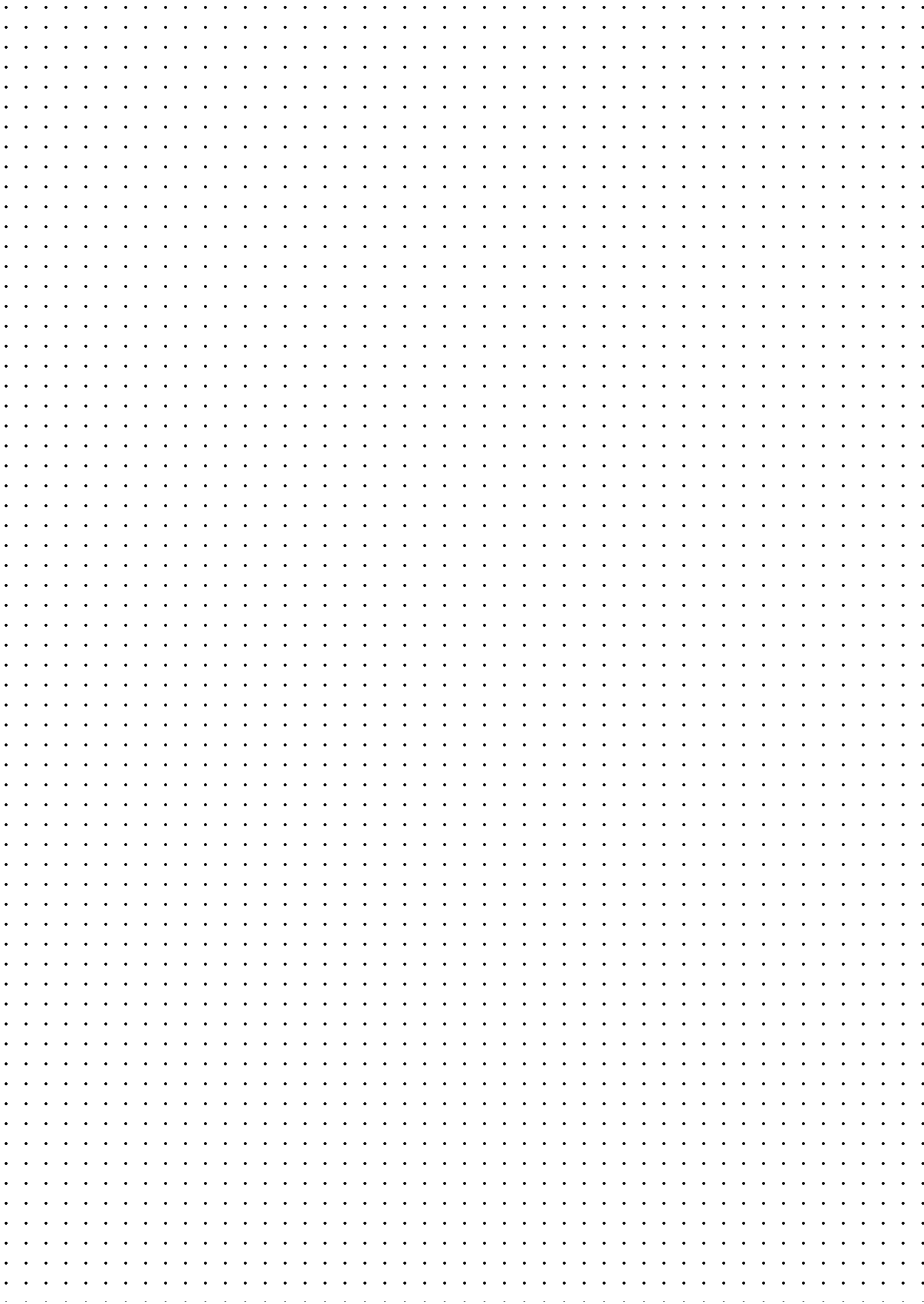
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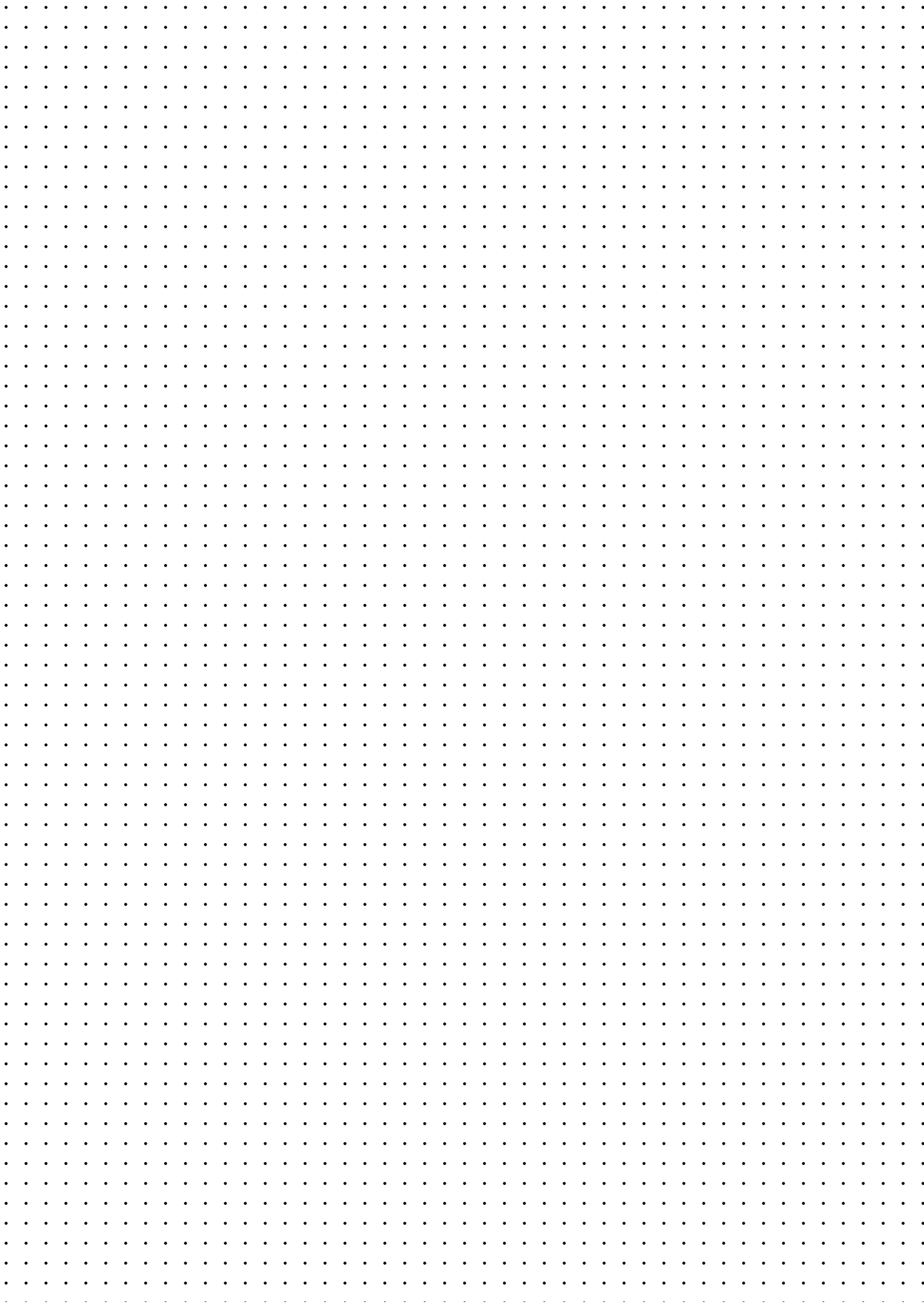
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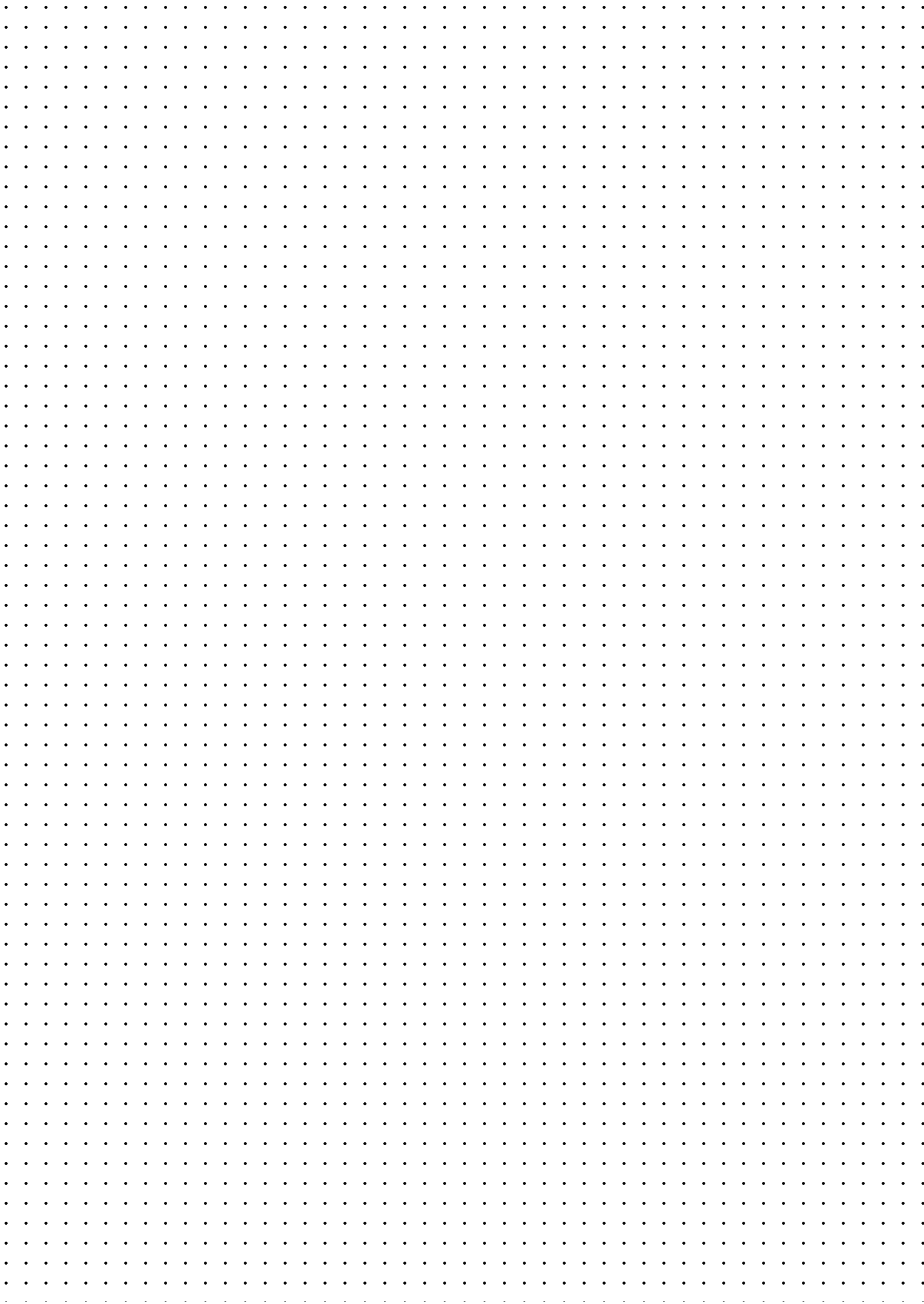
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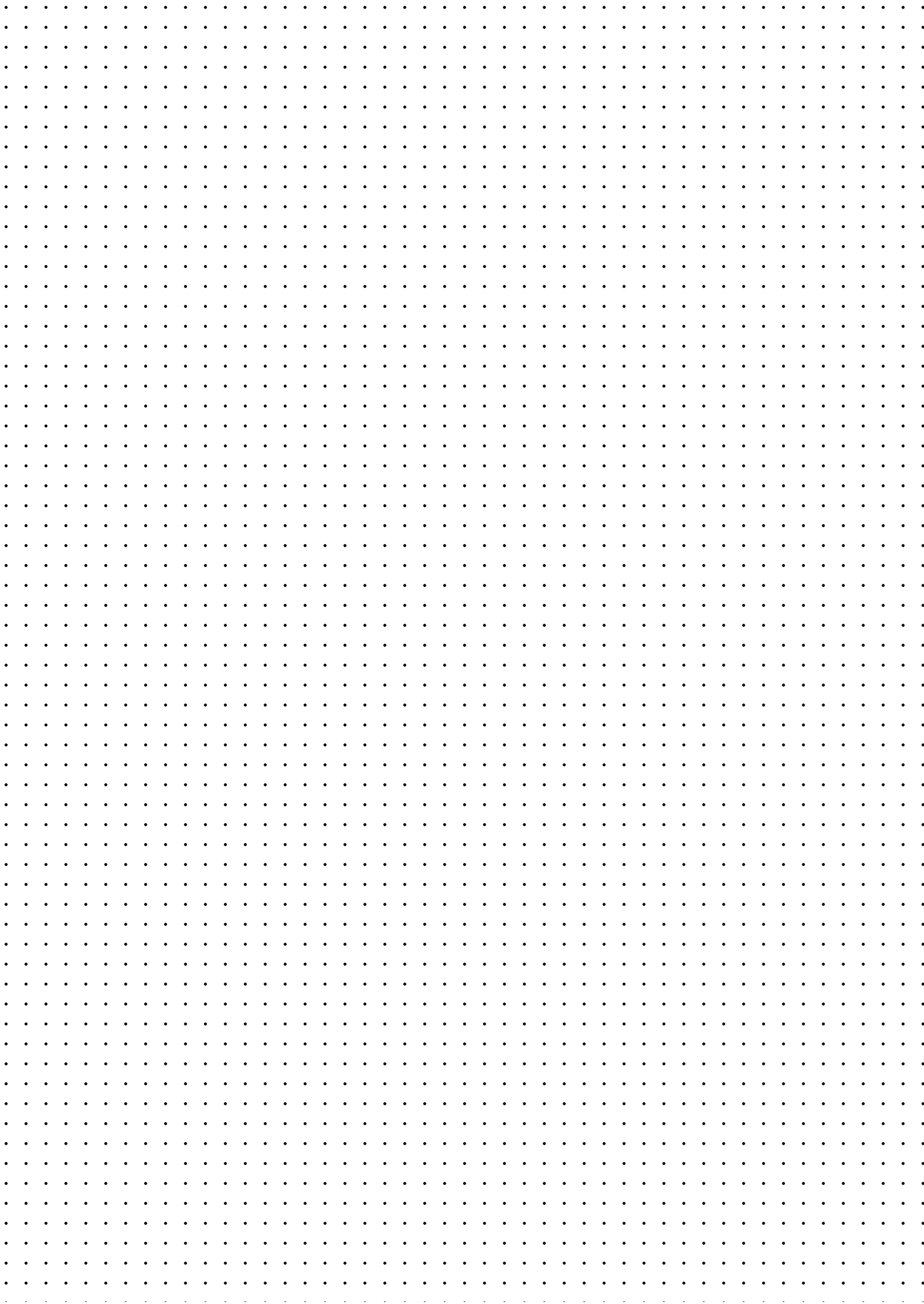
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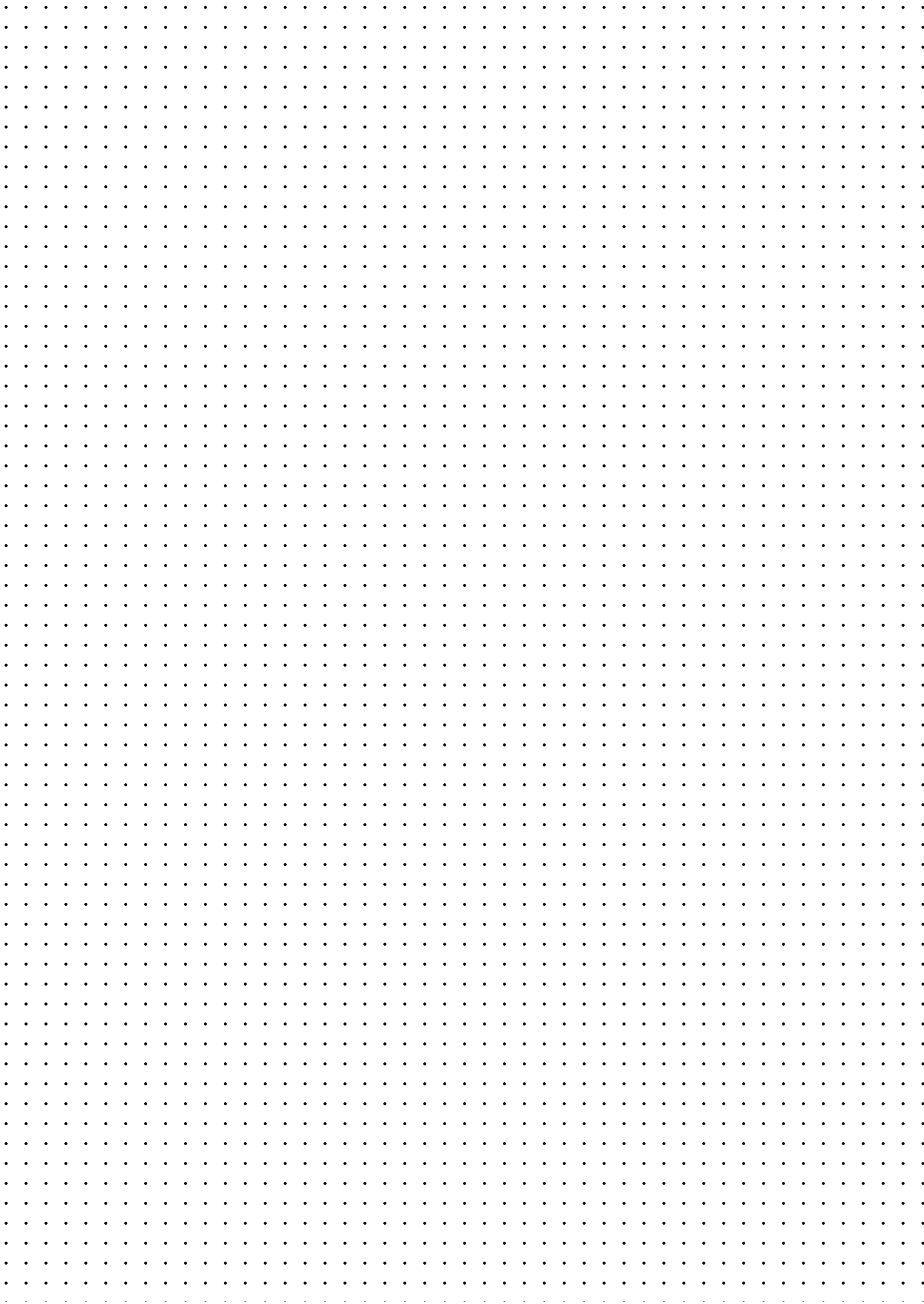
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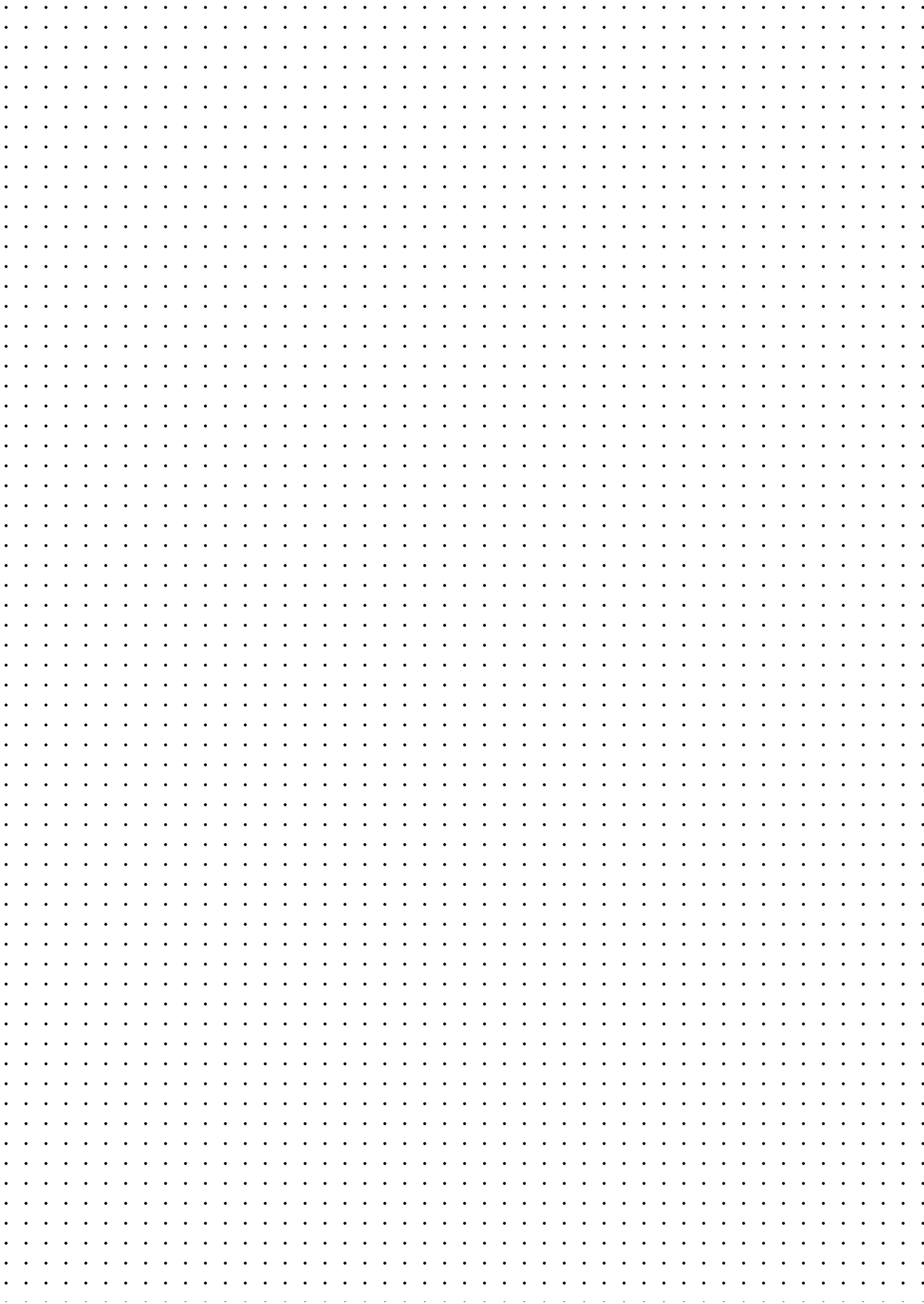
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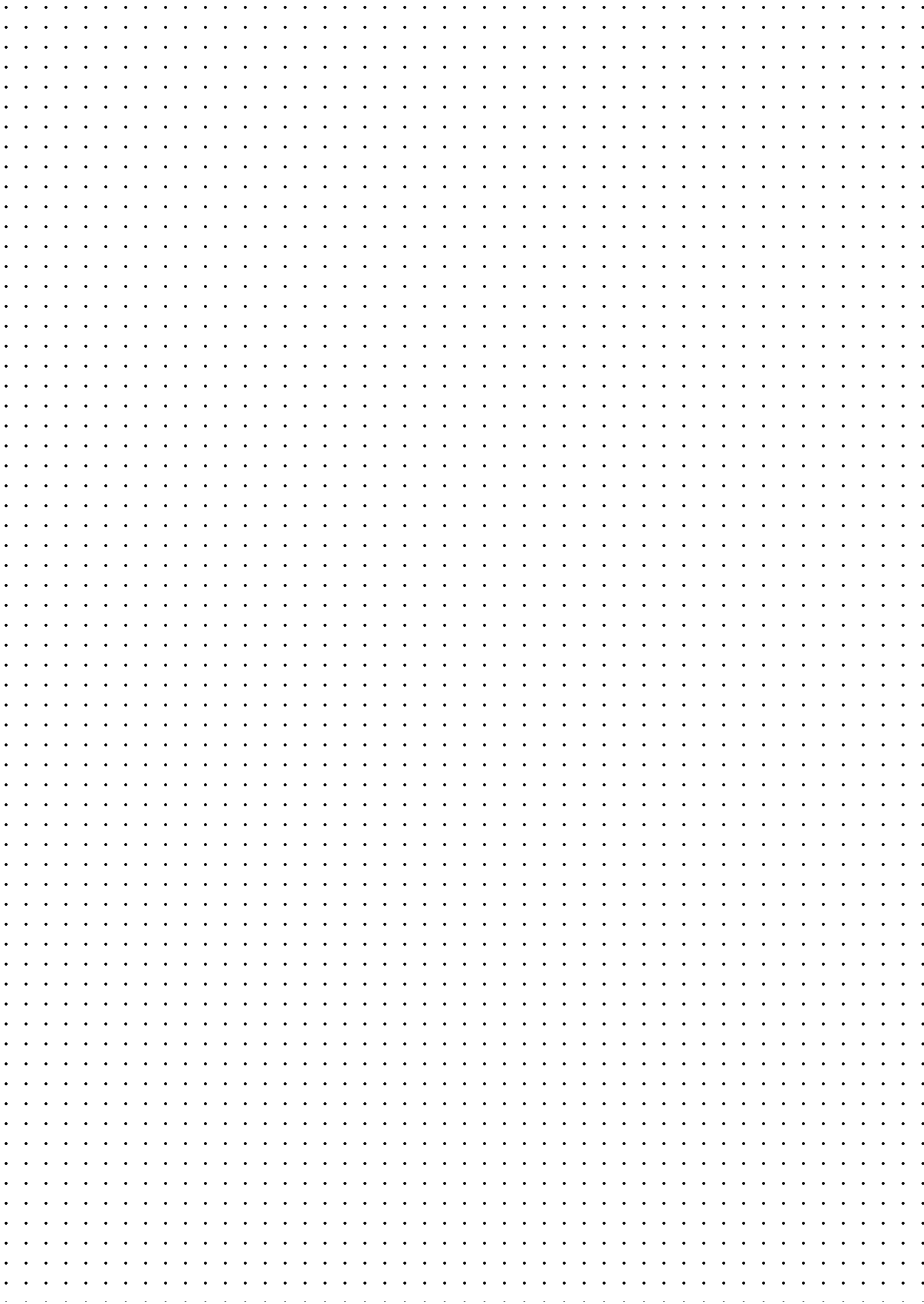
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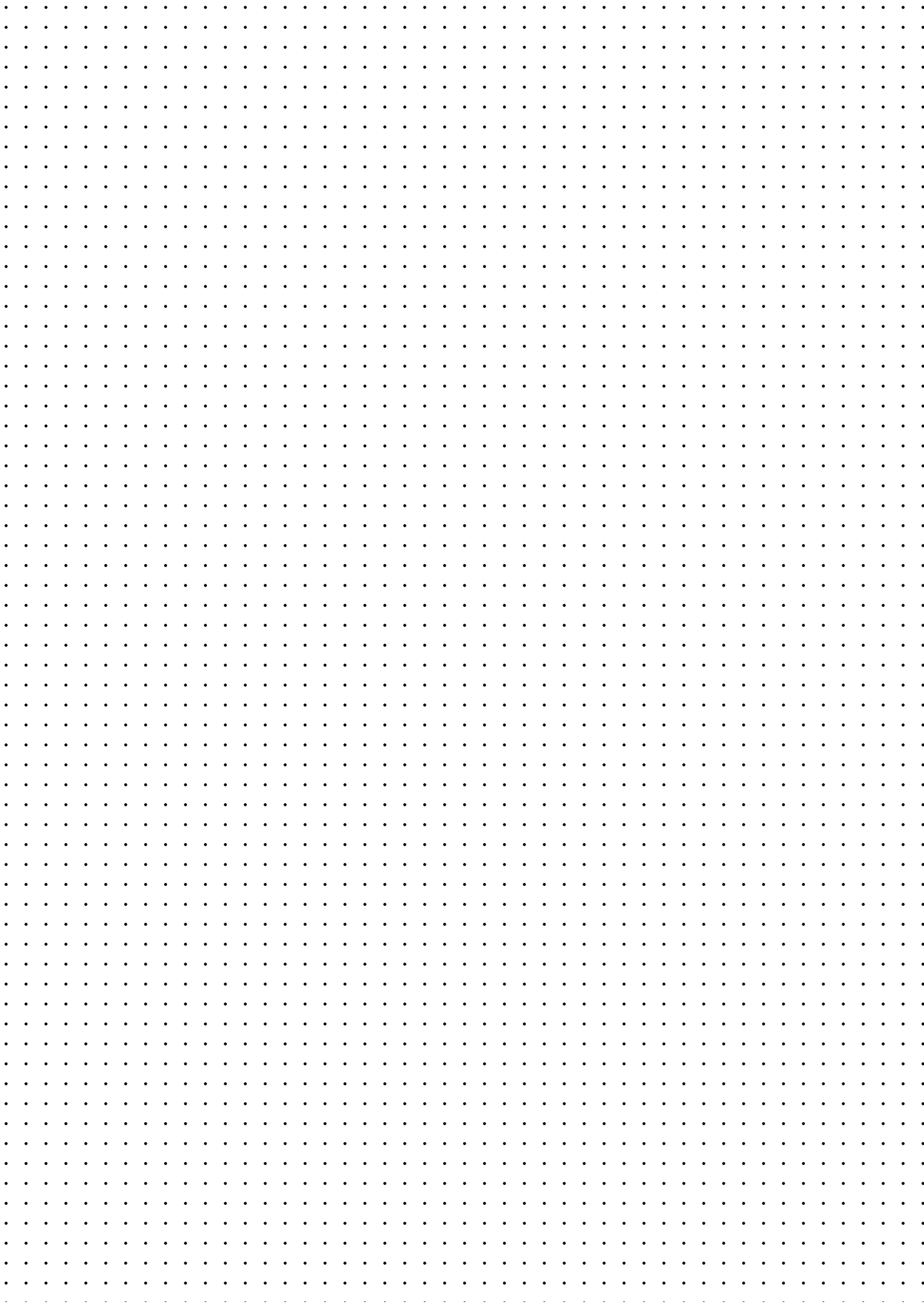
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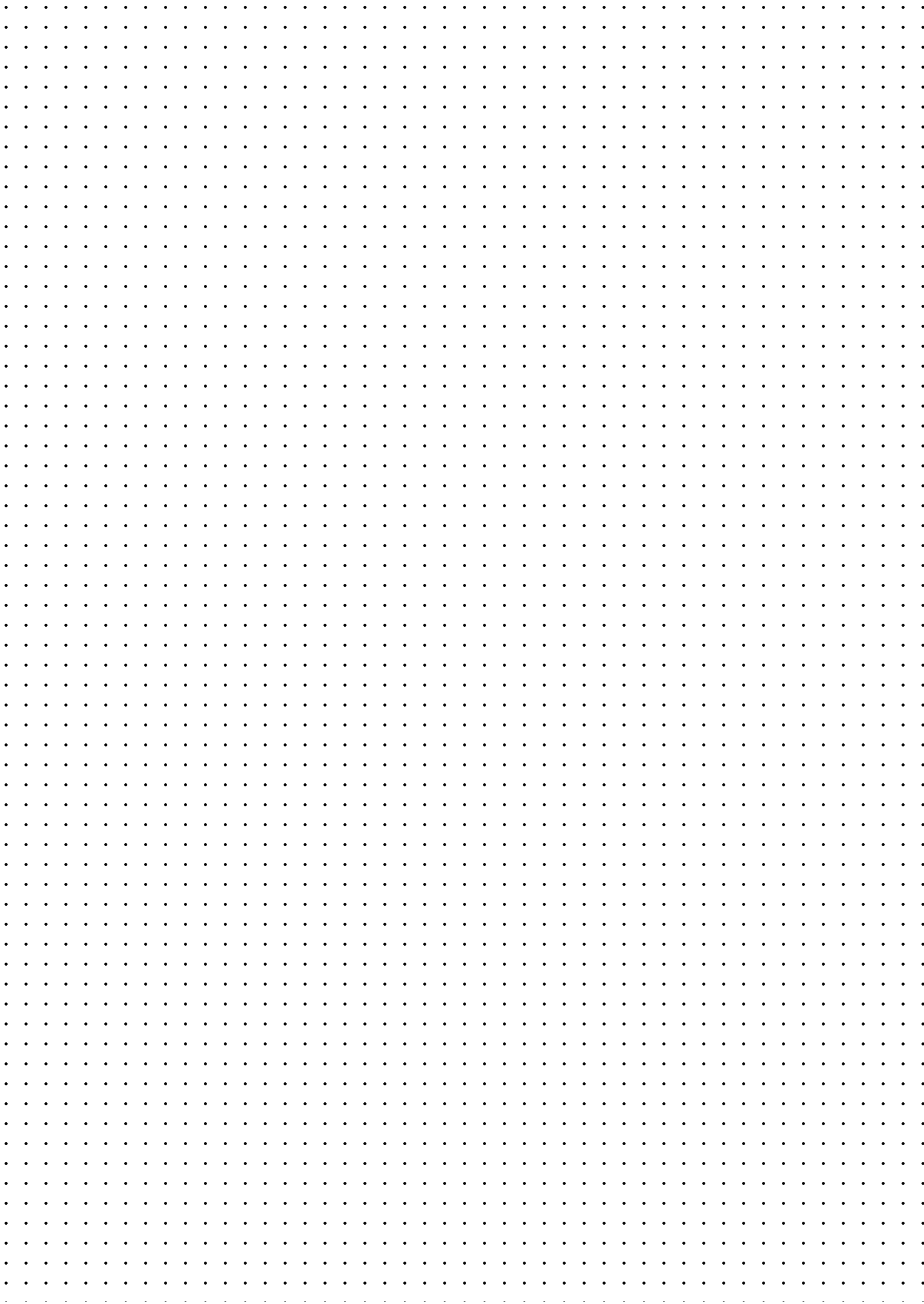
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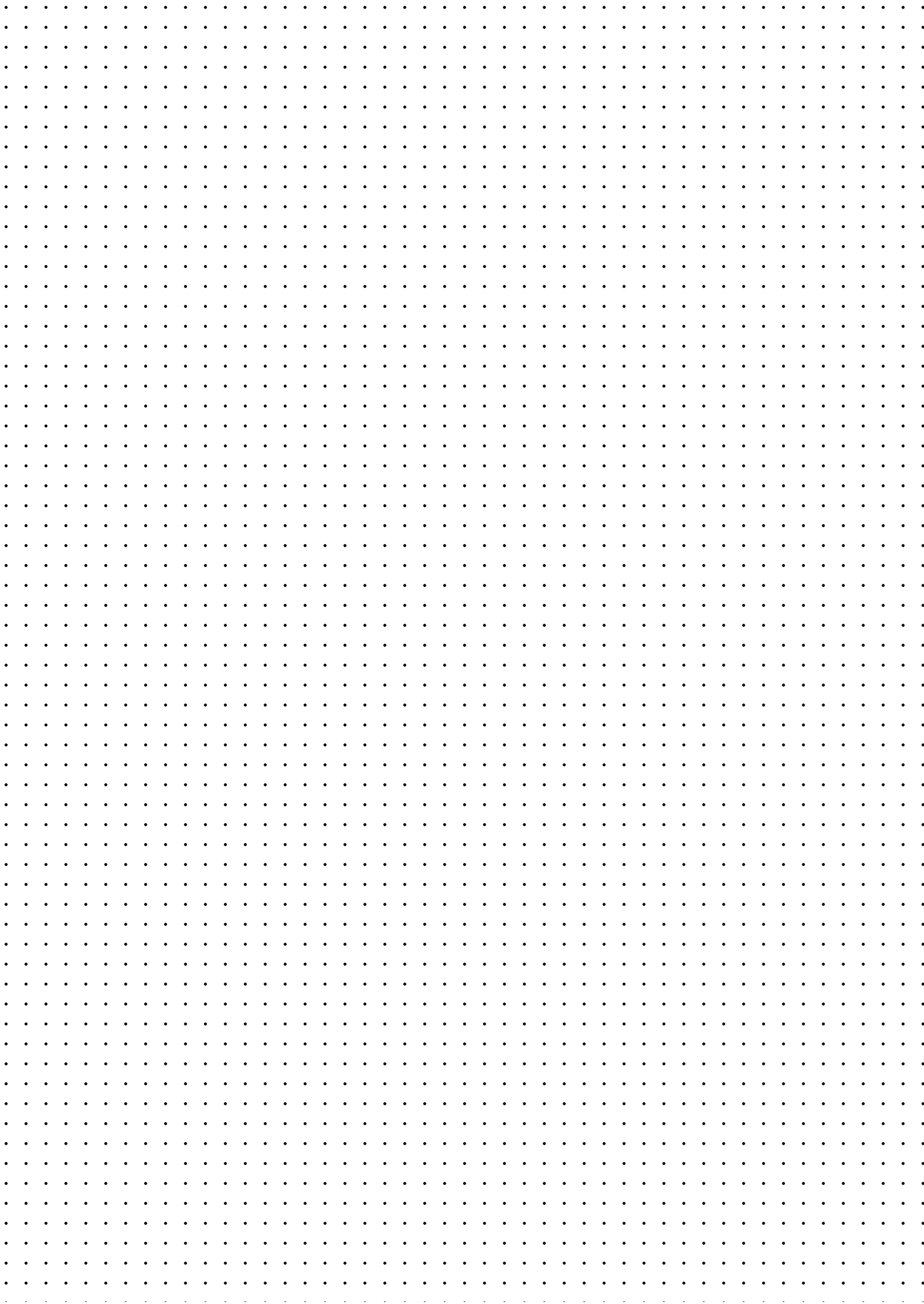
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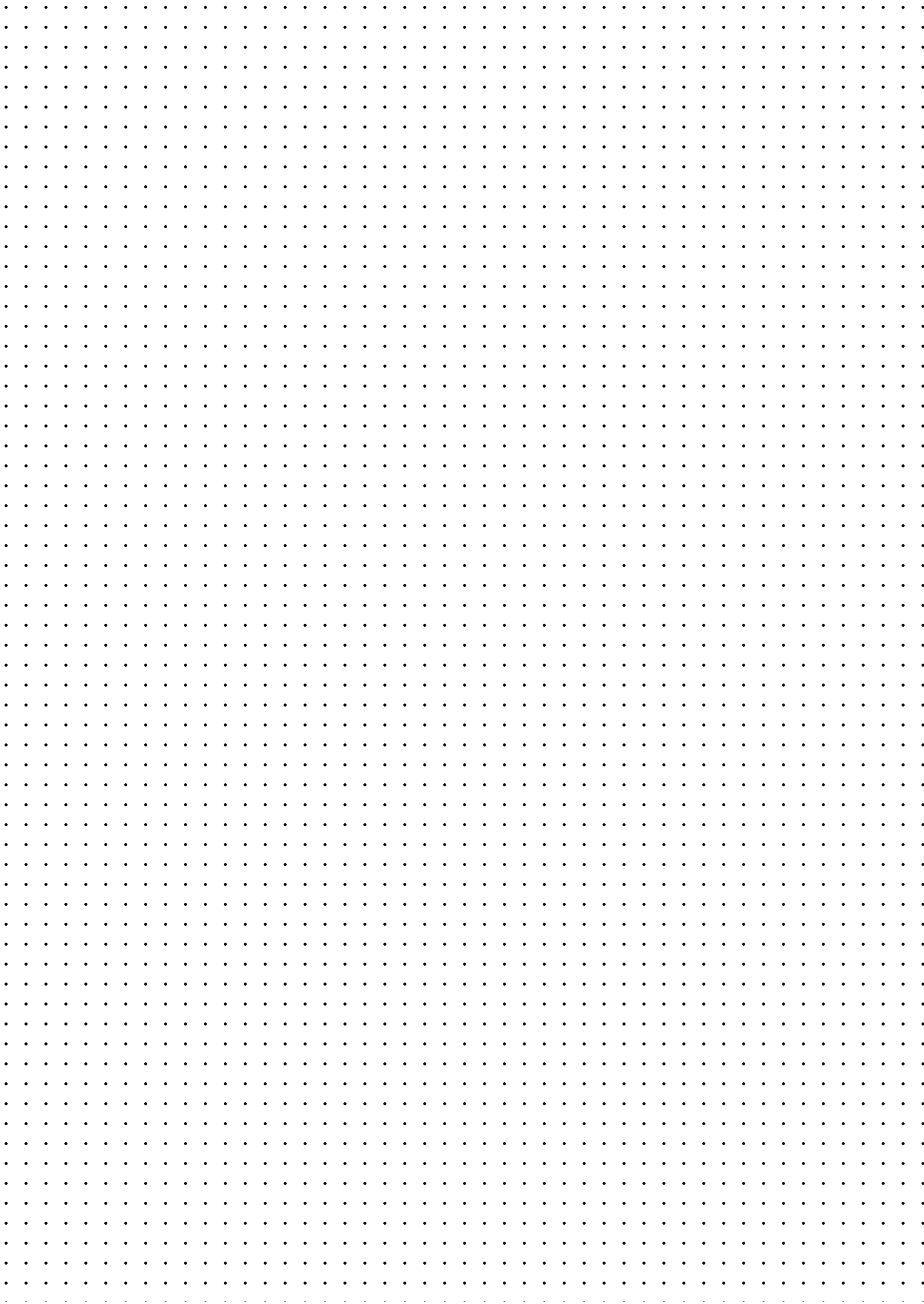
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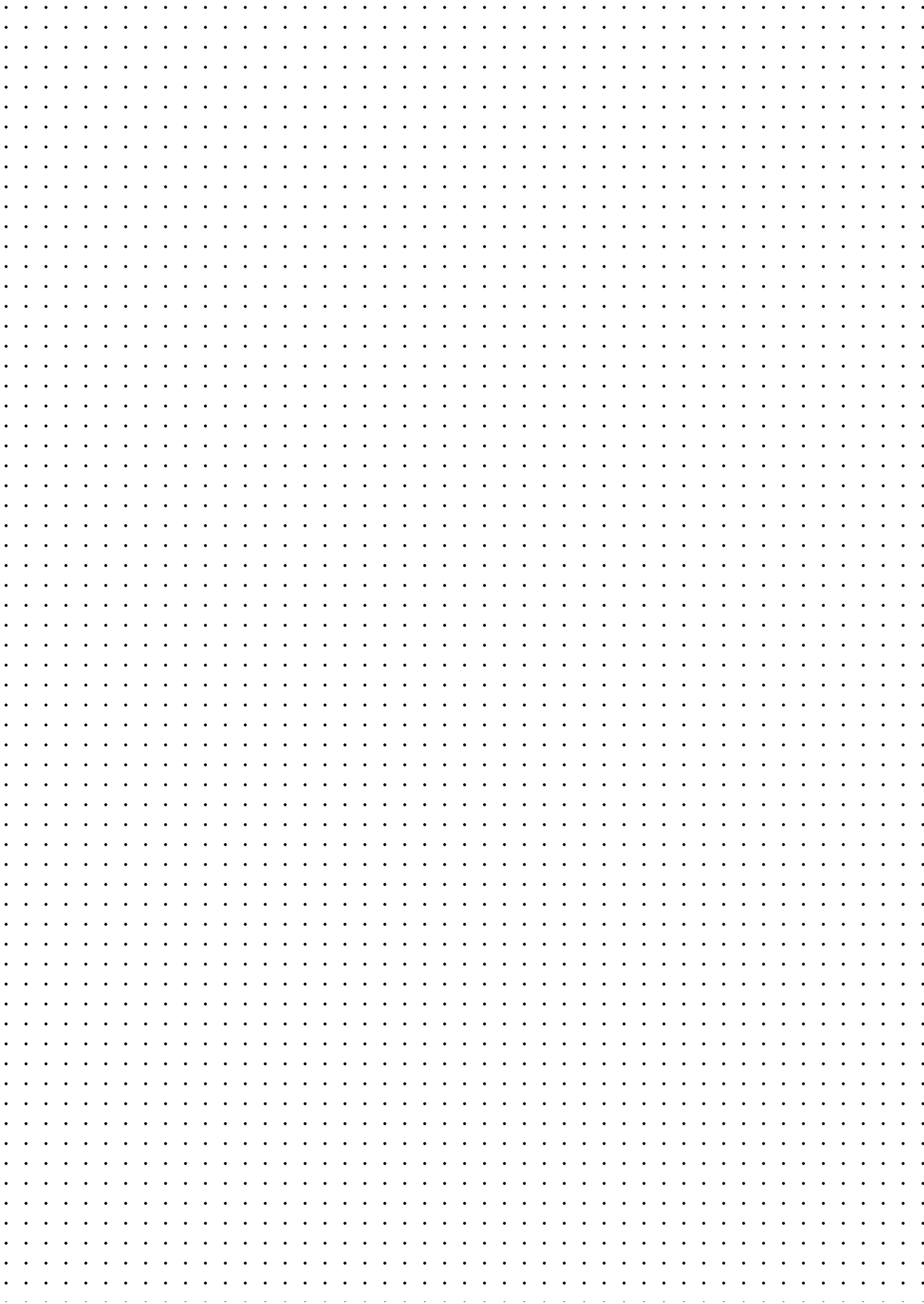
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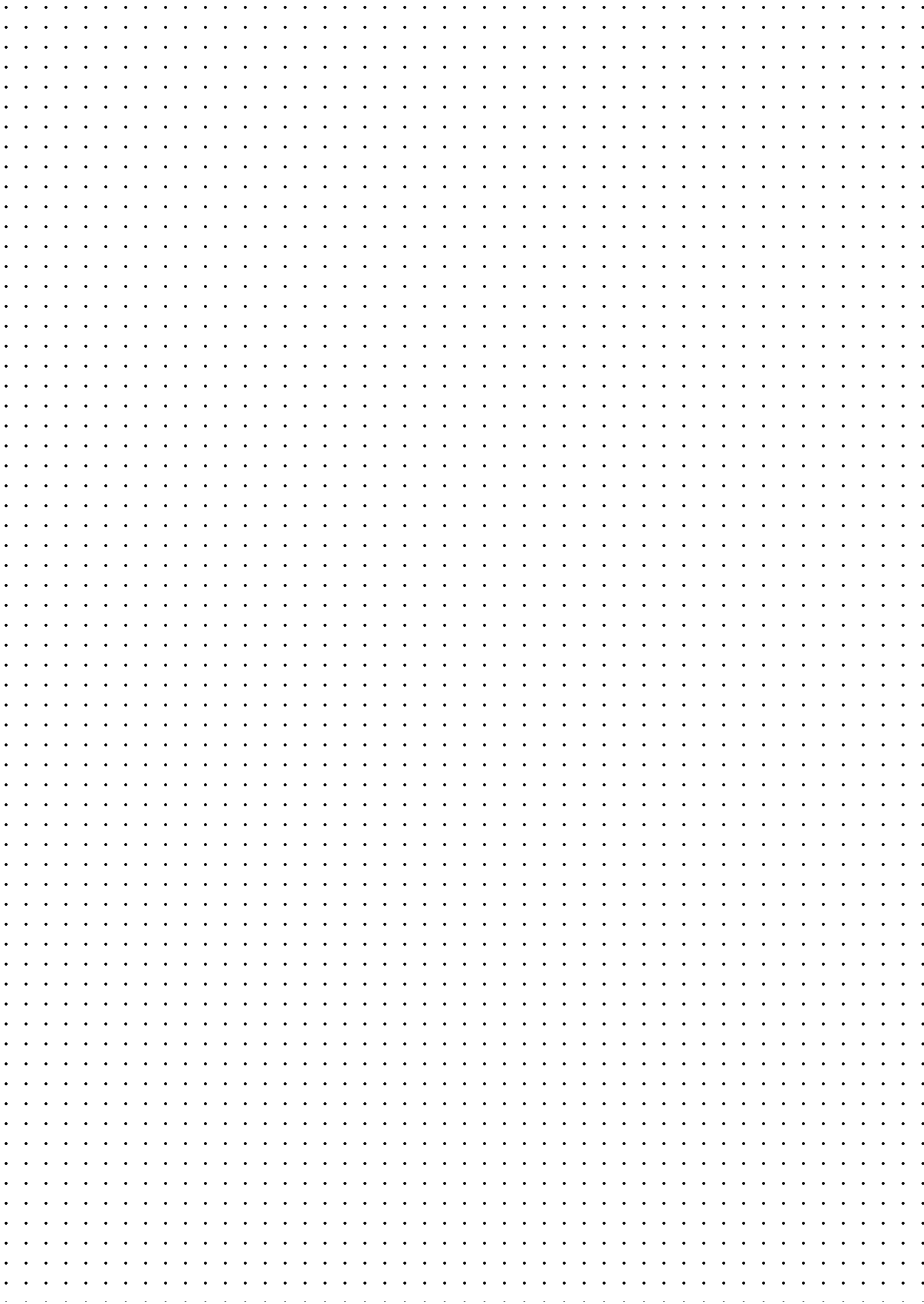
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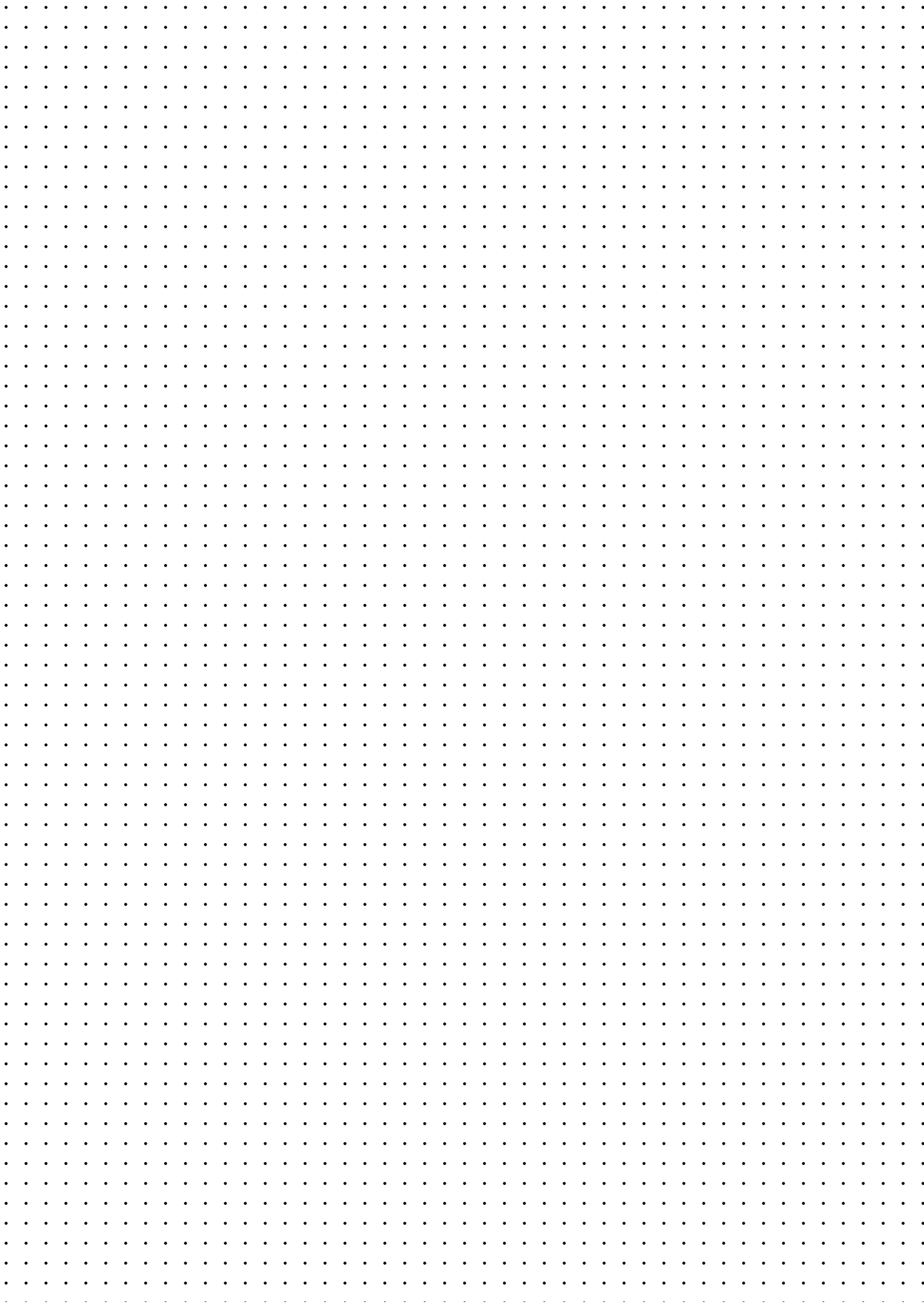
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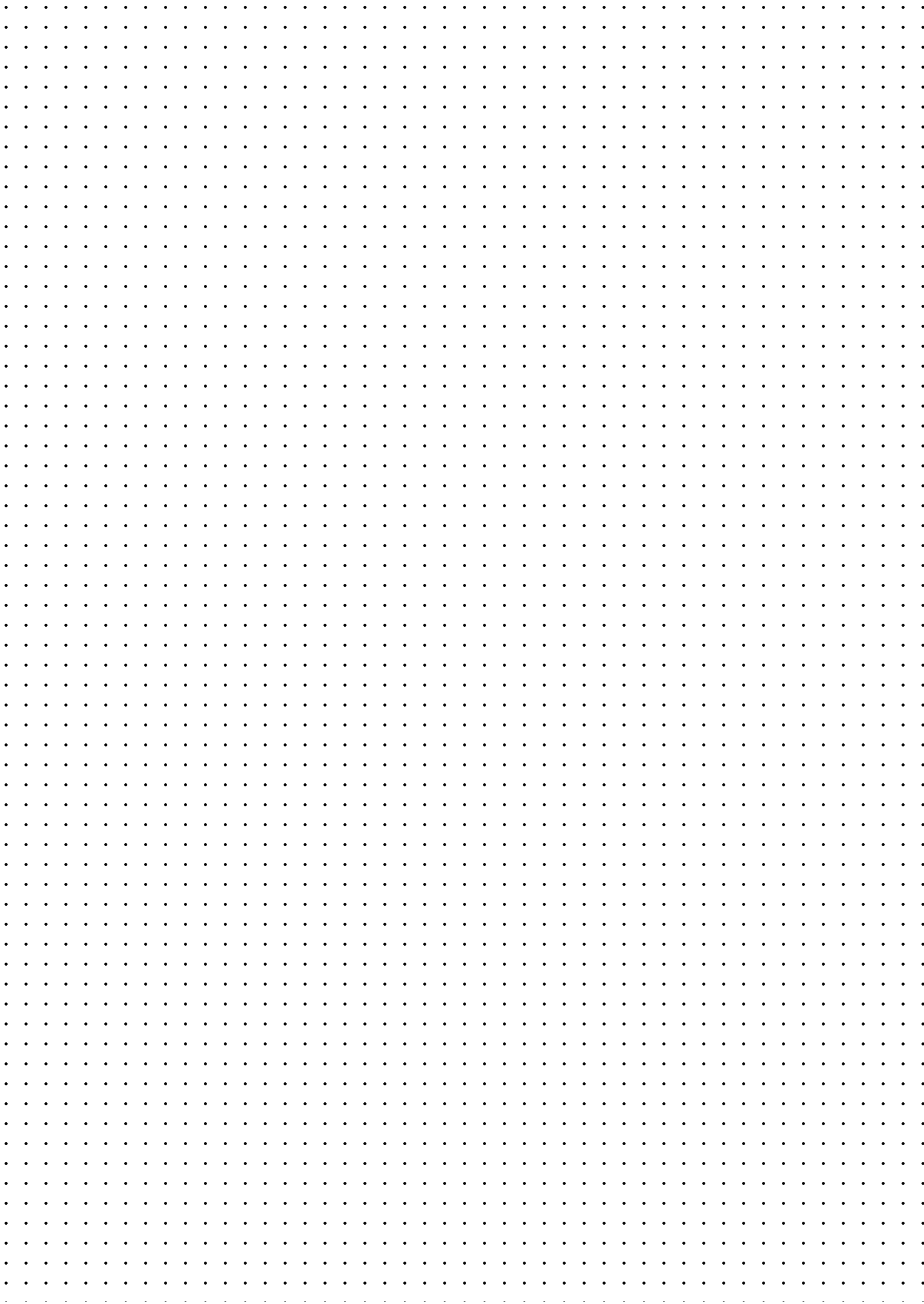
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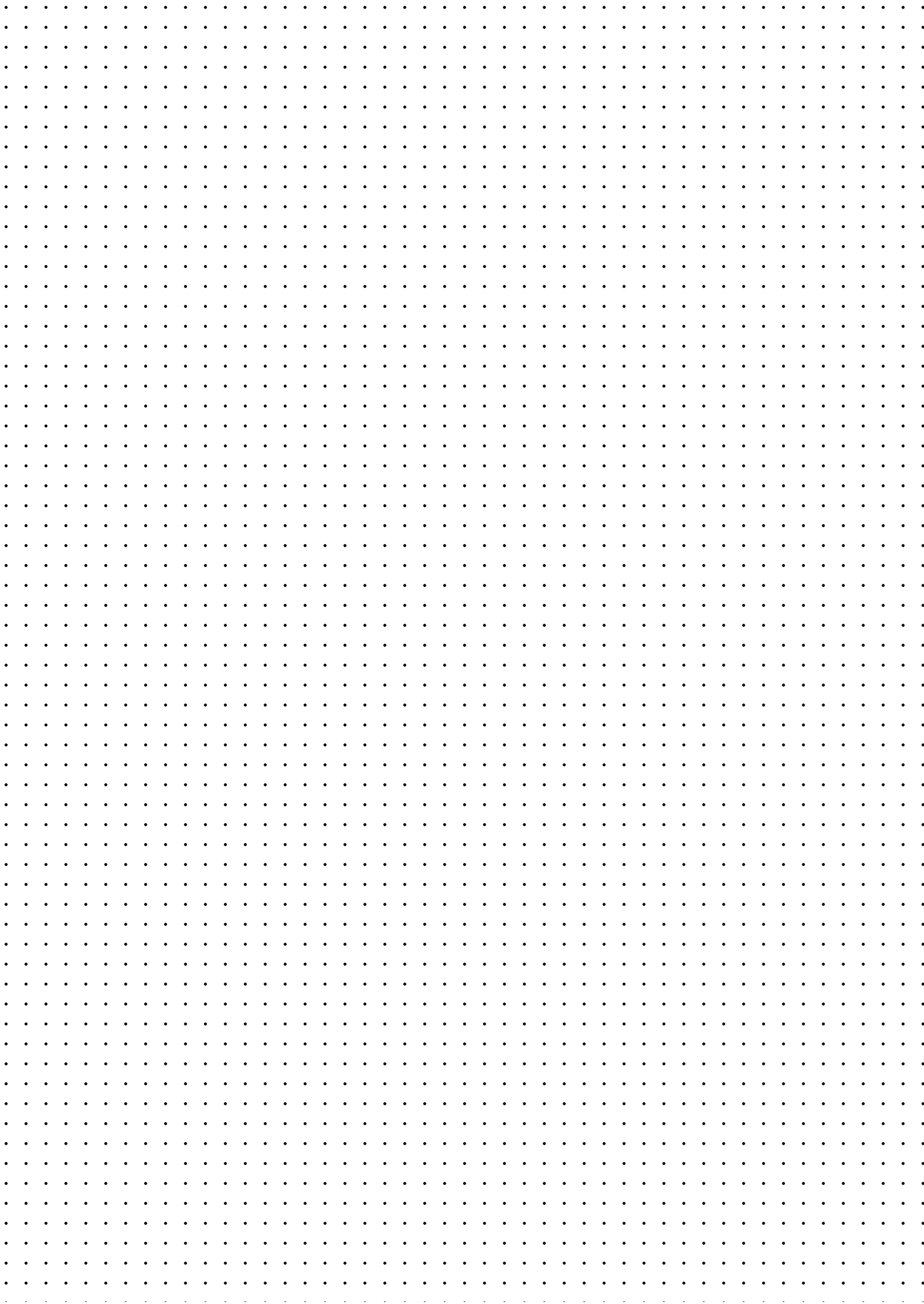
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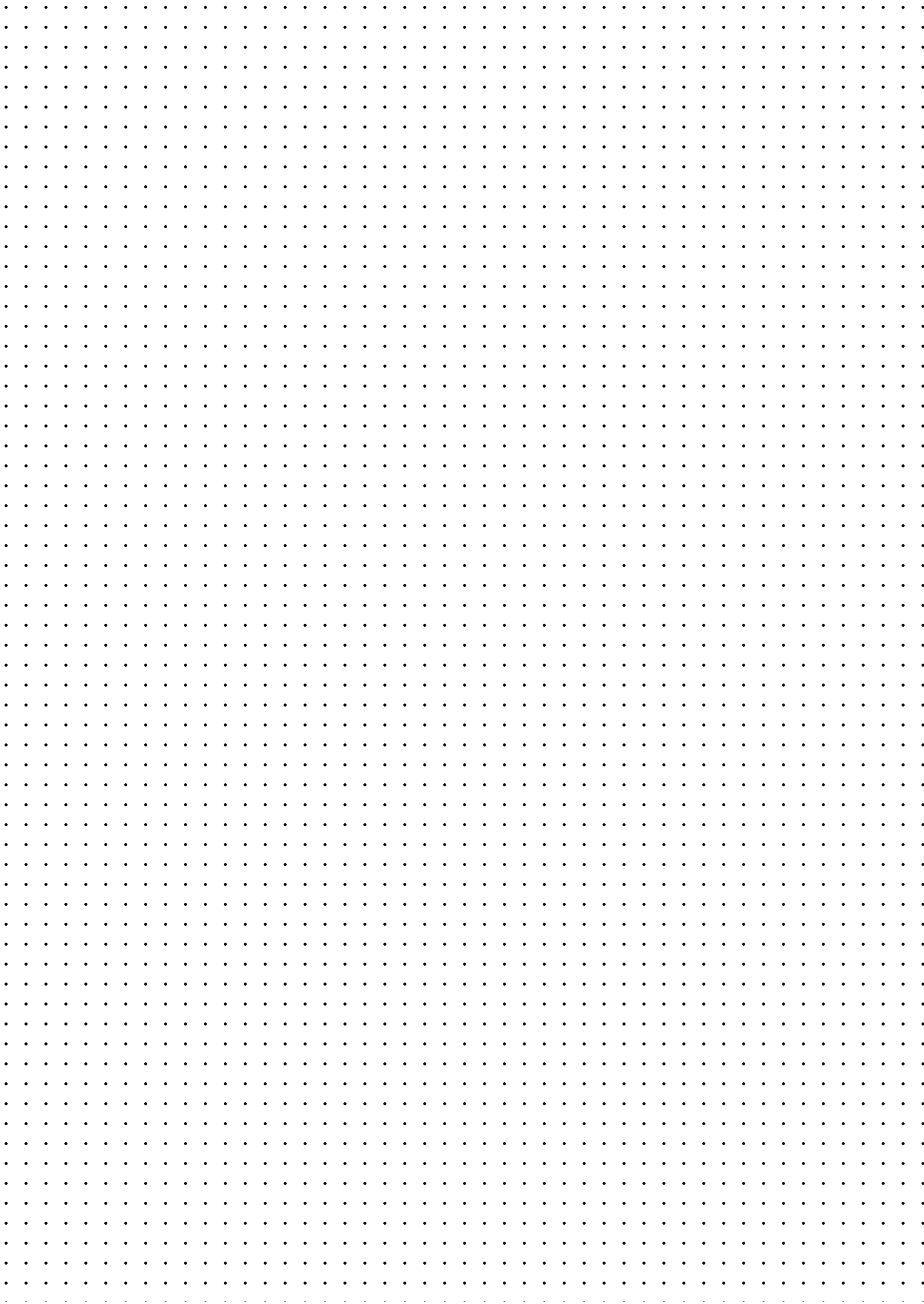
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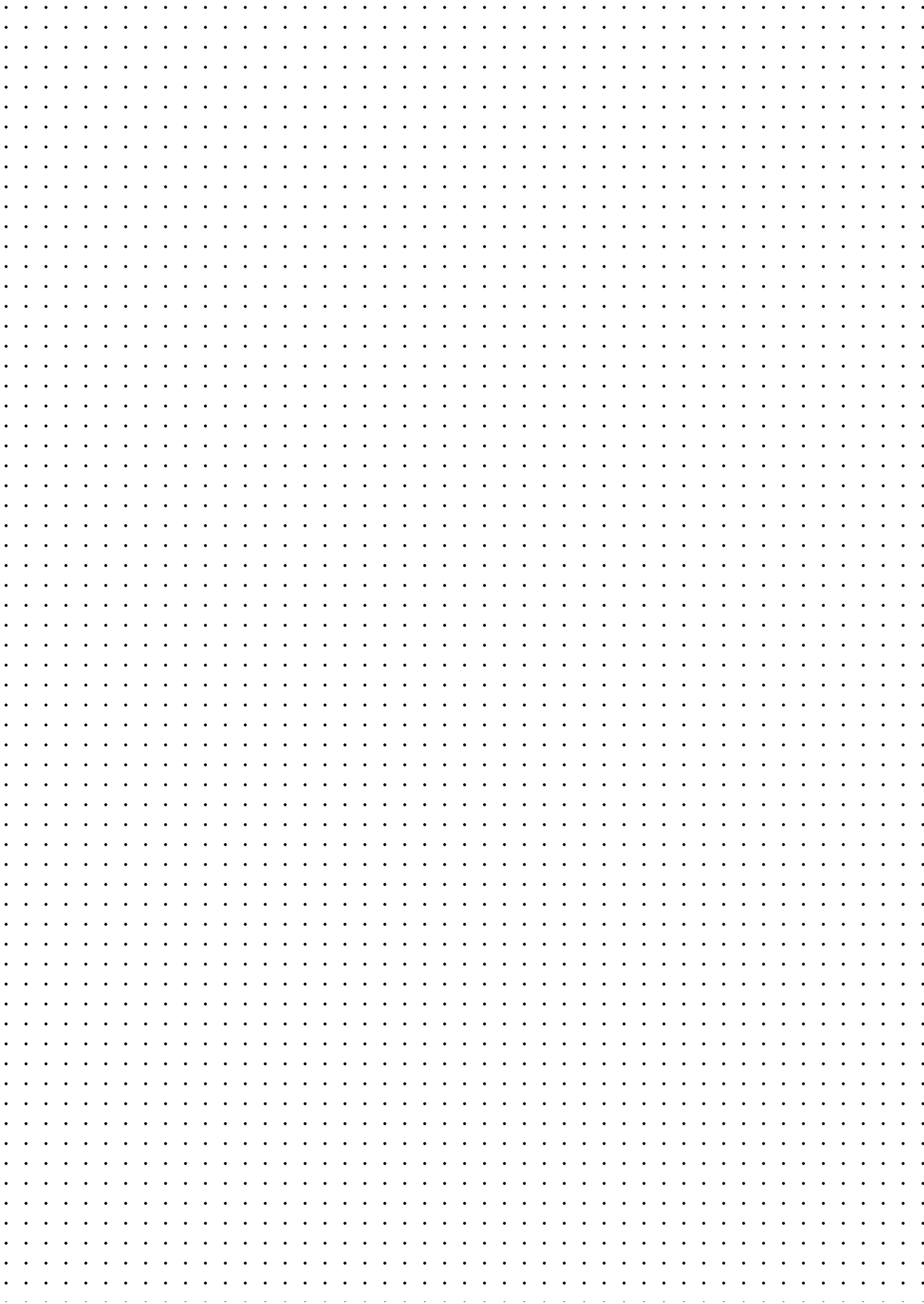
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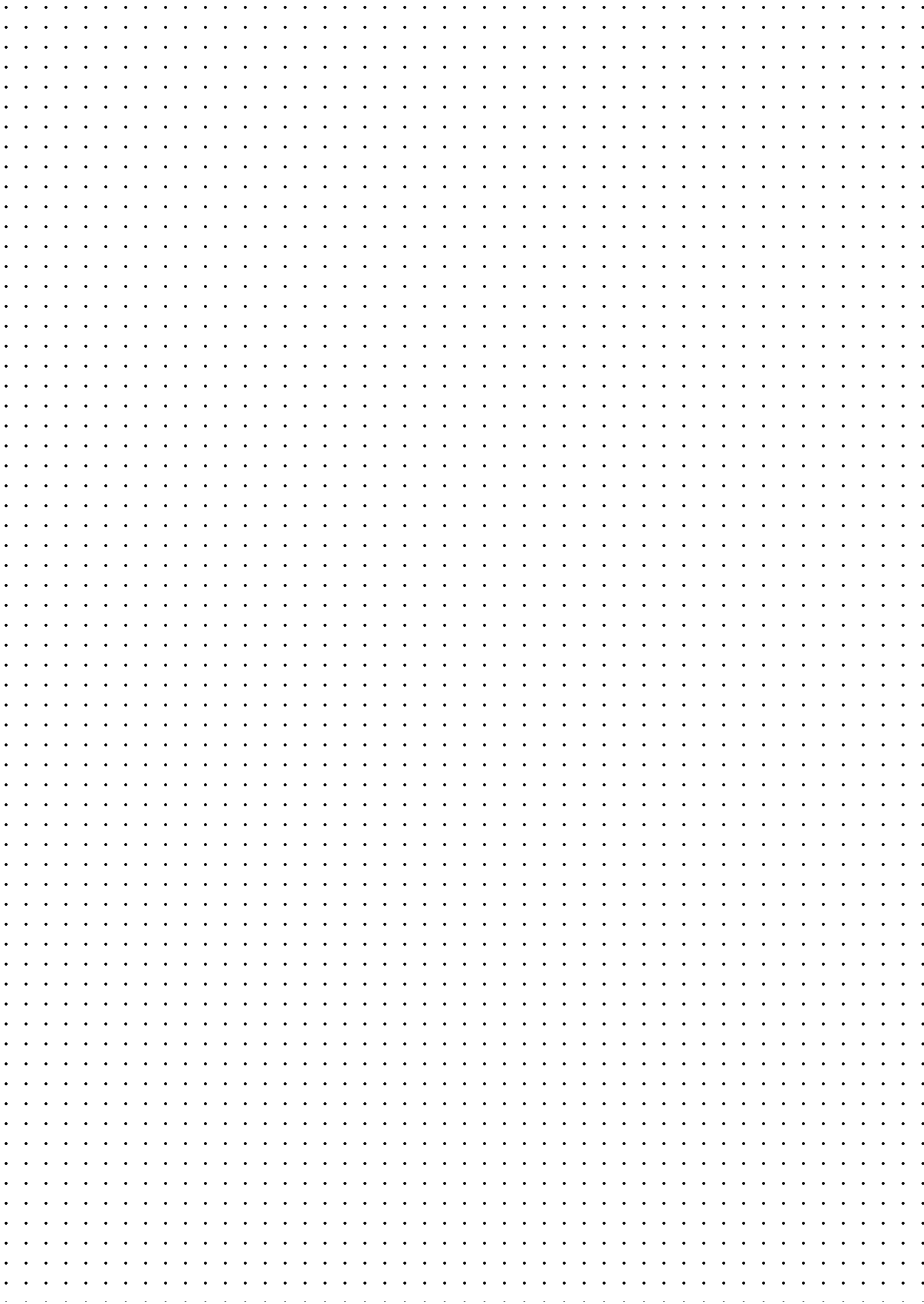
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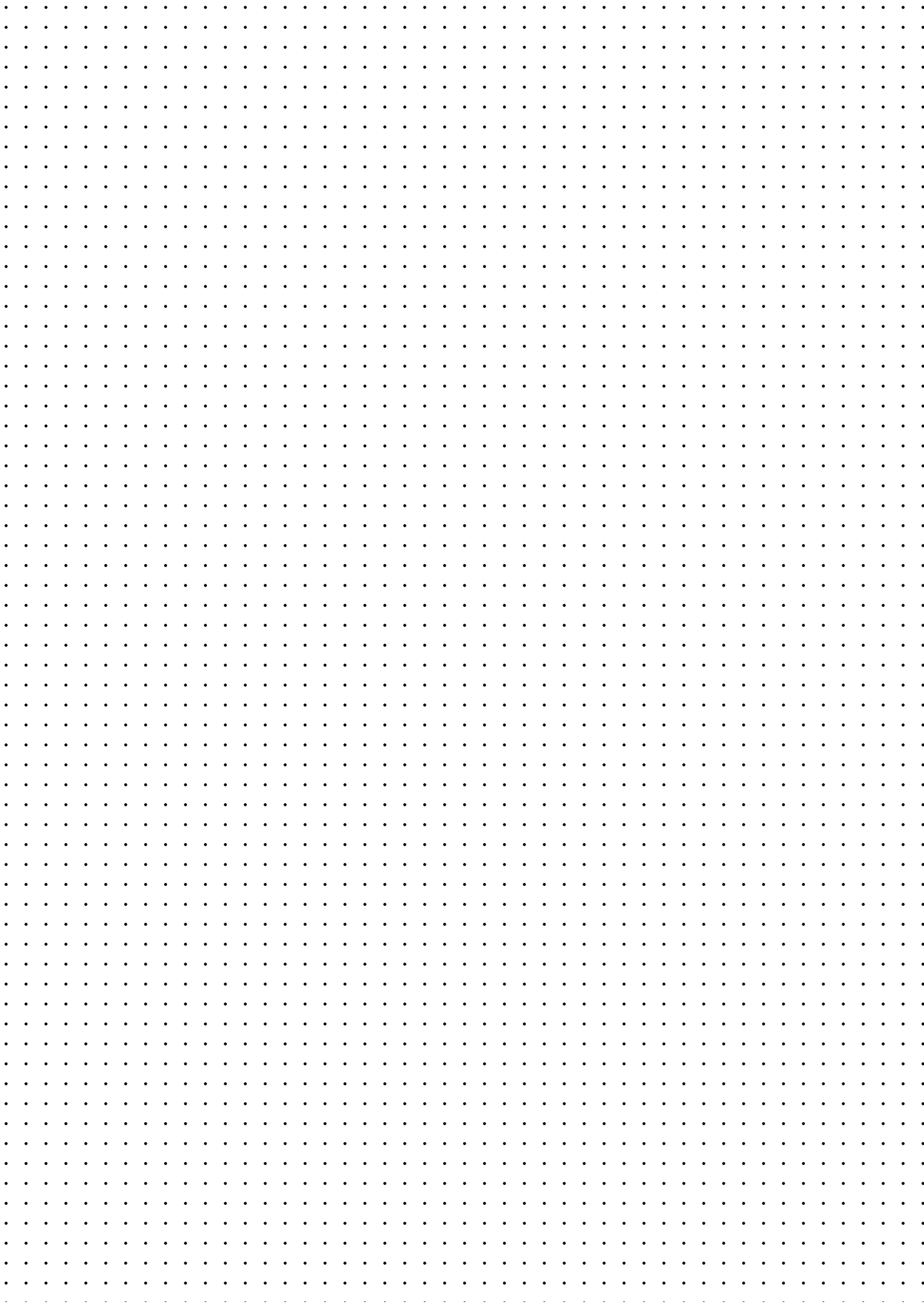
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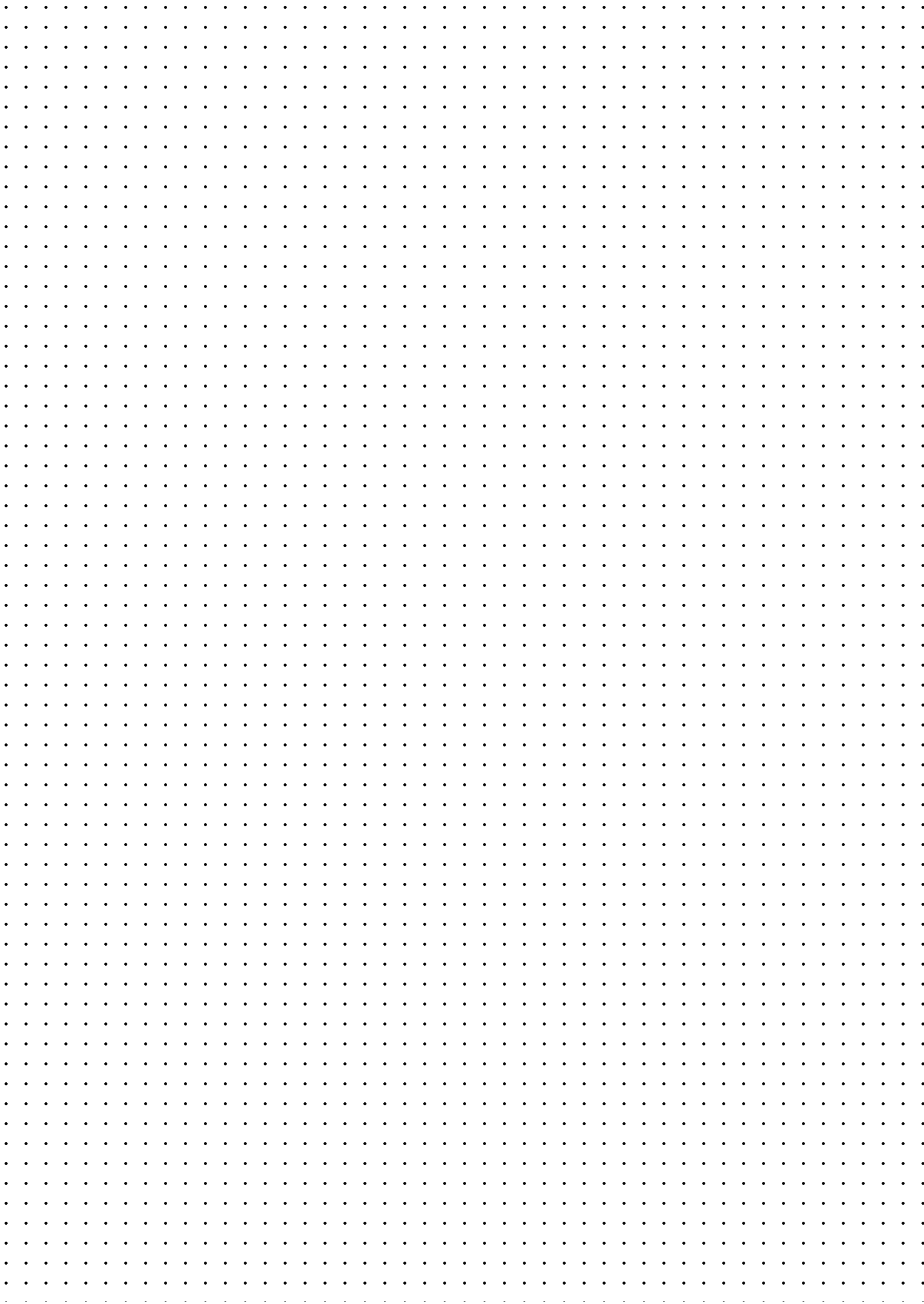
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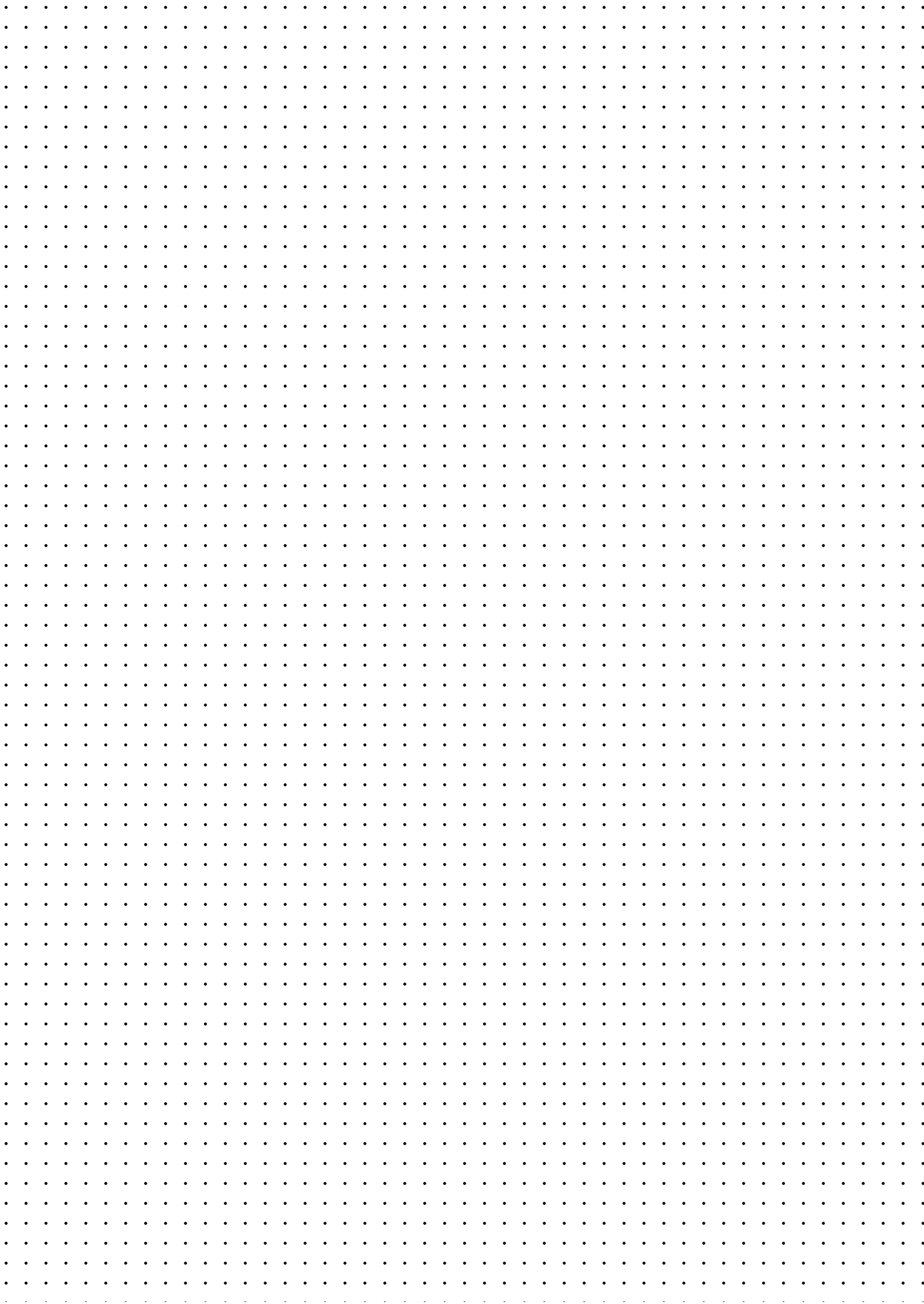
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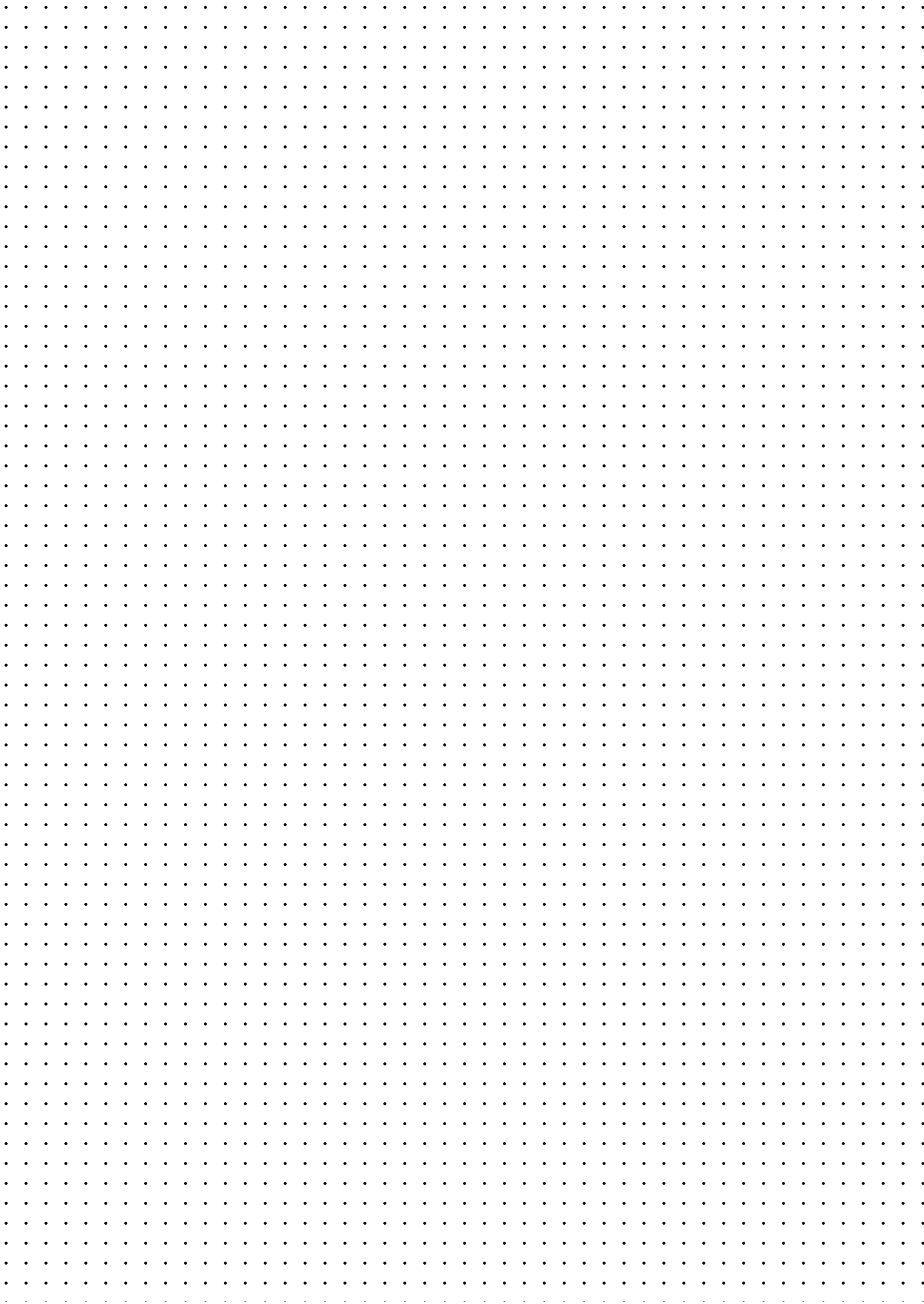
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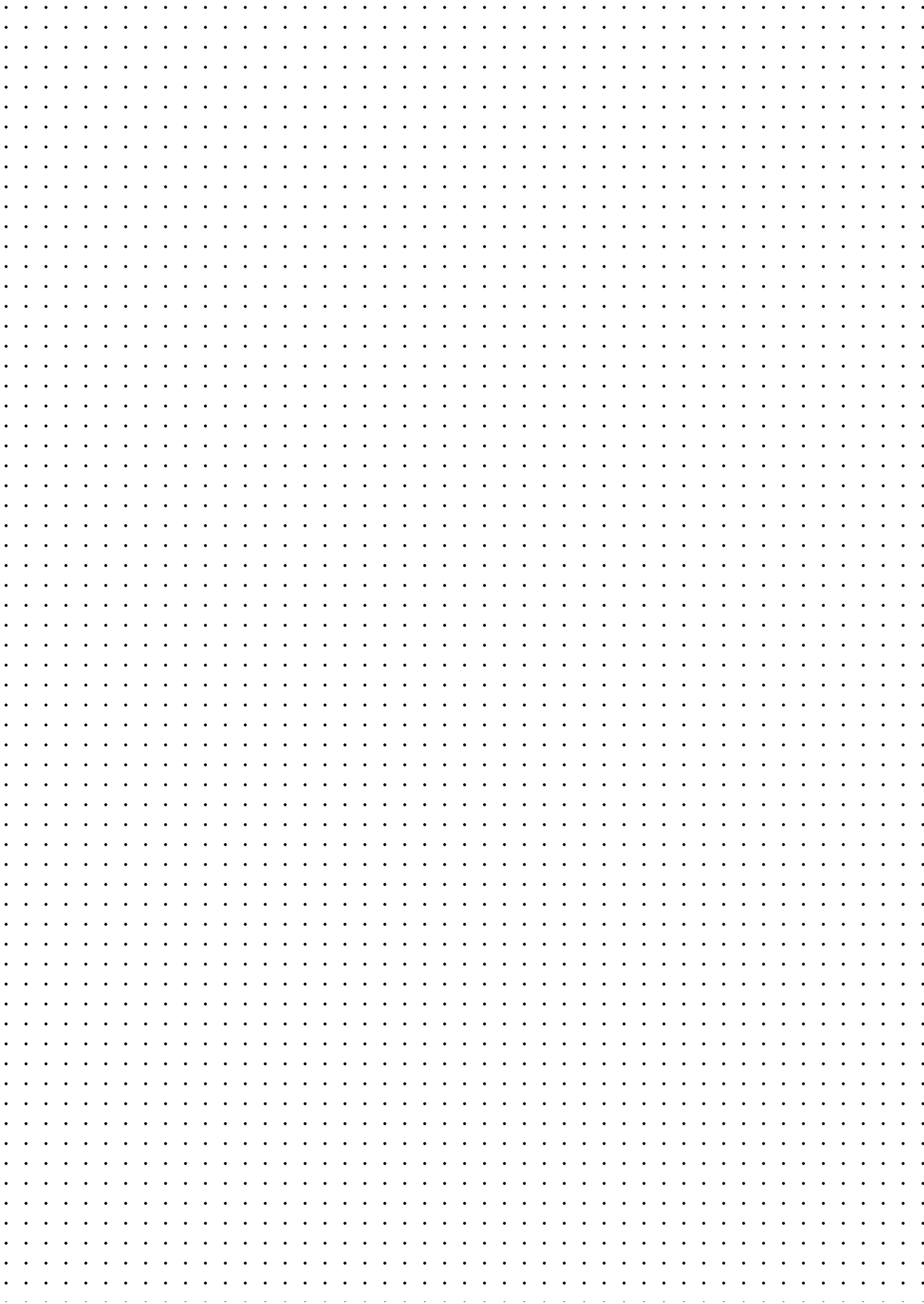
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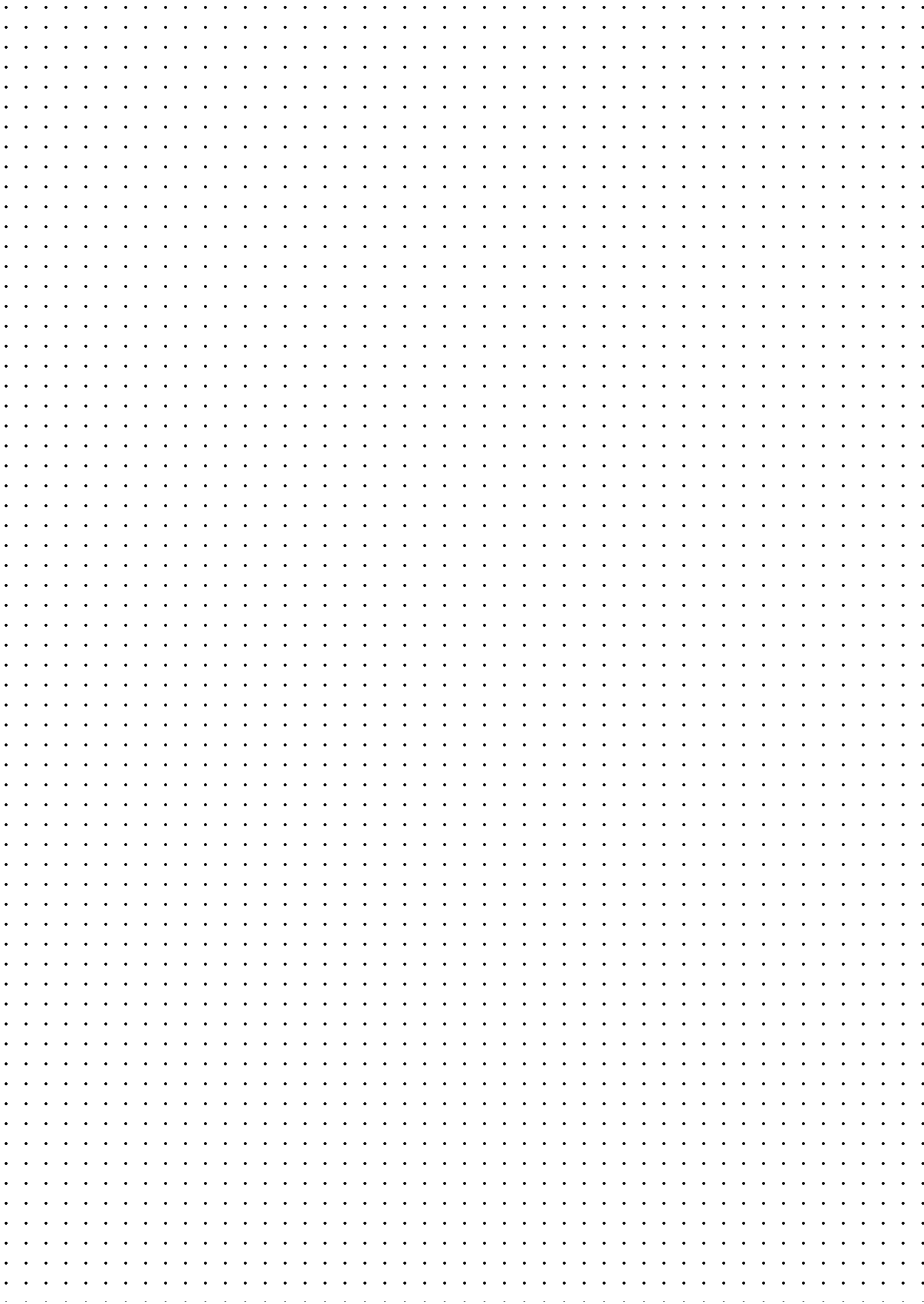
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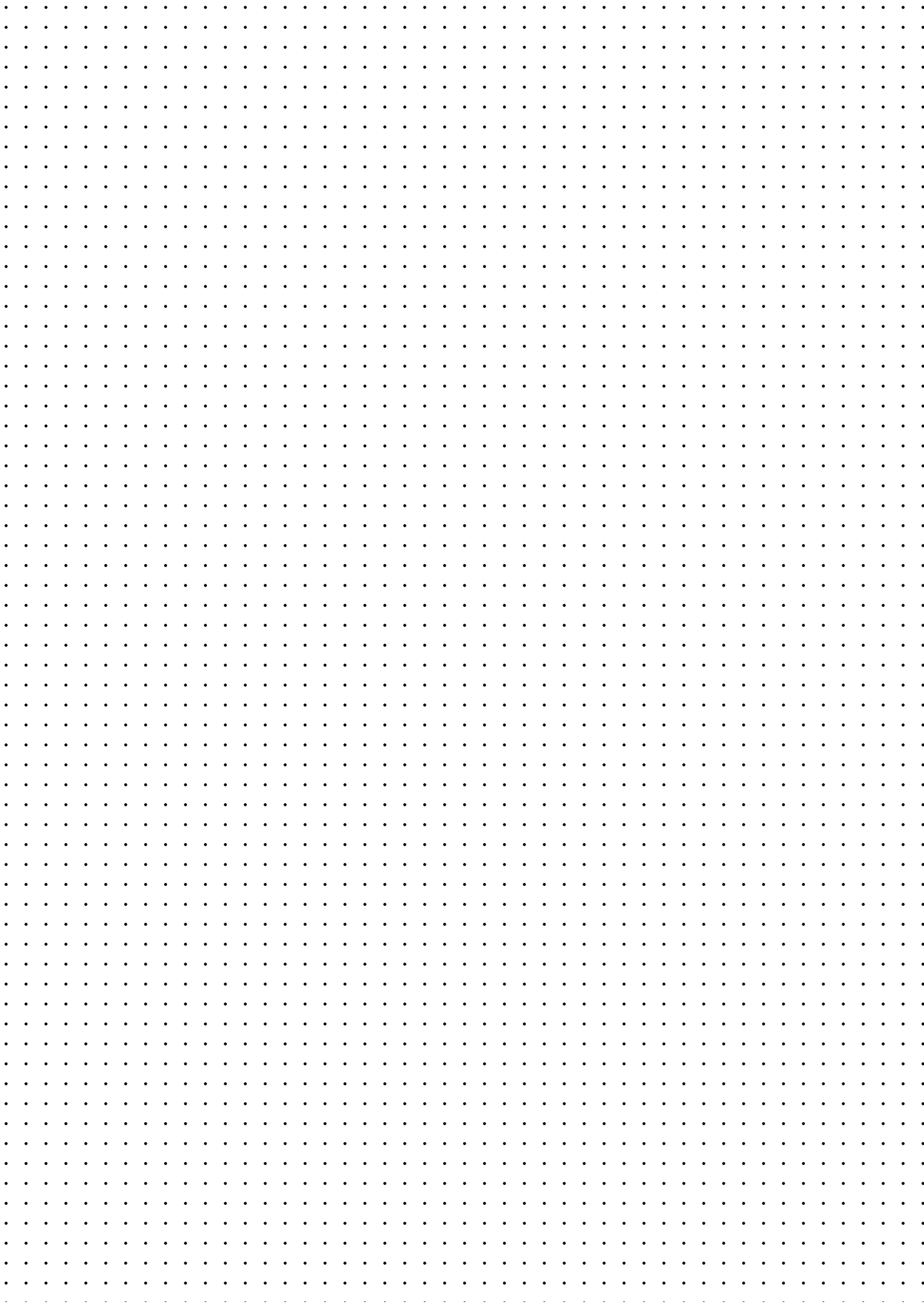
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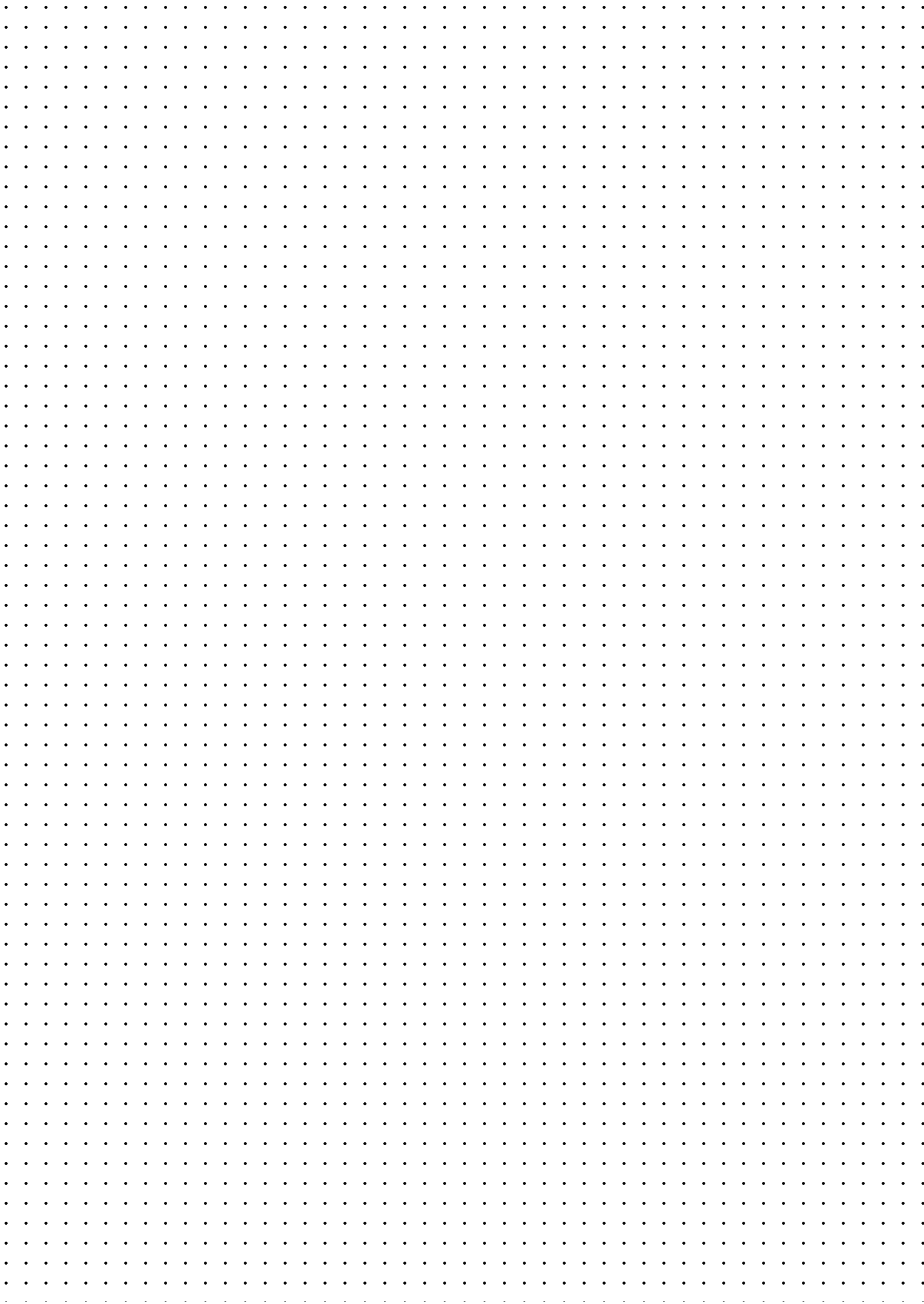
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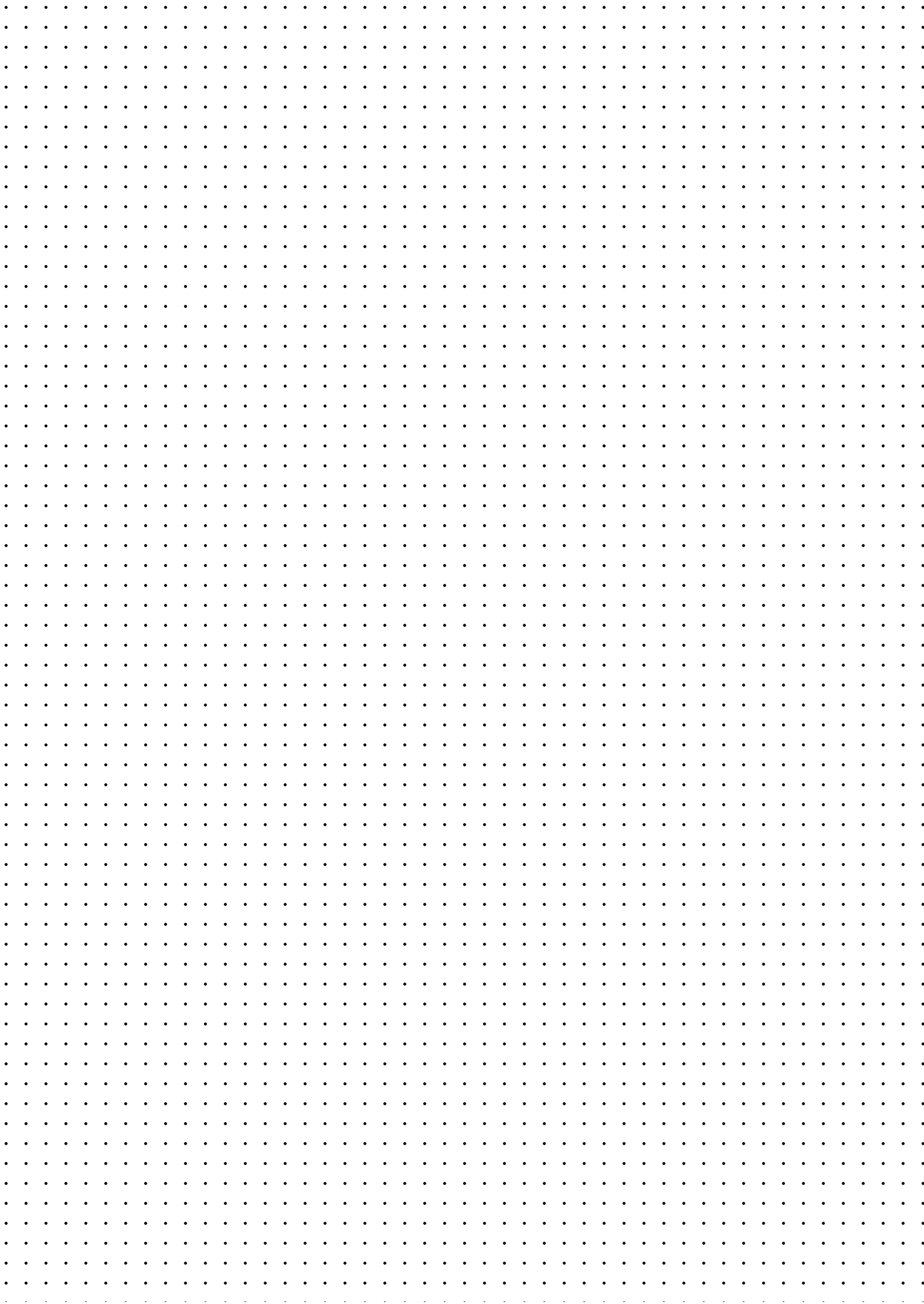
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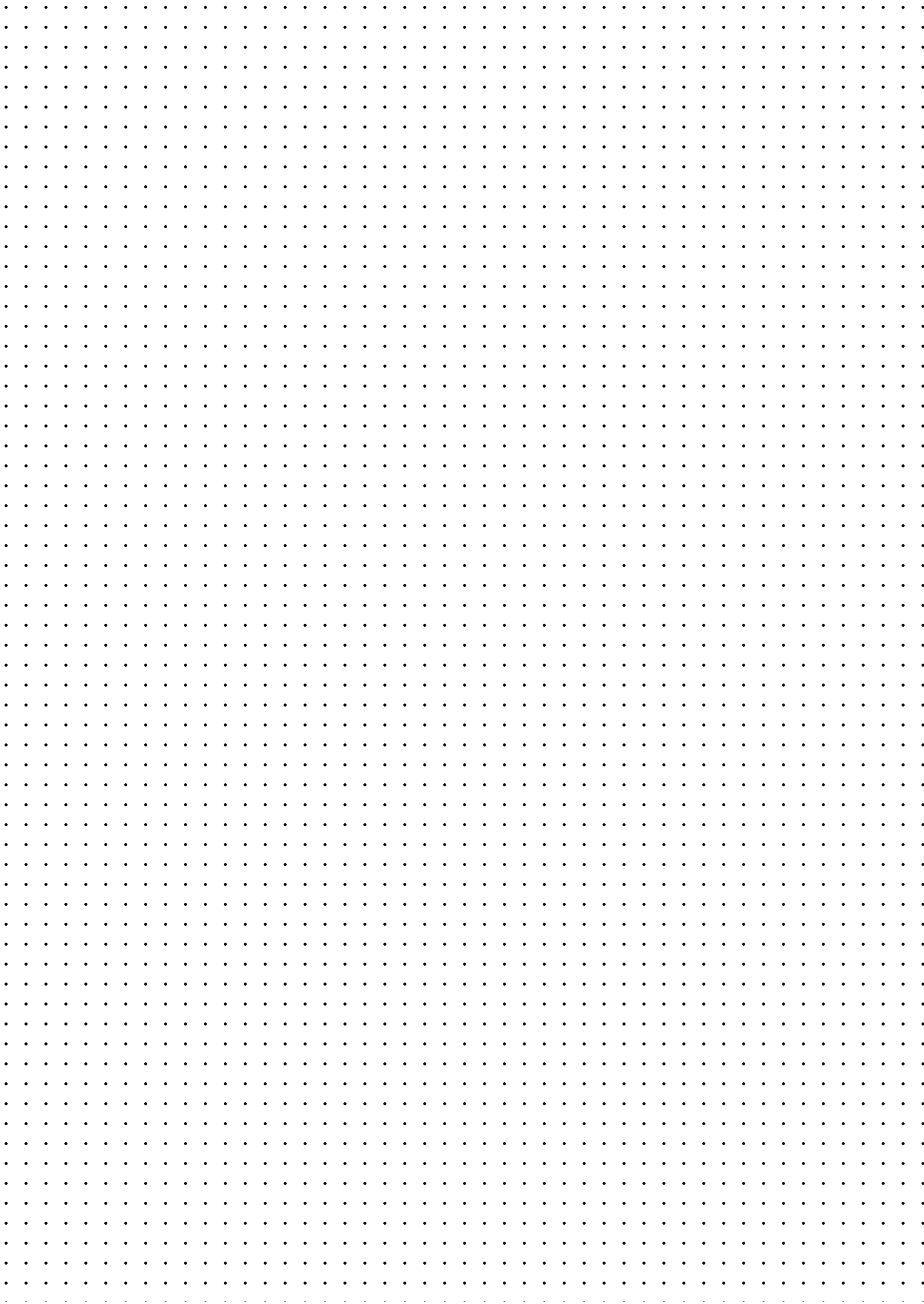
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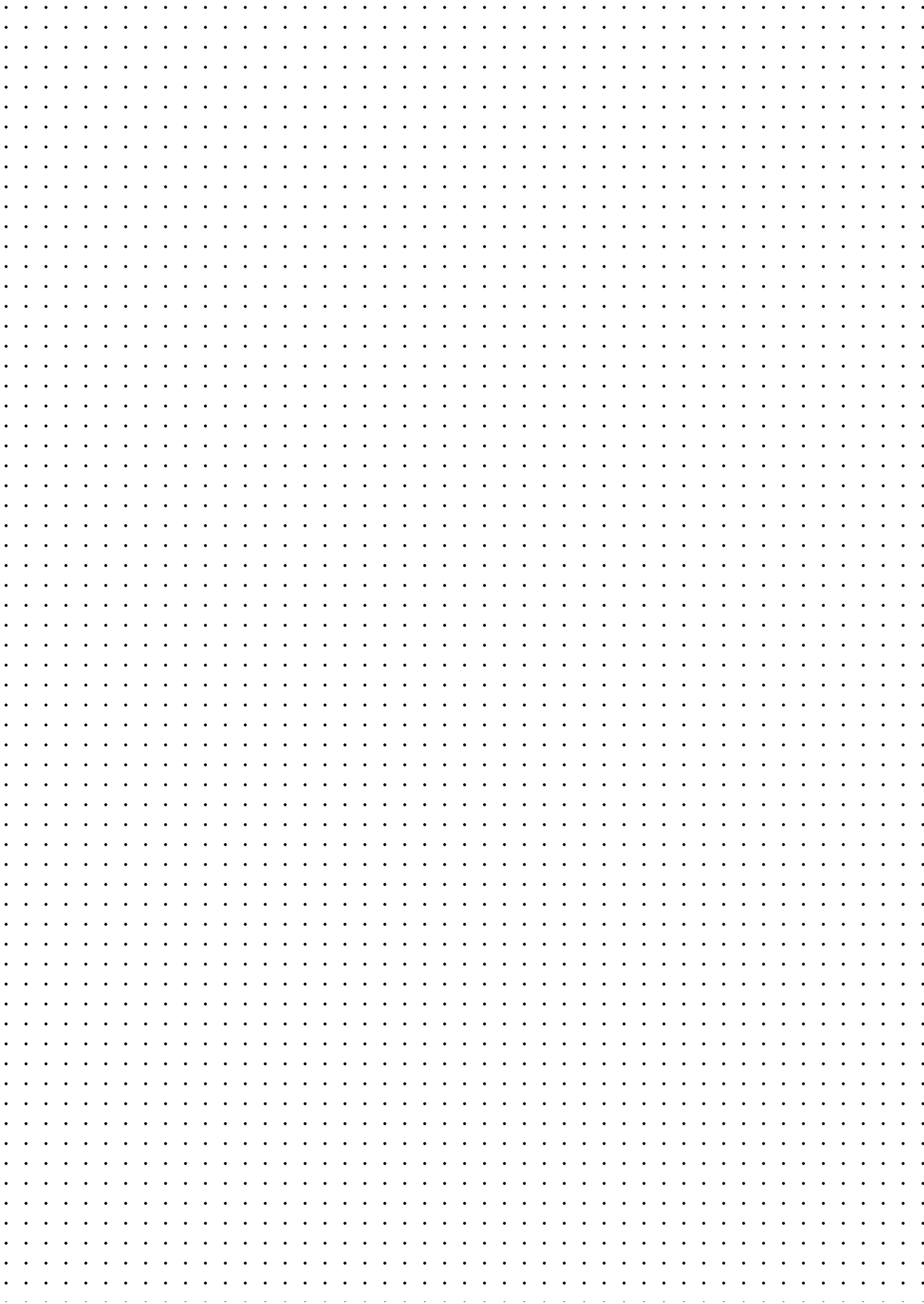
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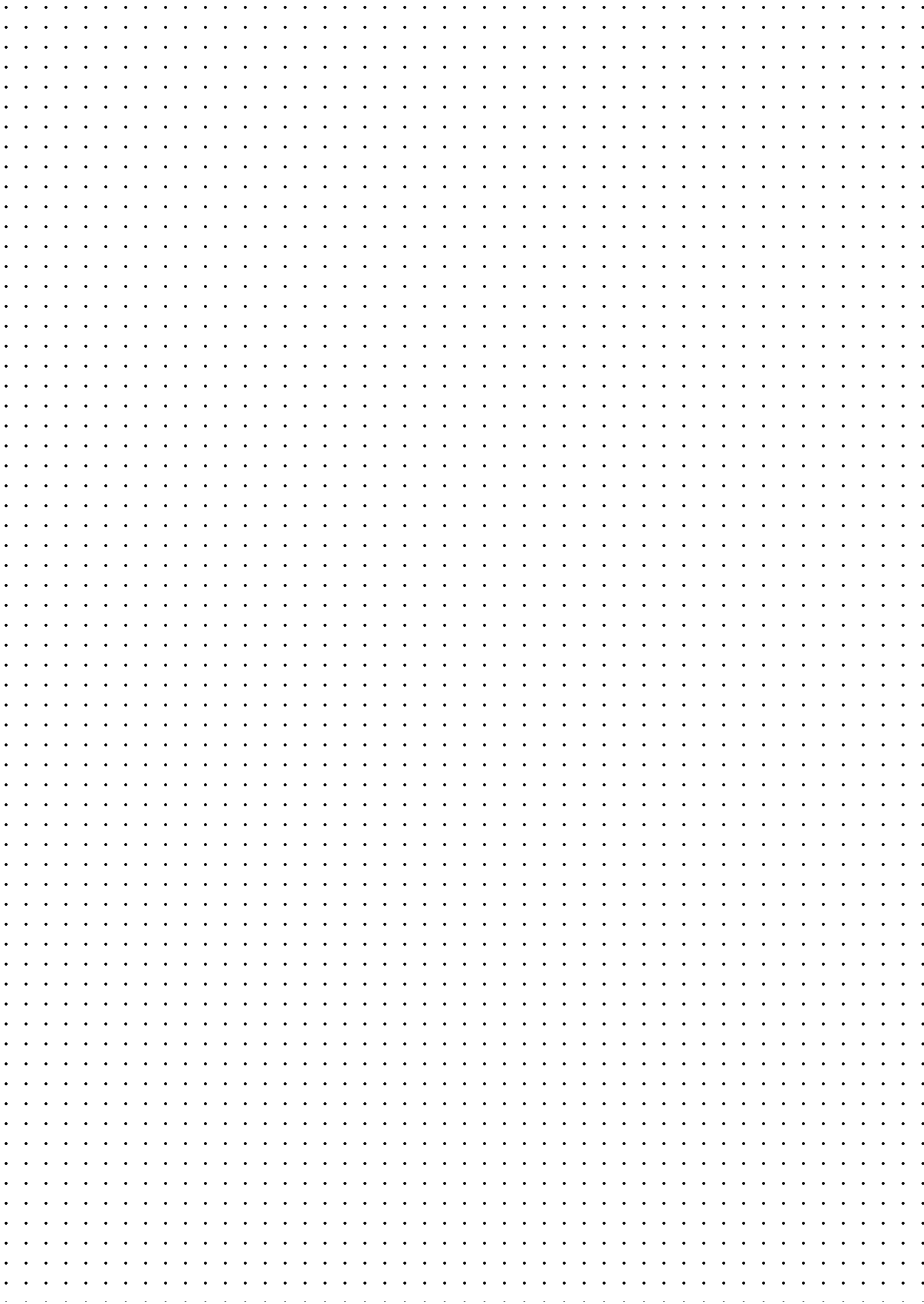
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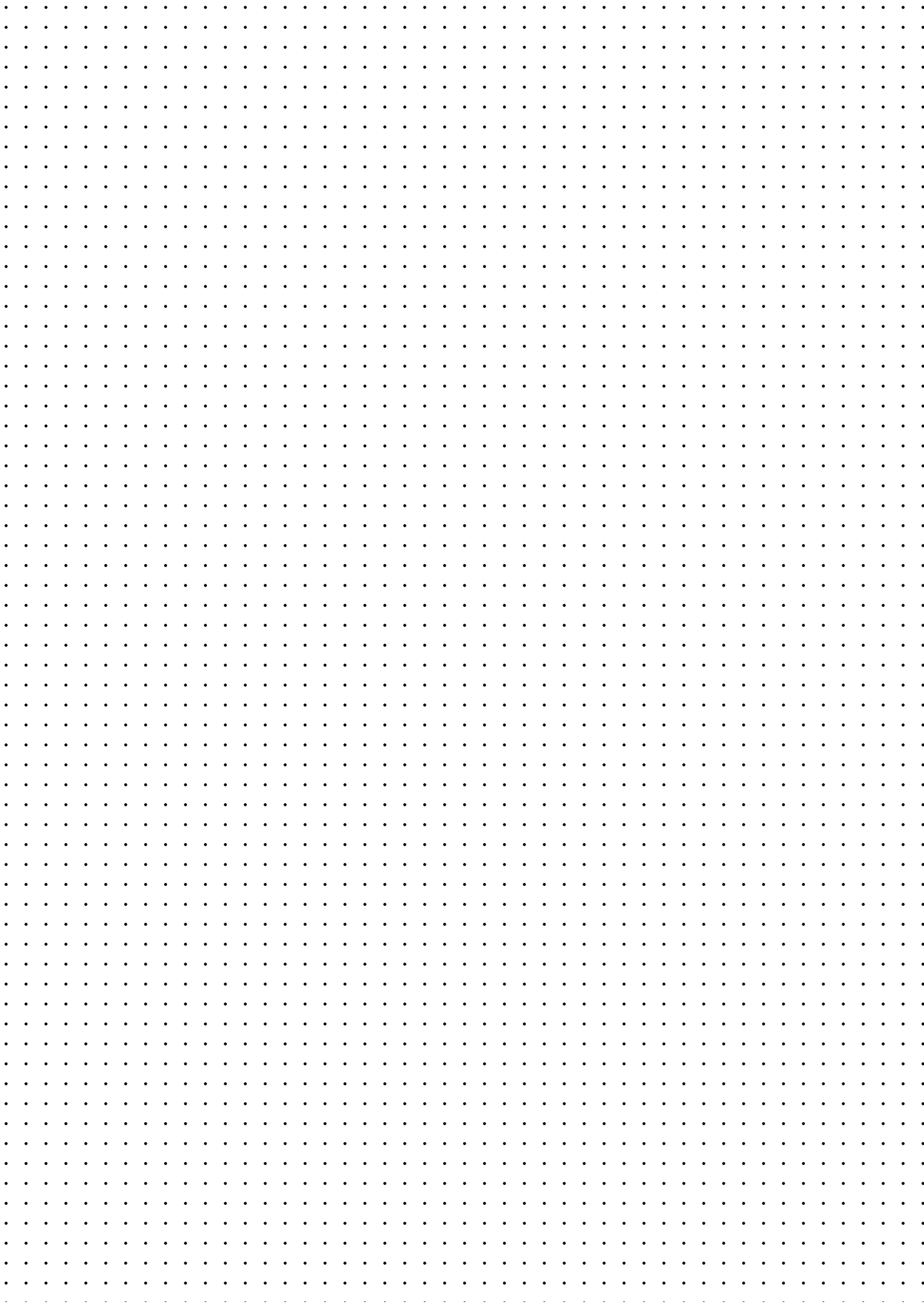
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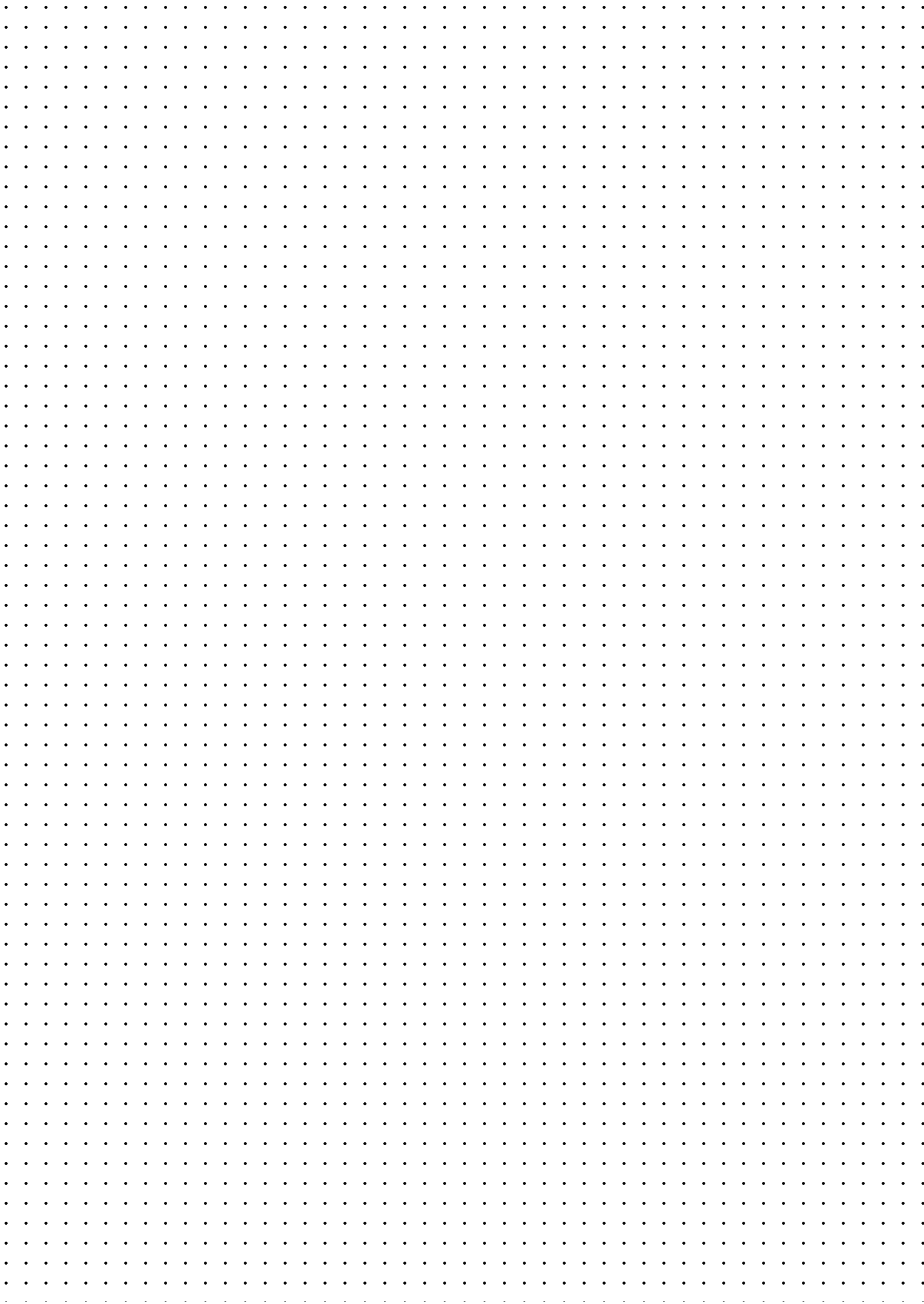
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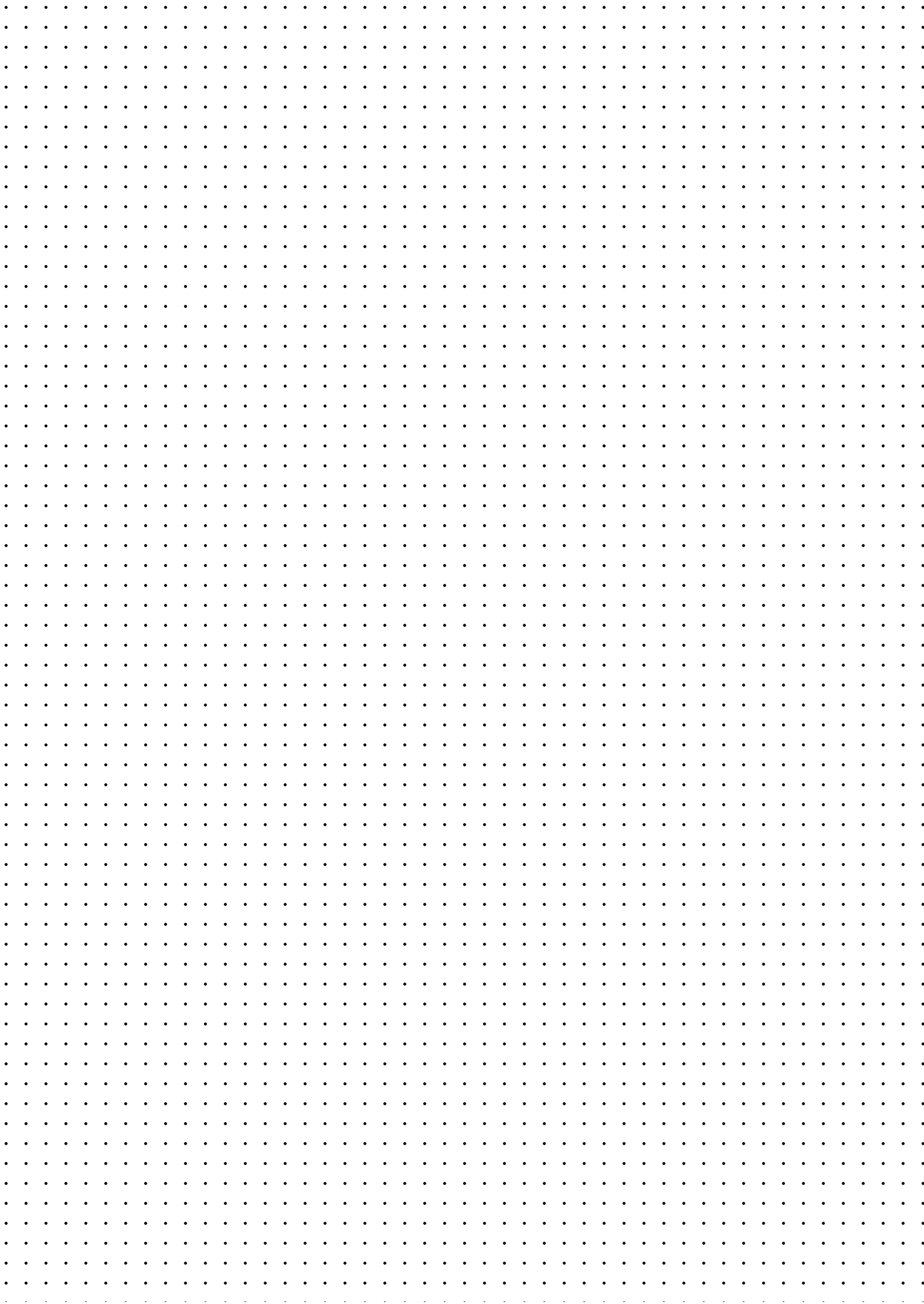
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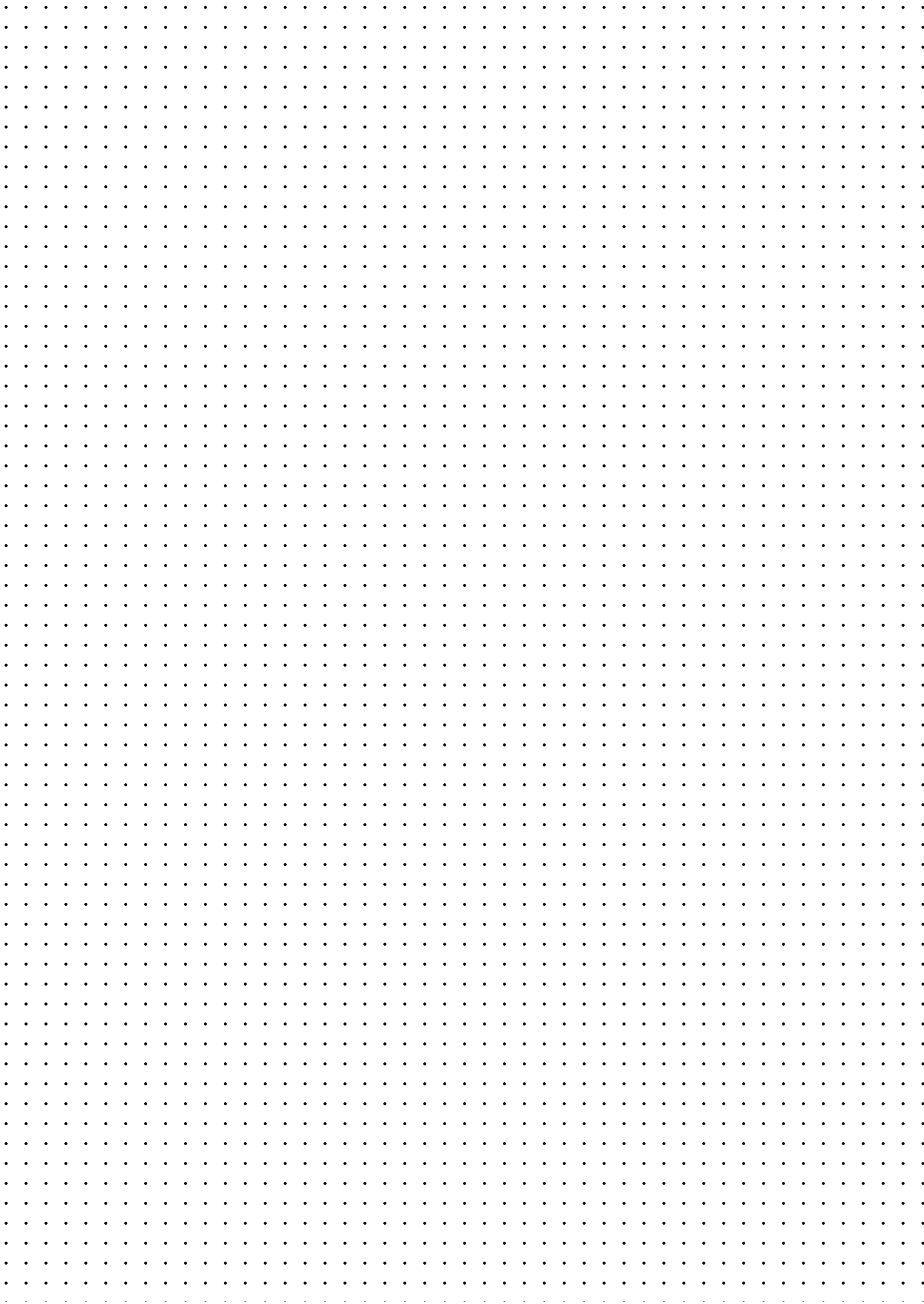
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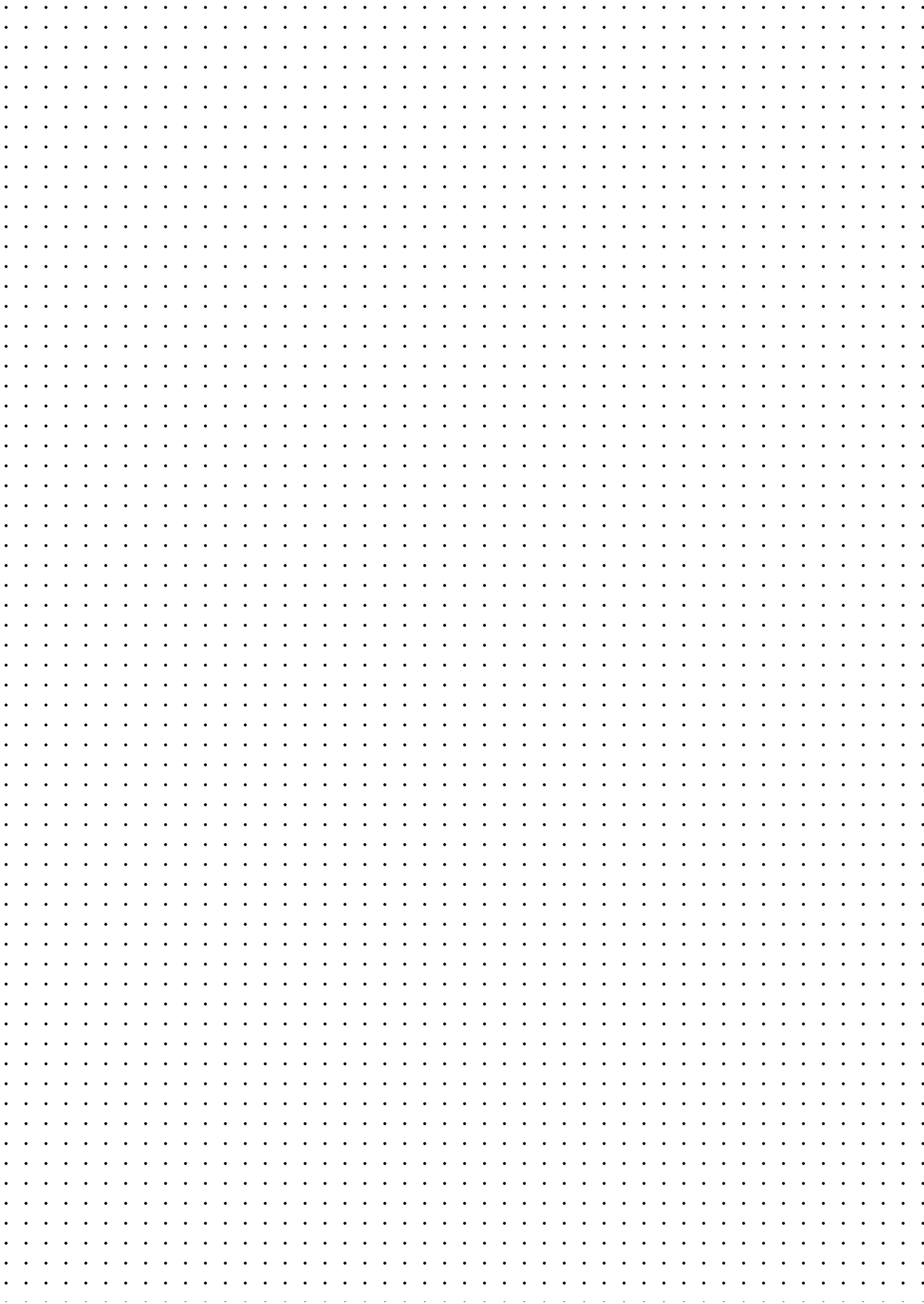
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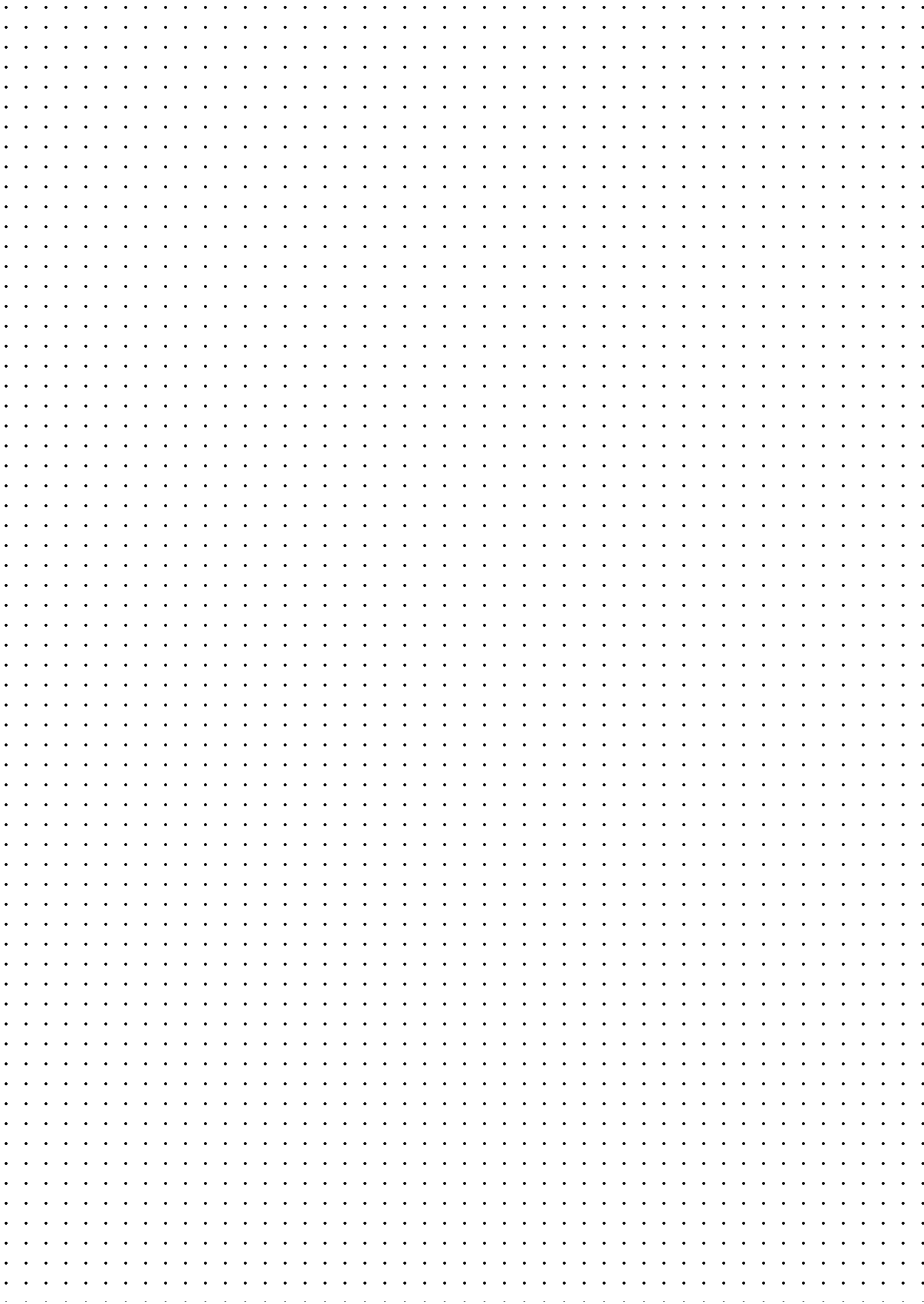
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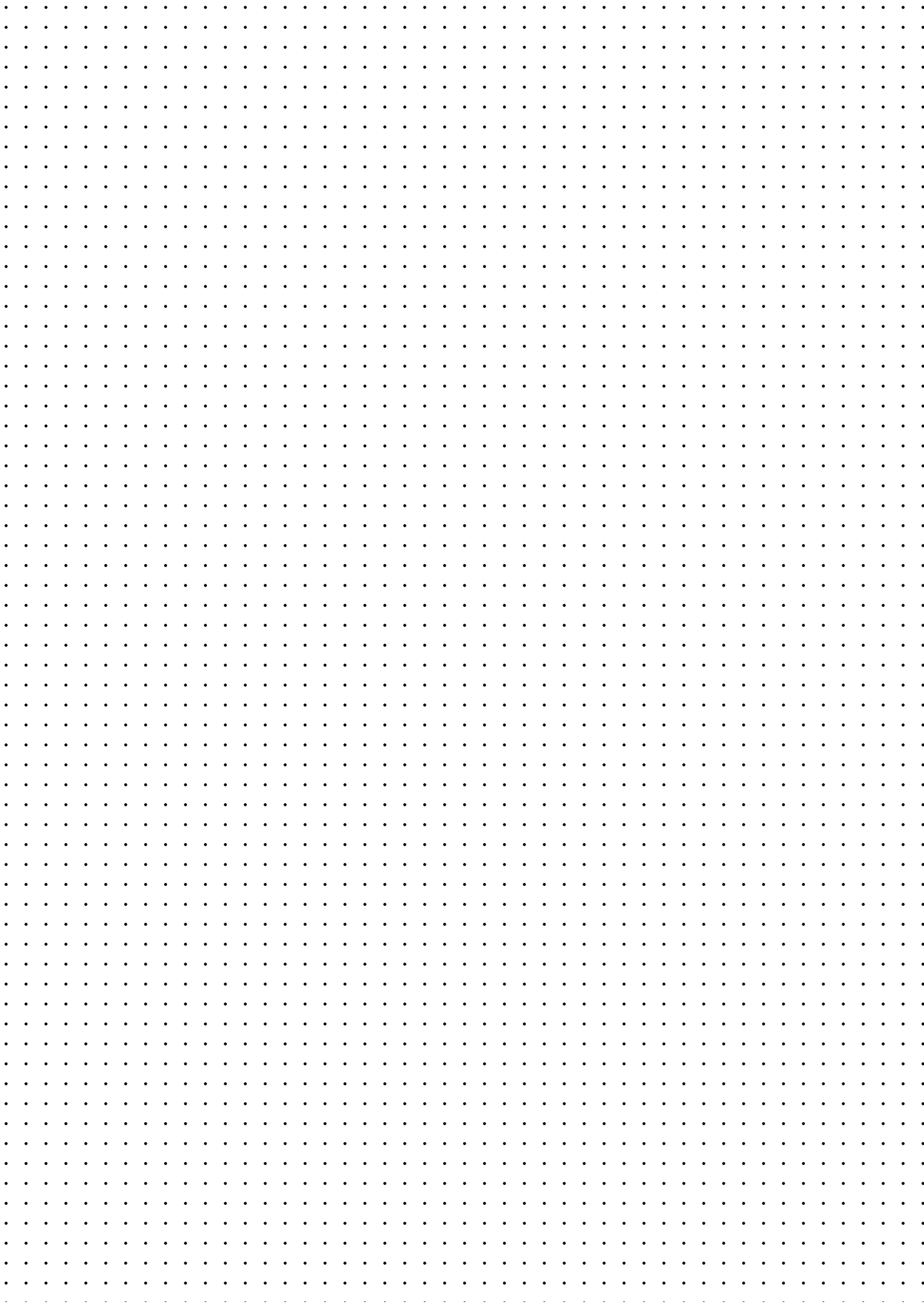
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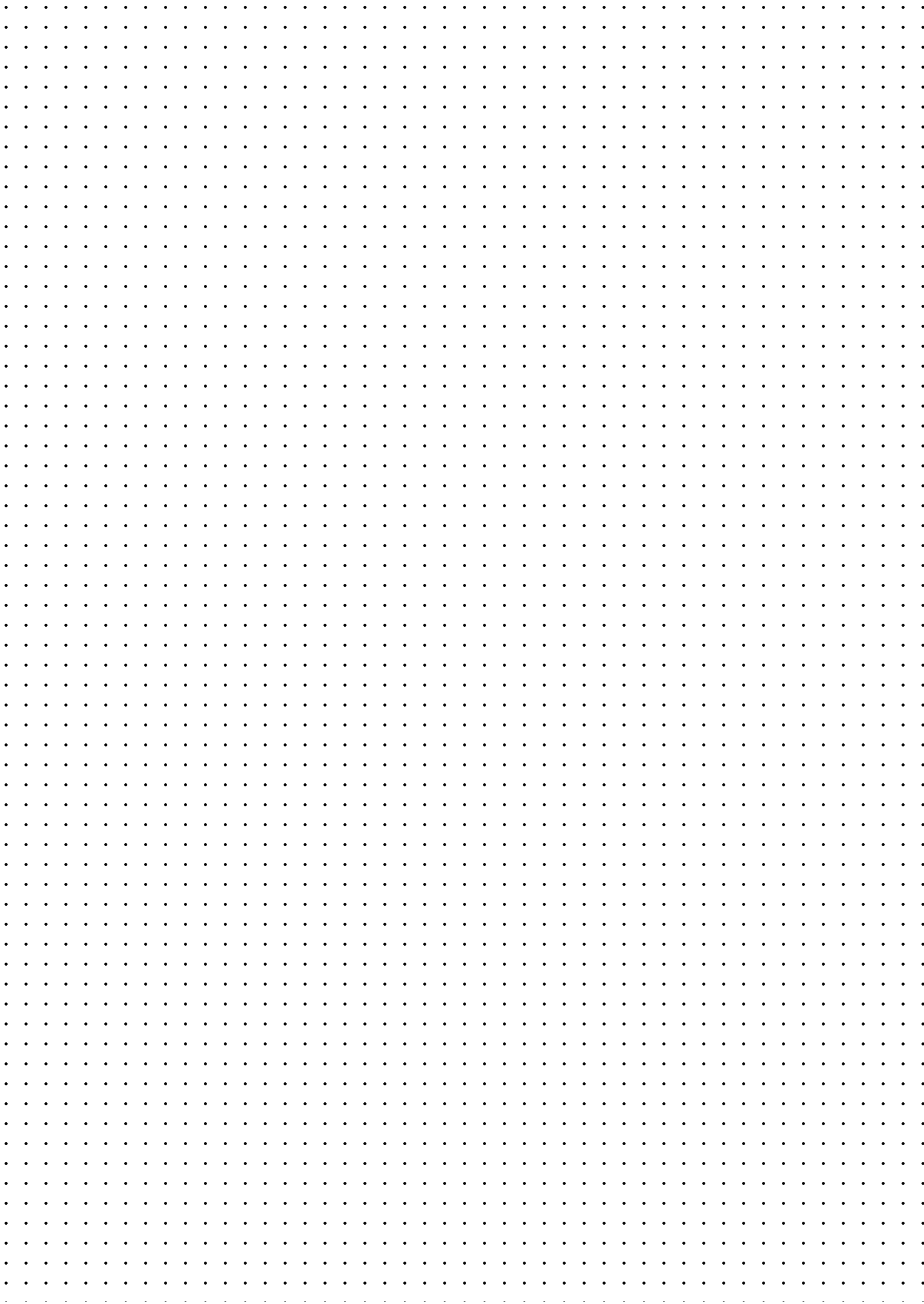
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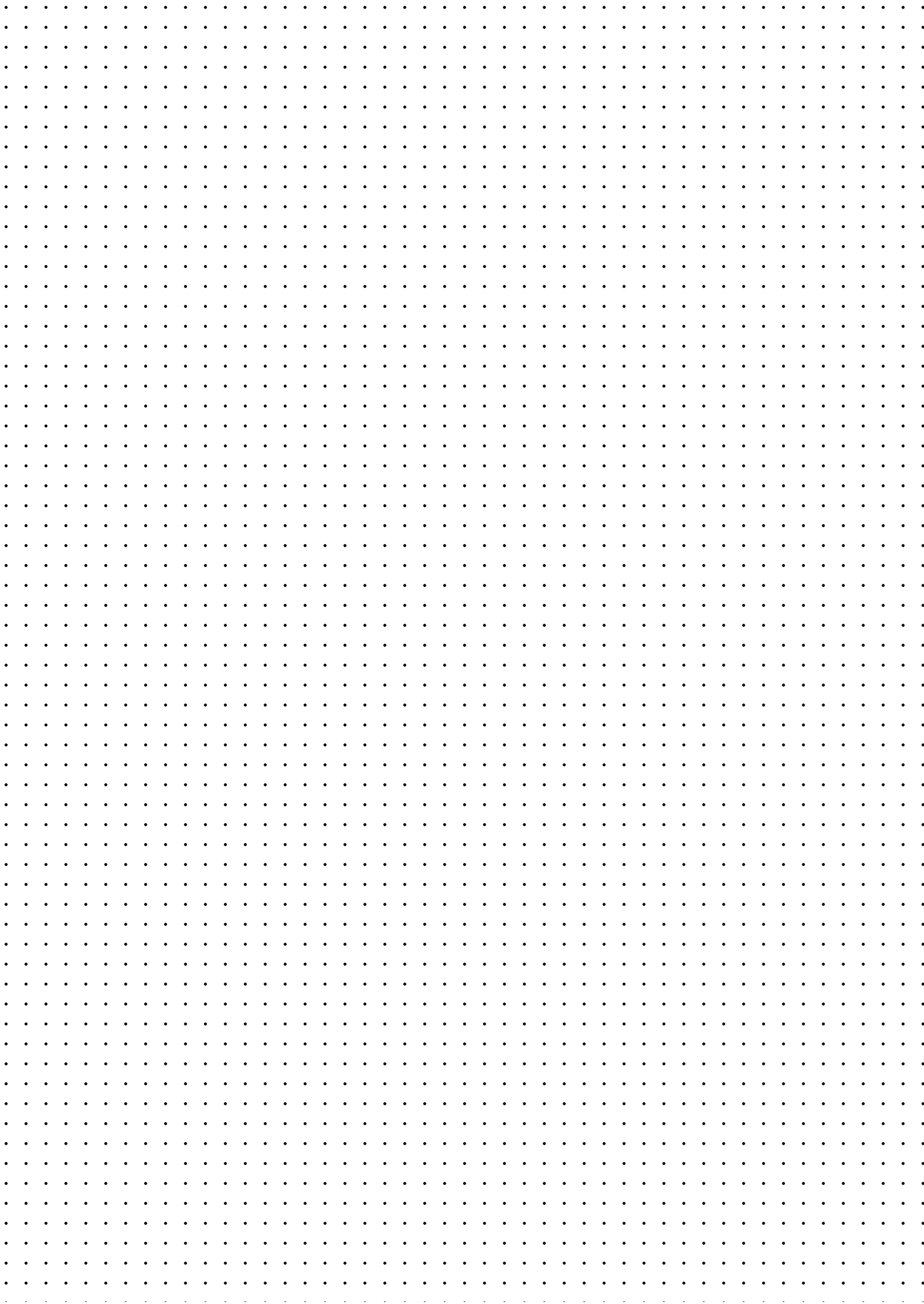
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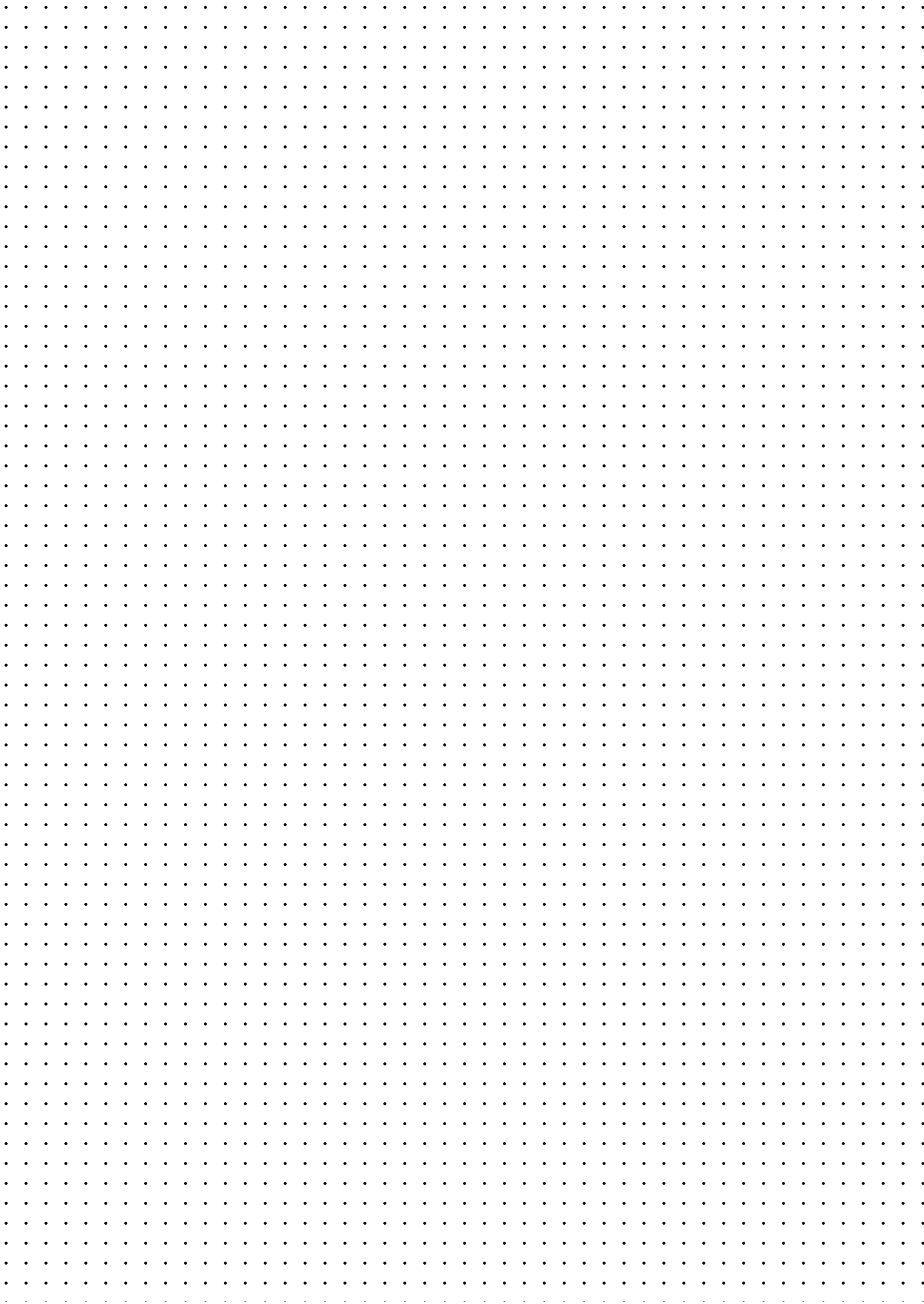
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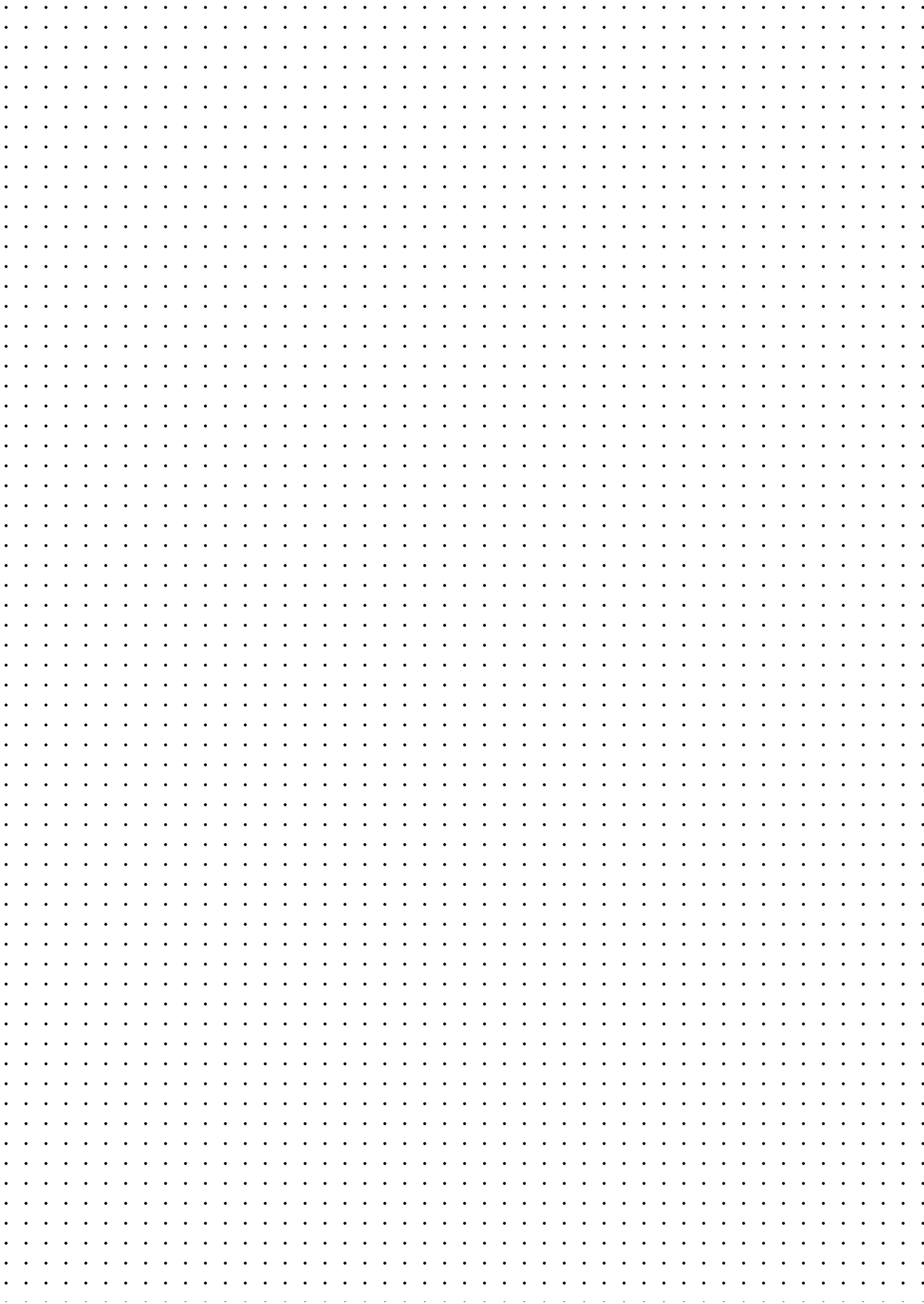
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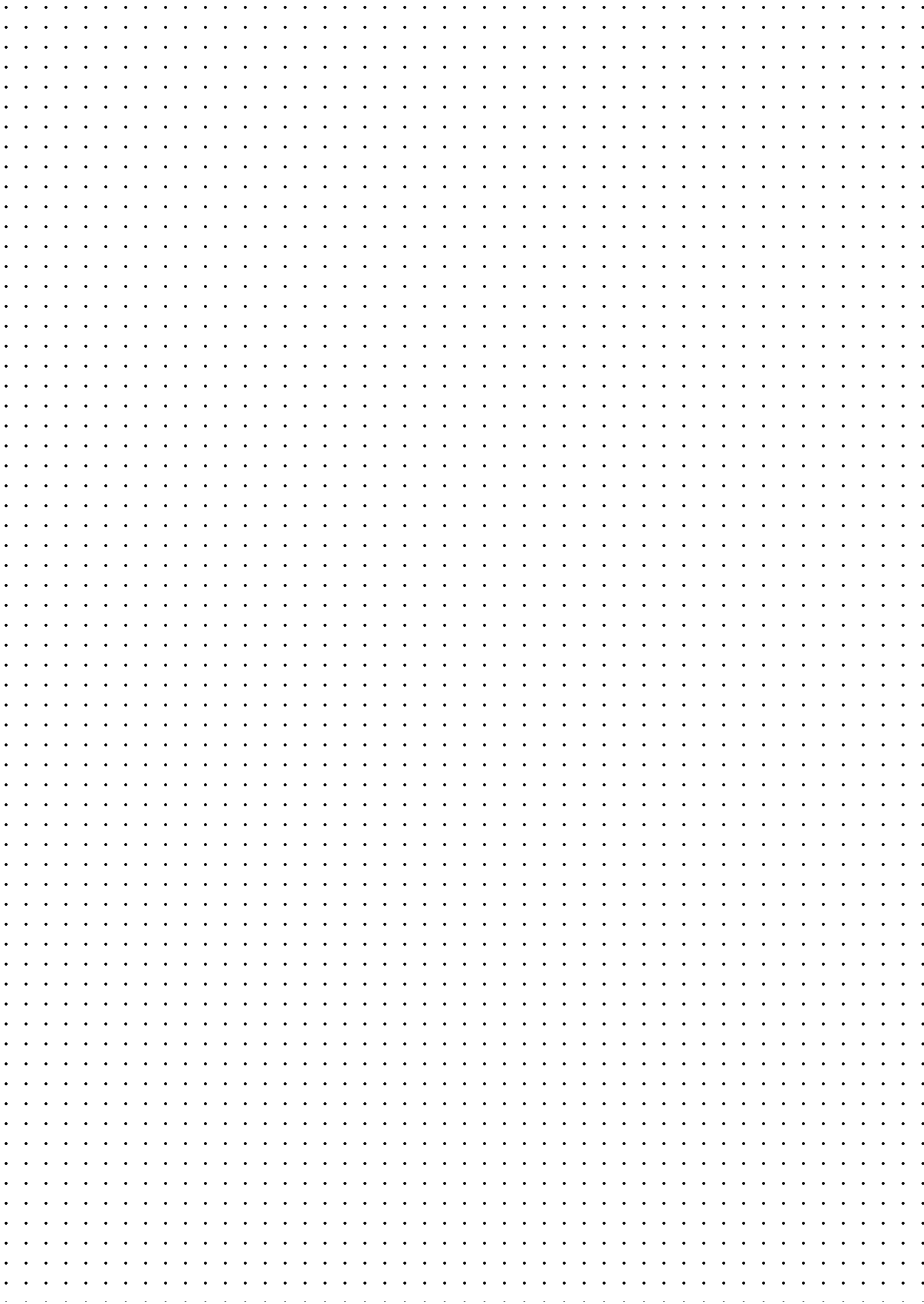
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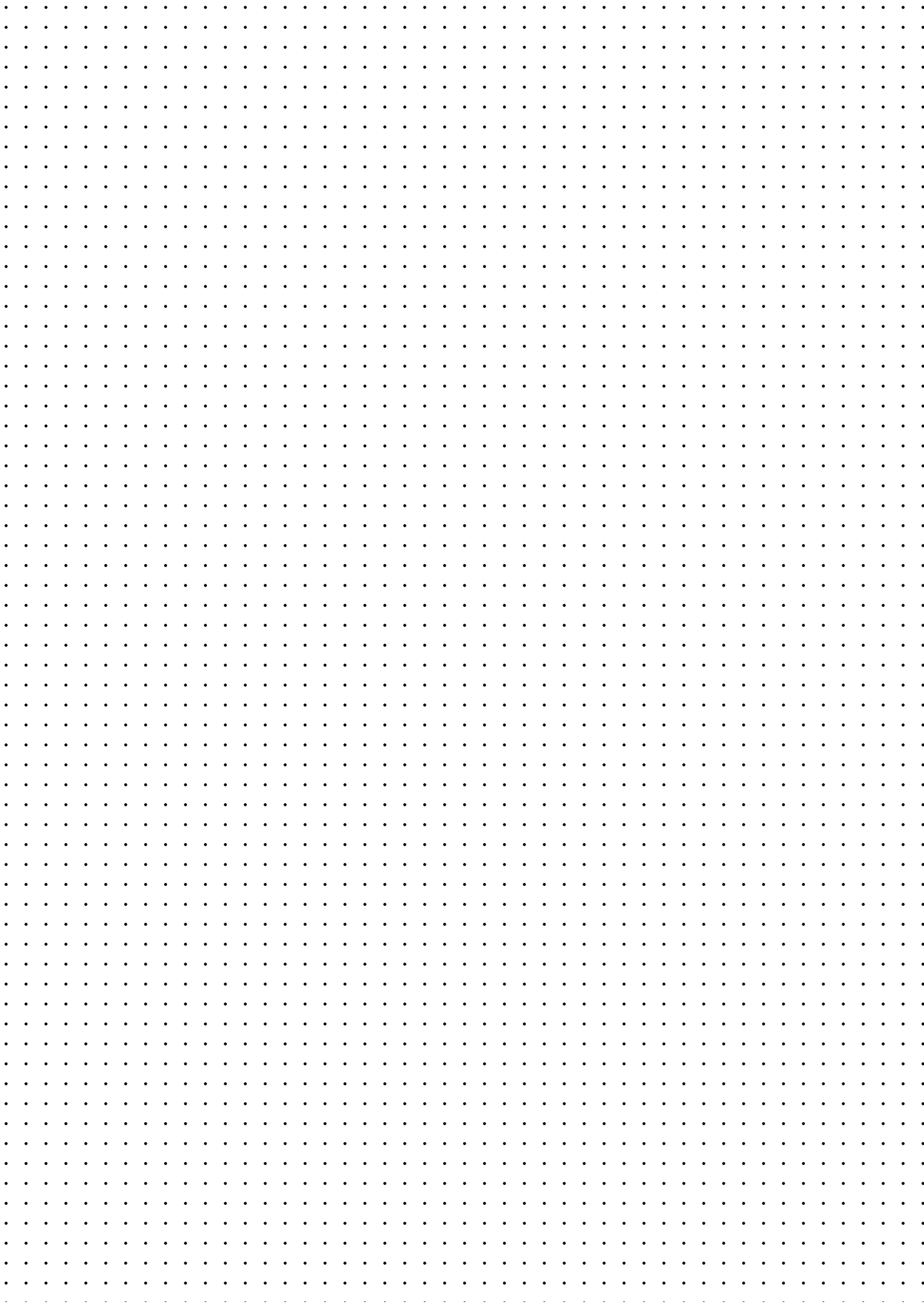
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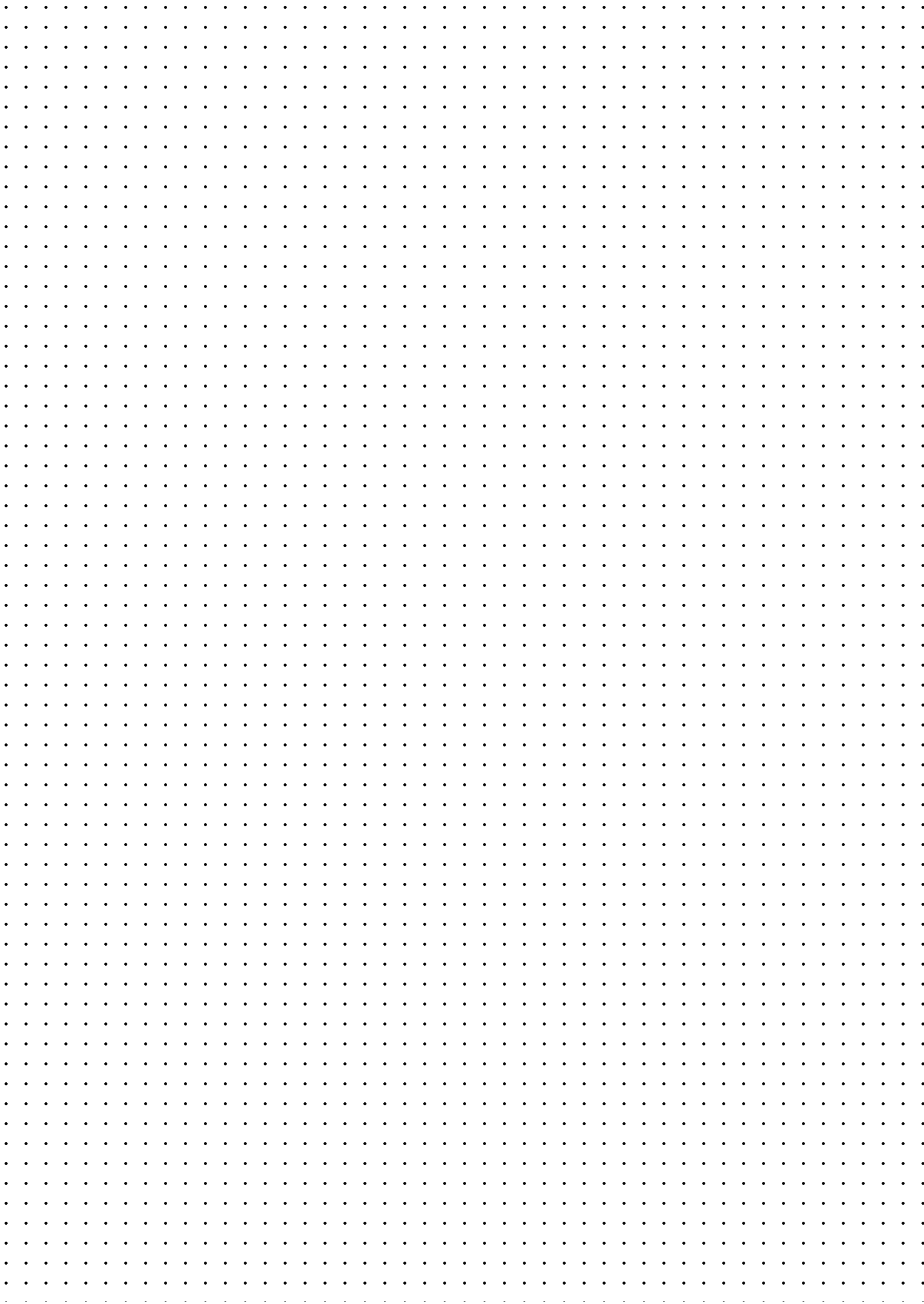
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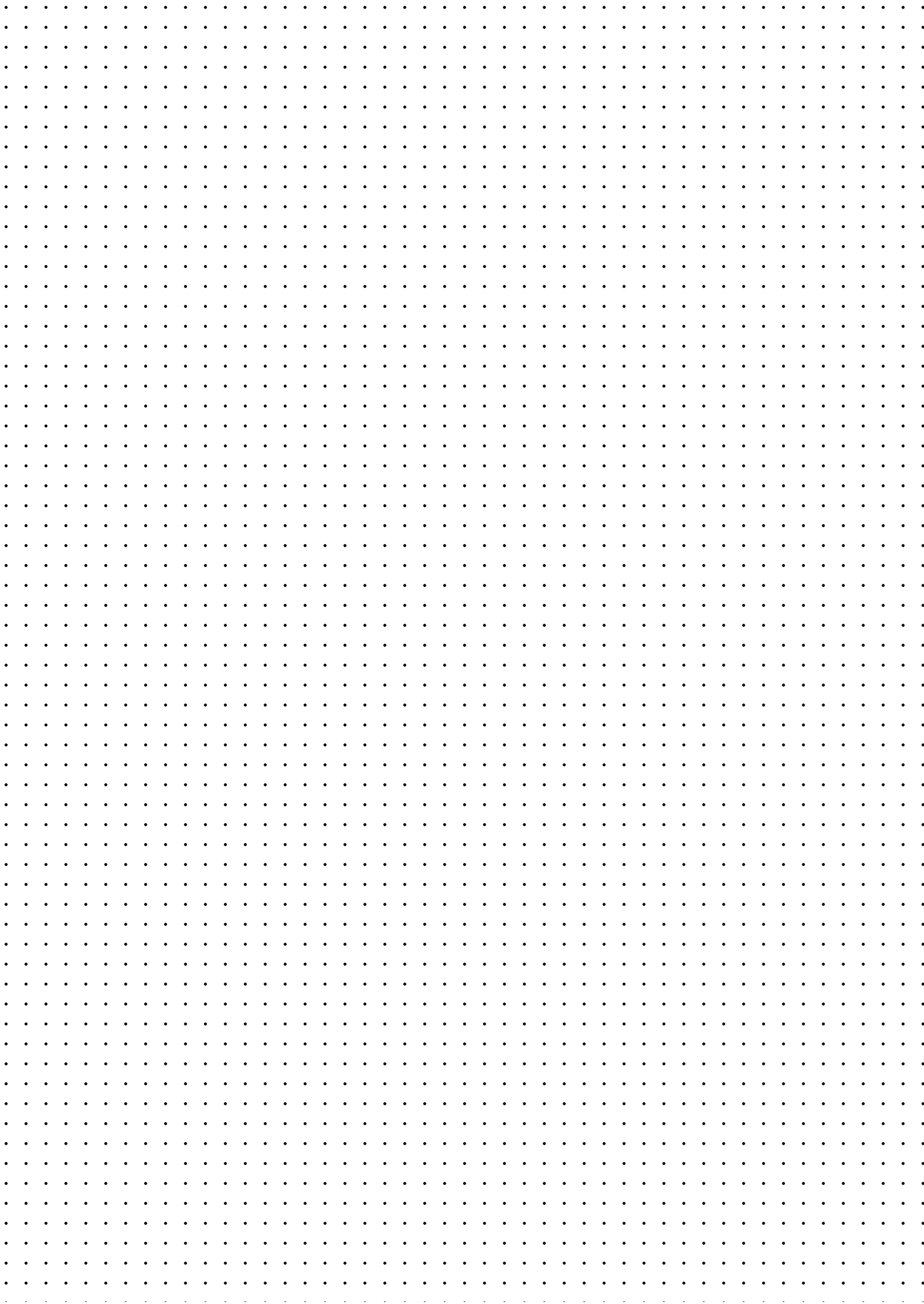
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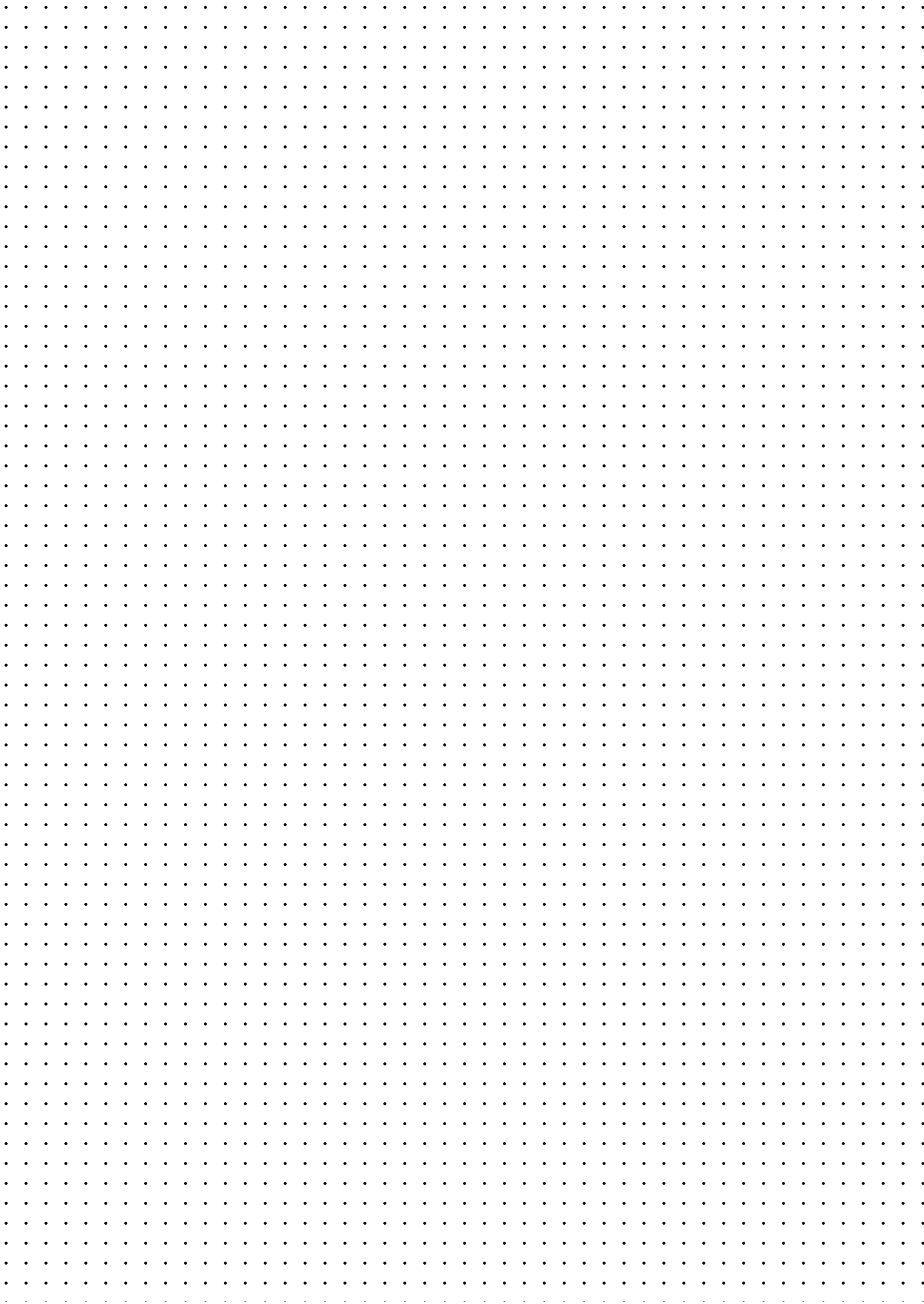
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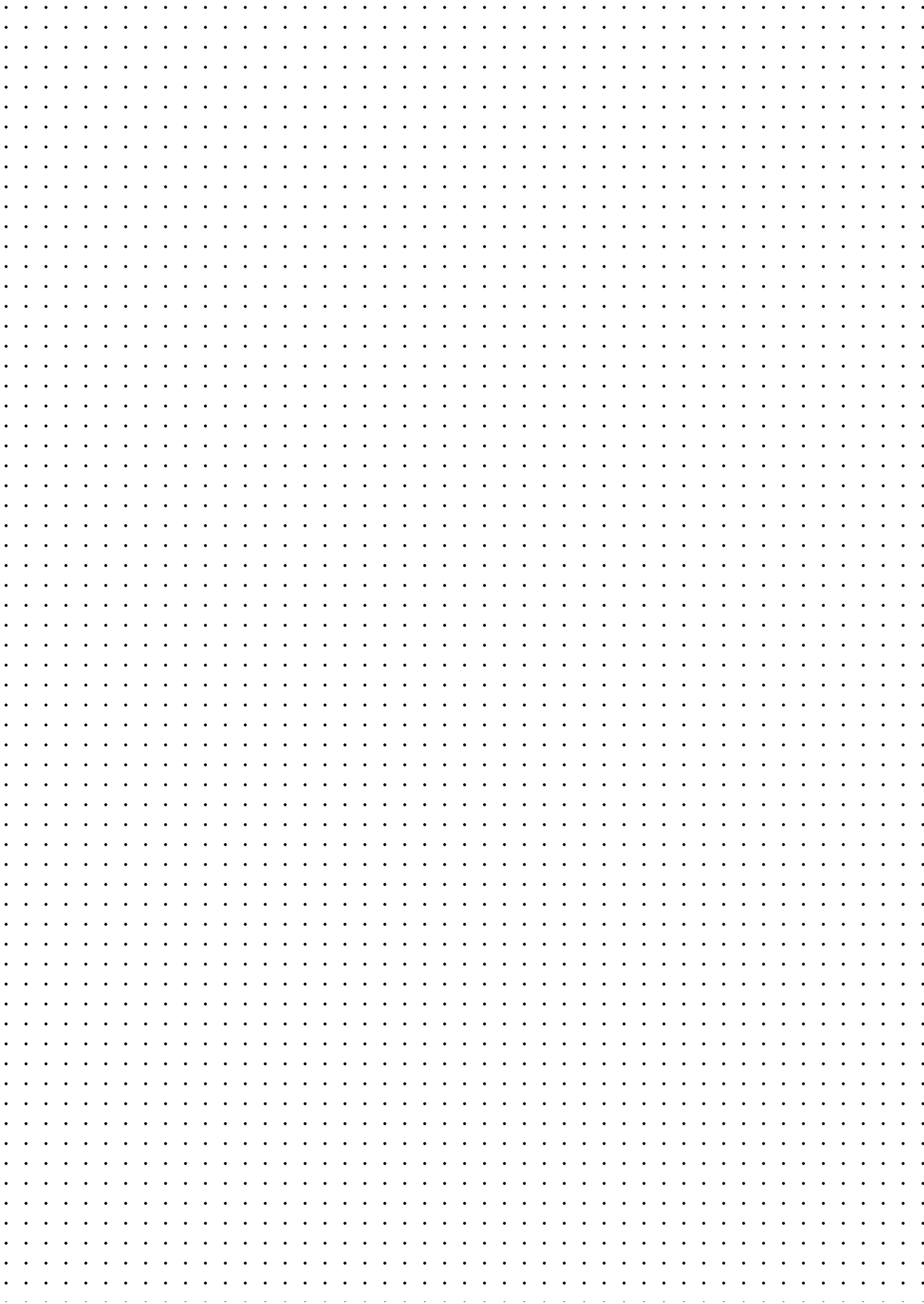
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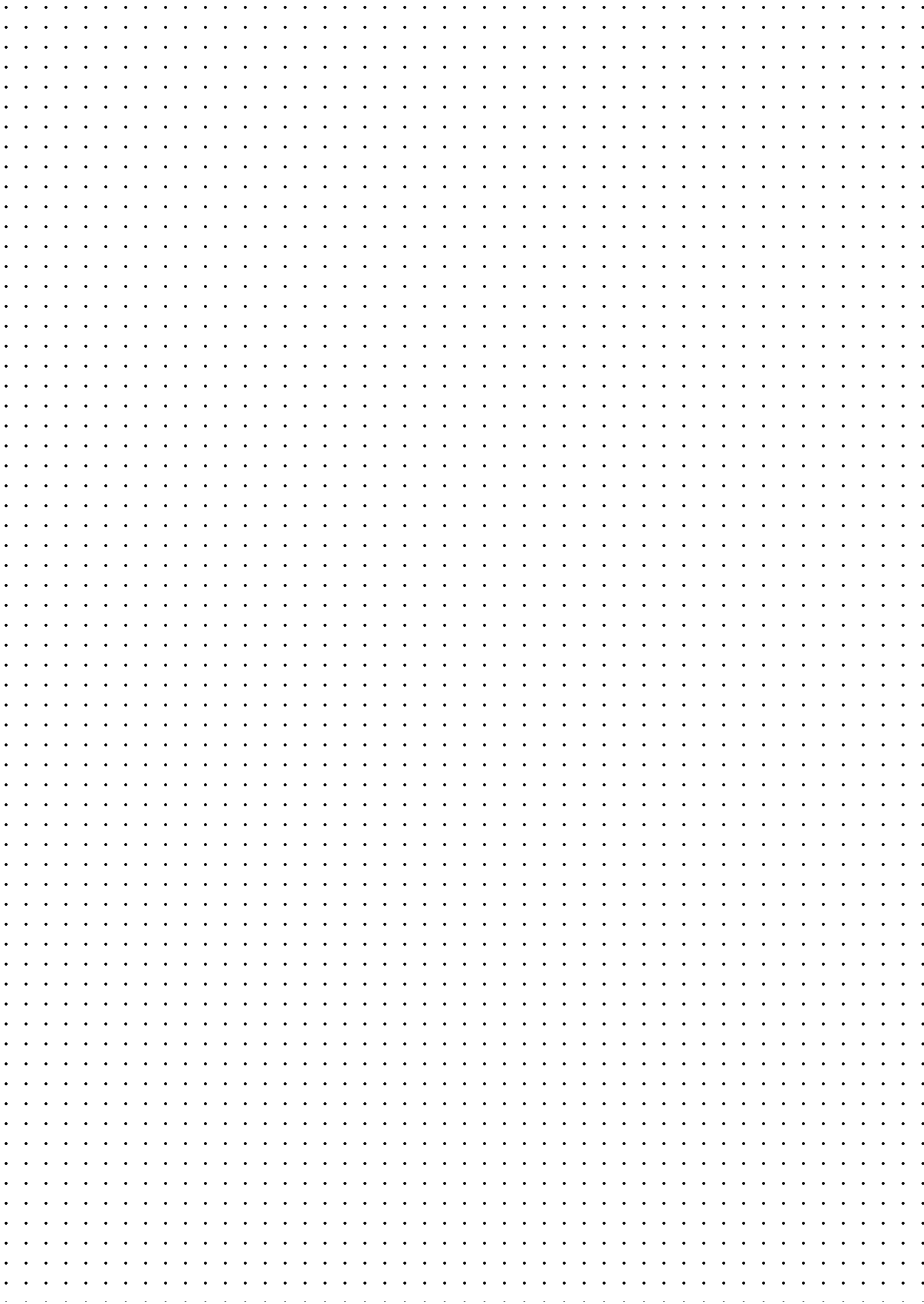
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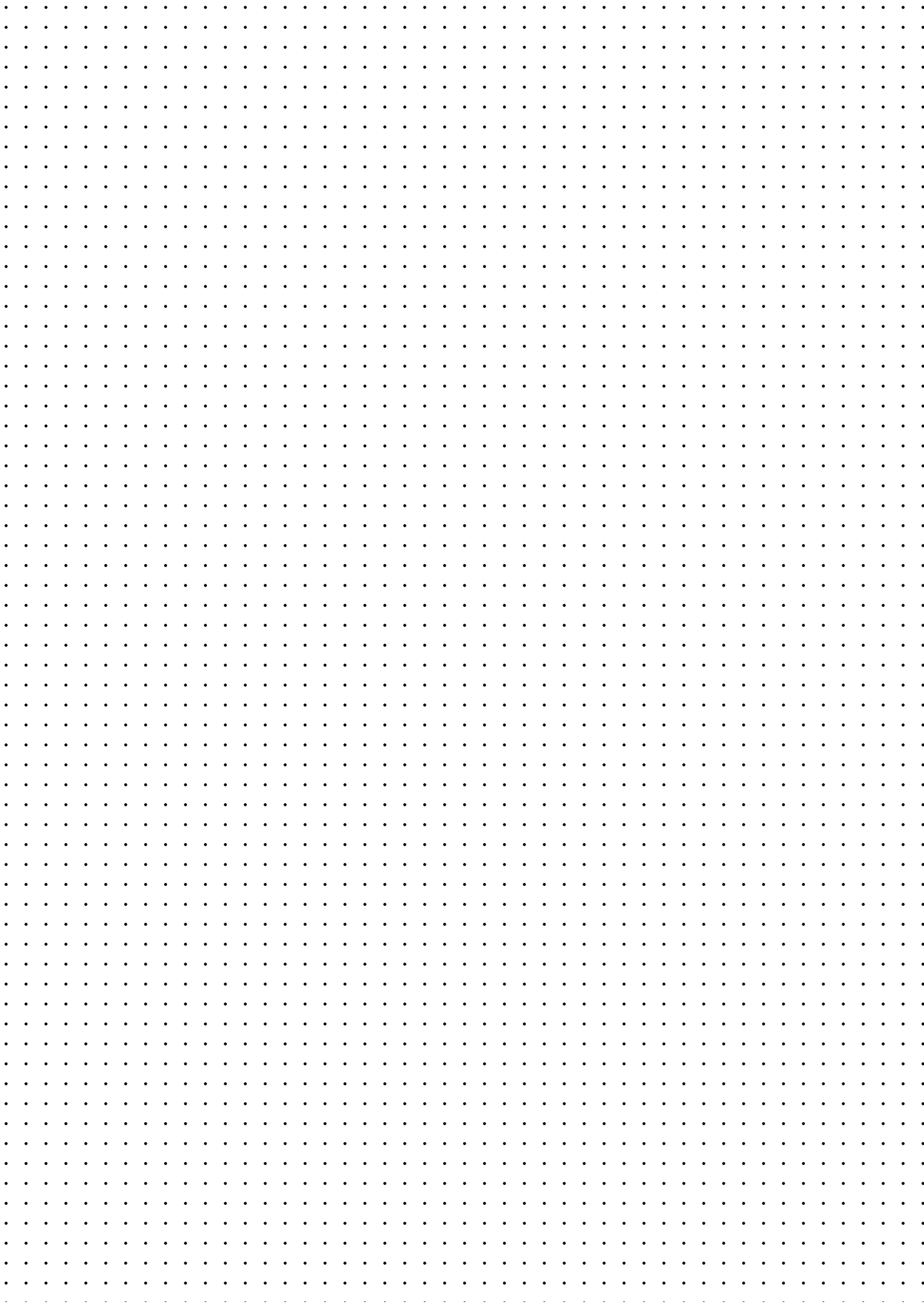
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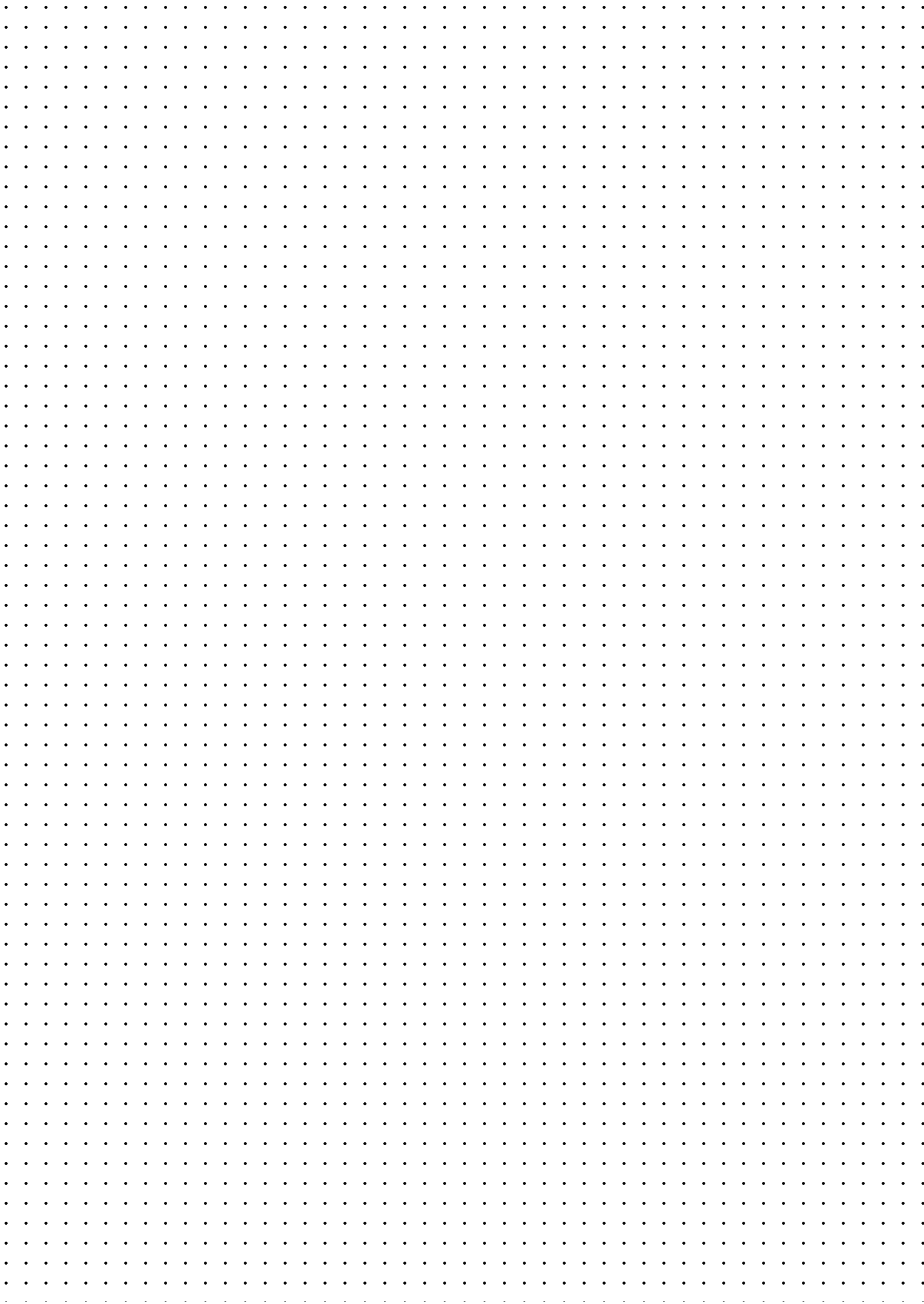
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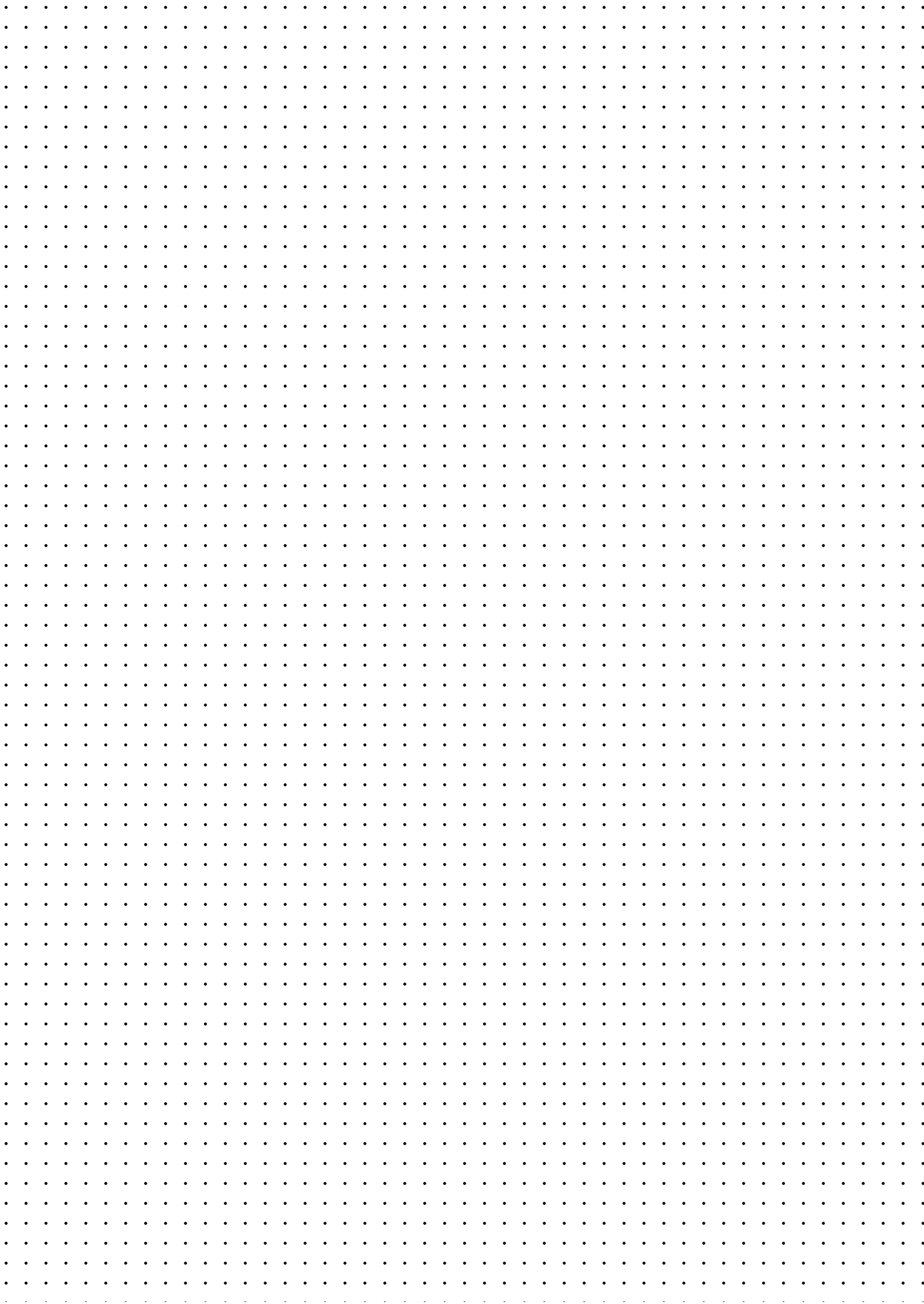
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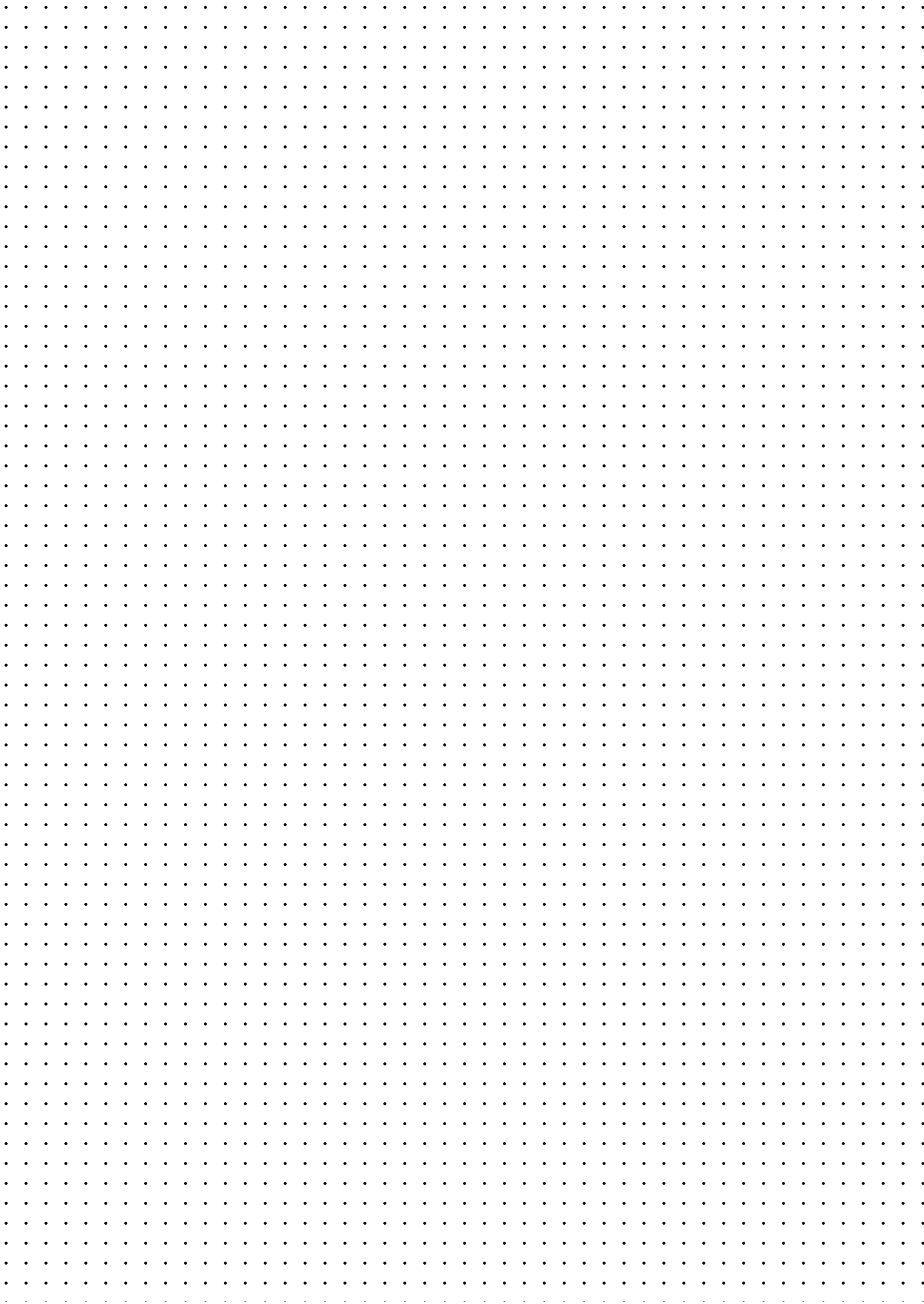
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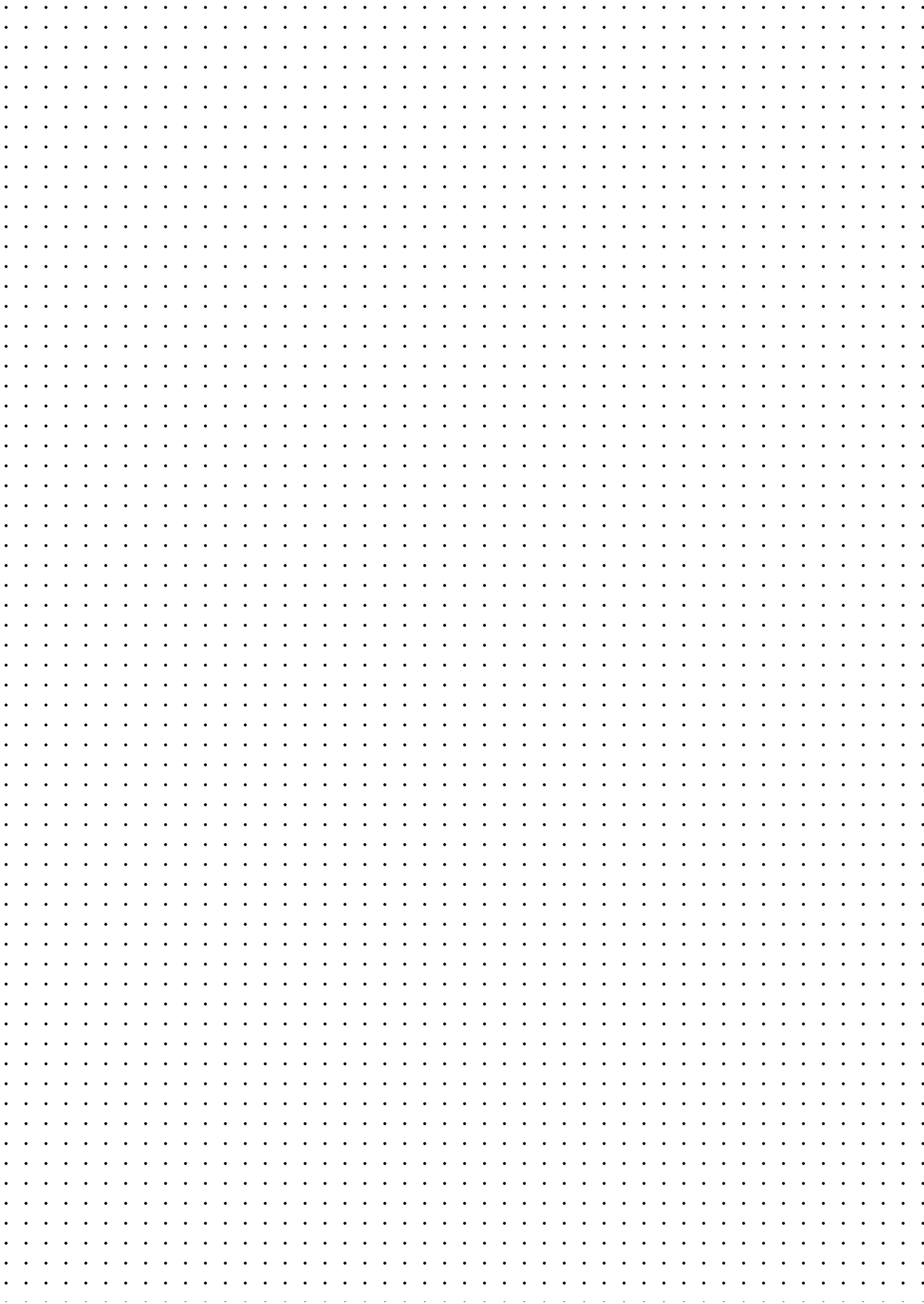
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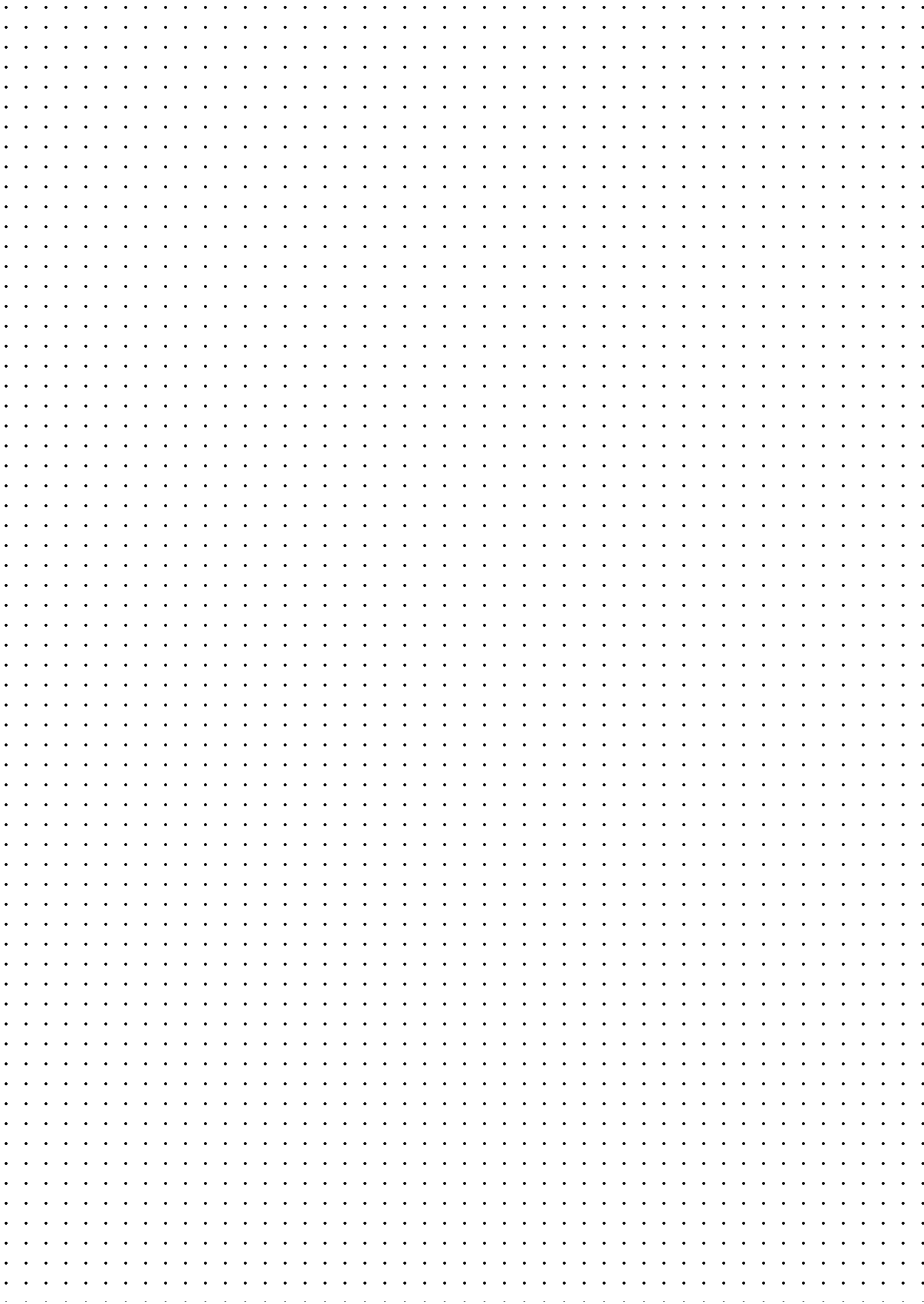
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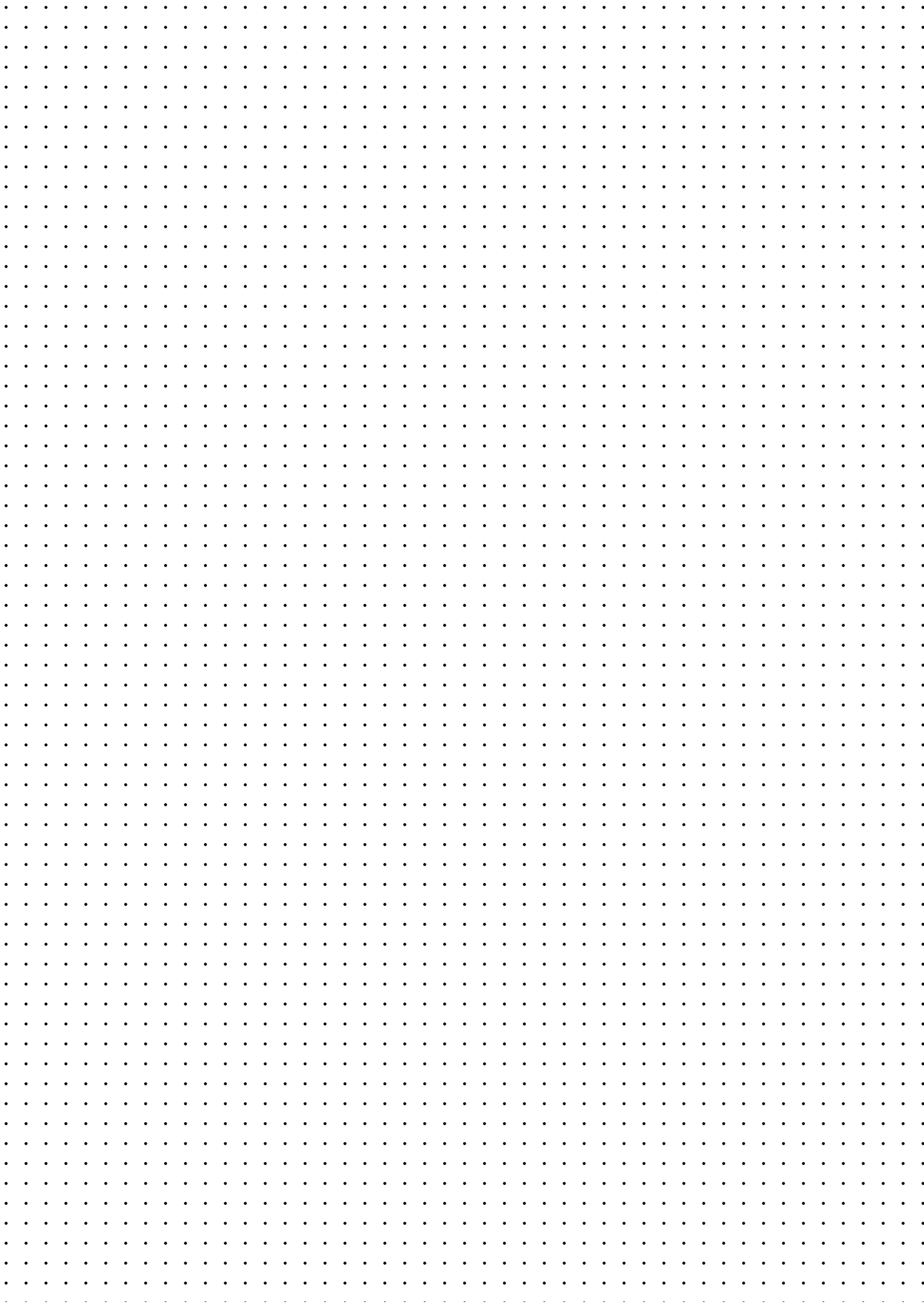
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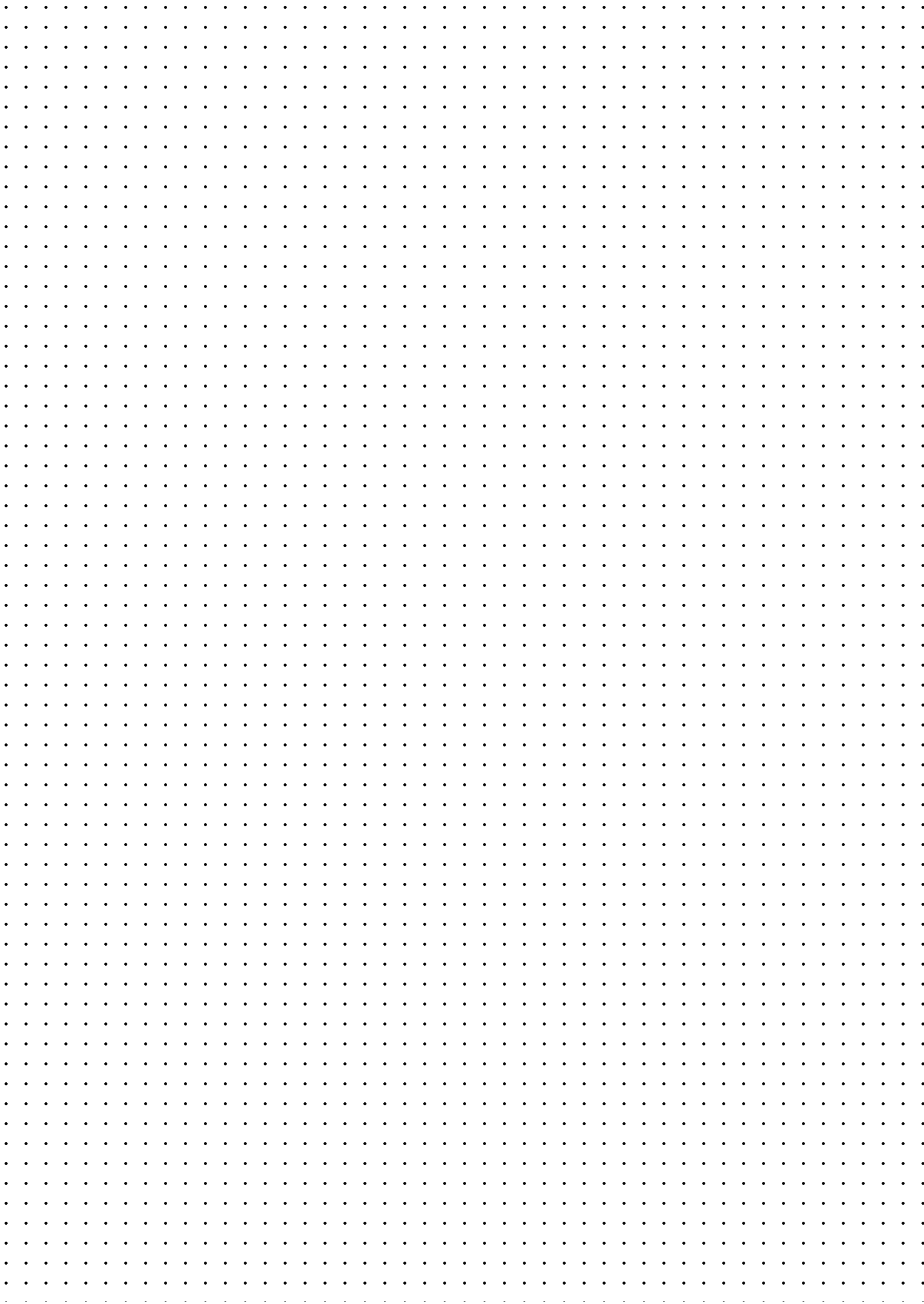
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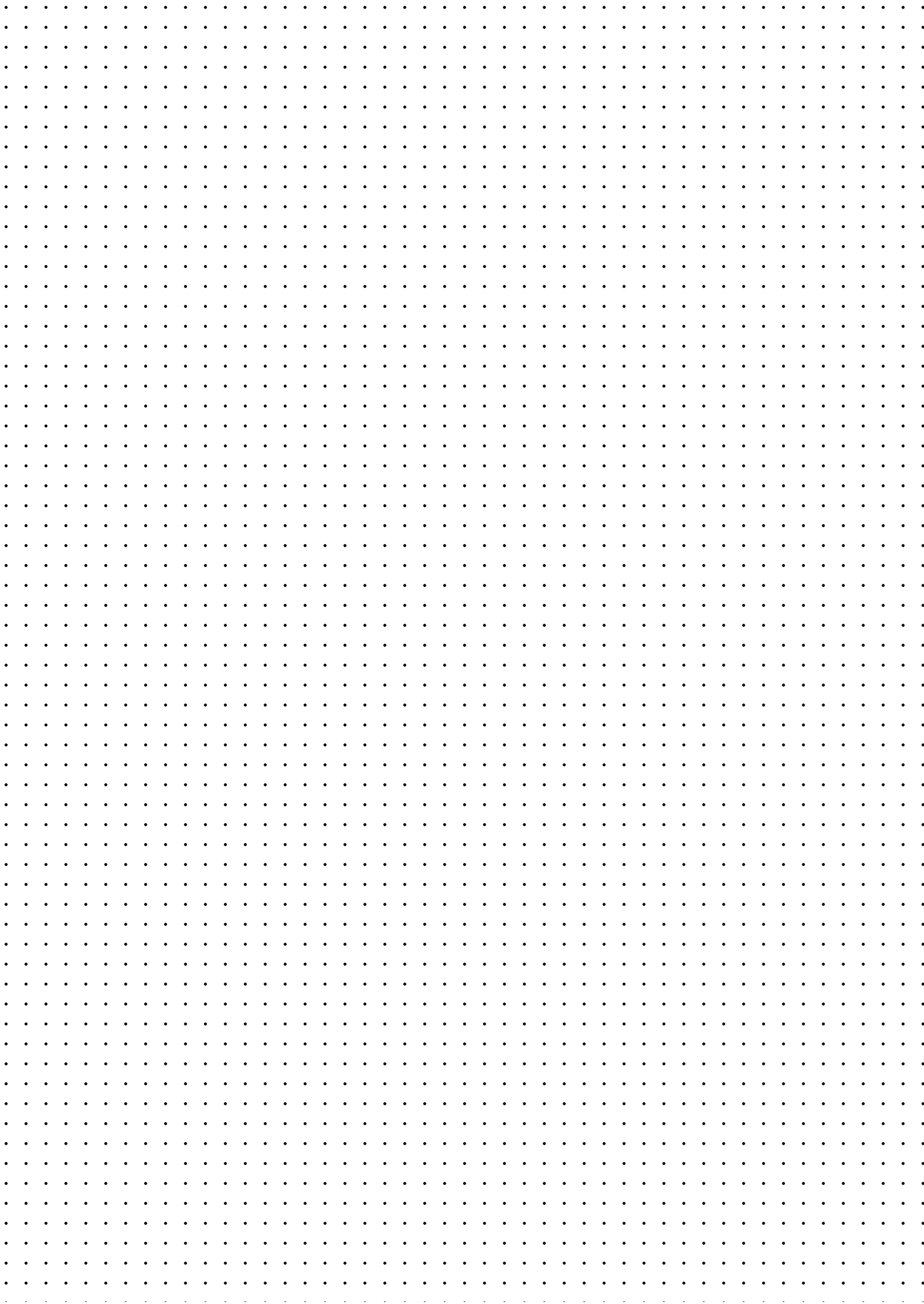
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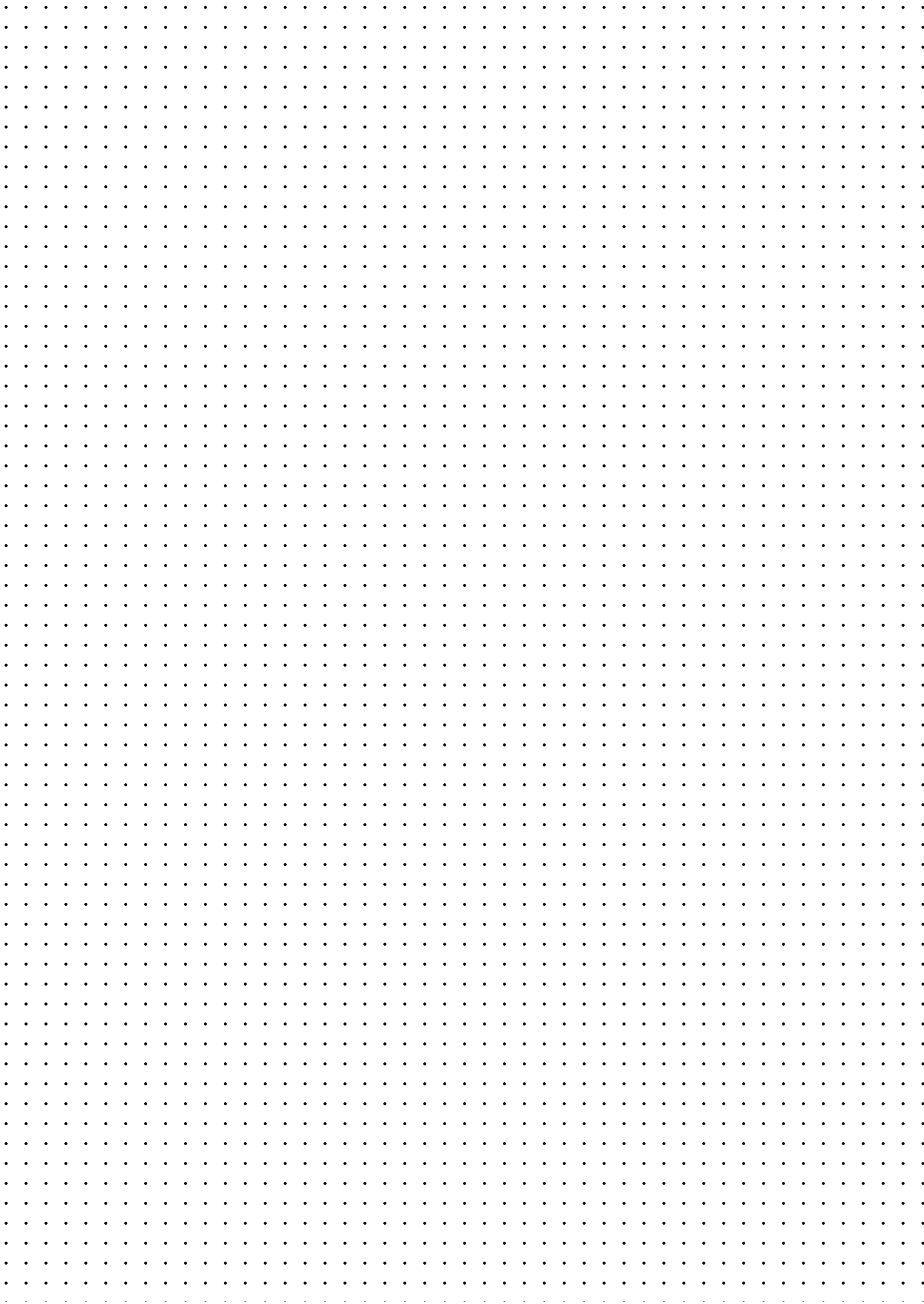
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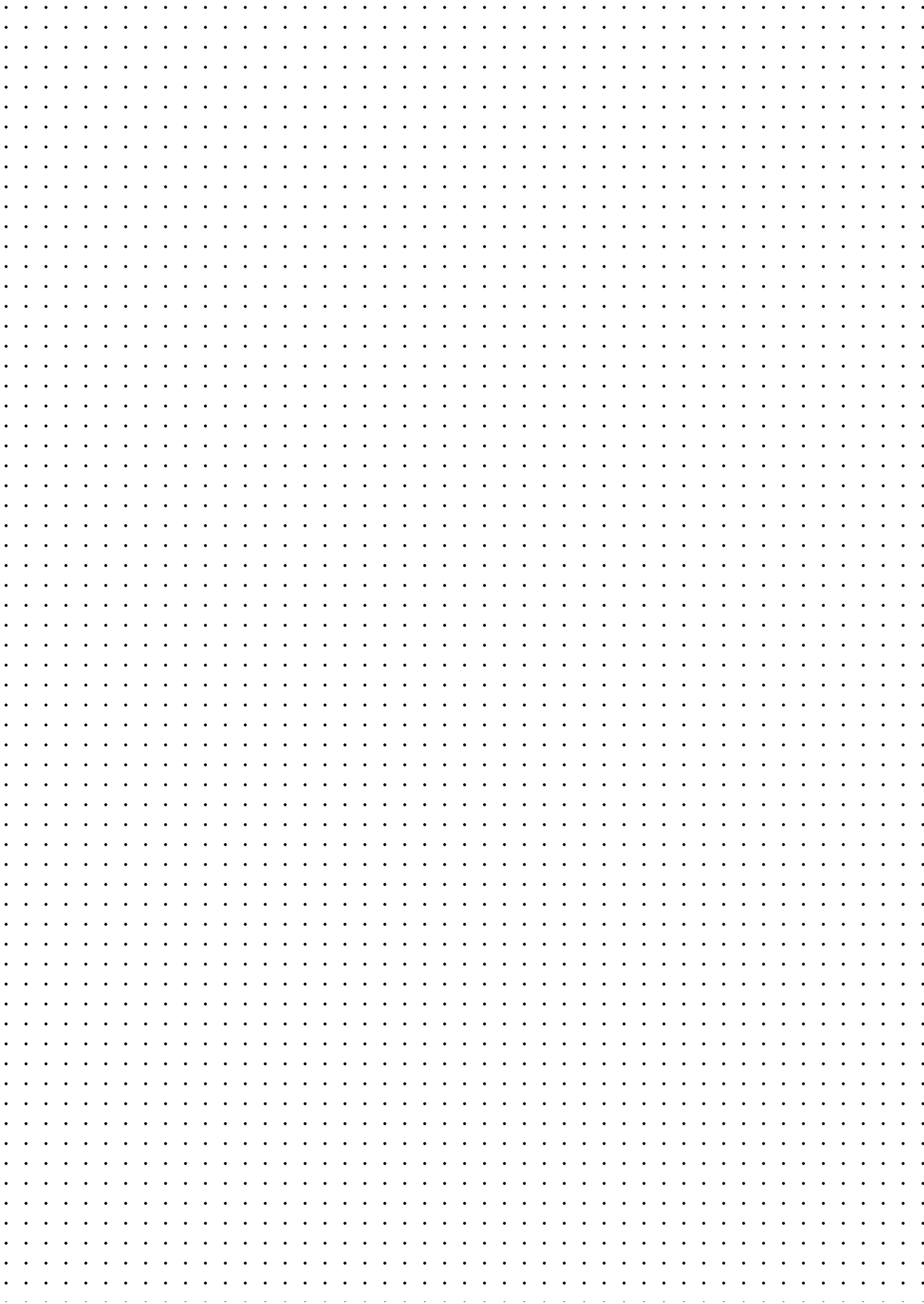
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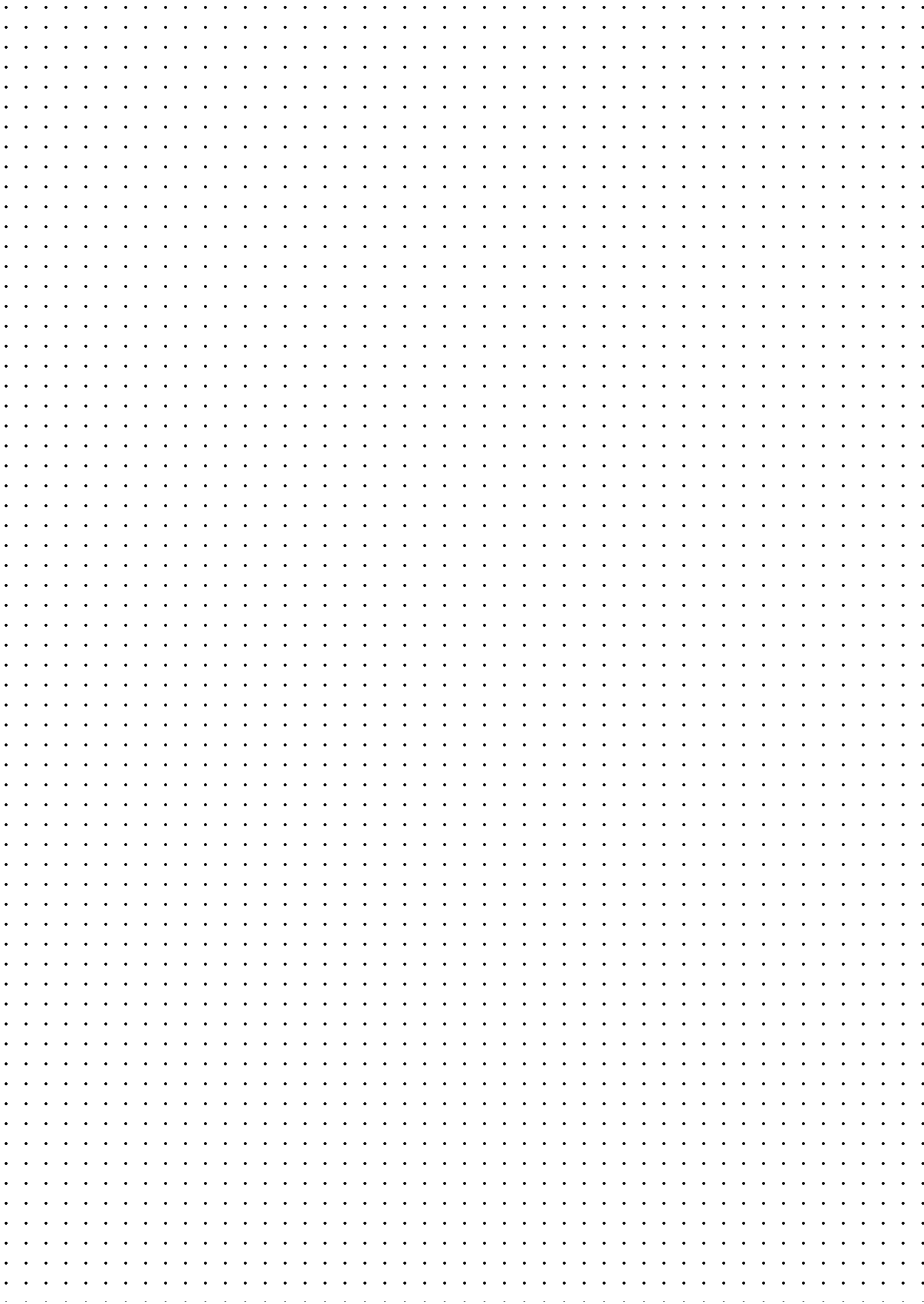
Raw Data File & Location

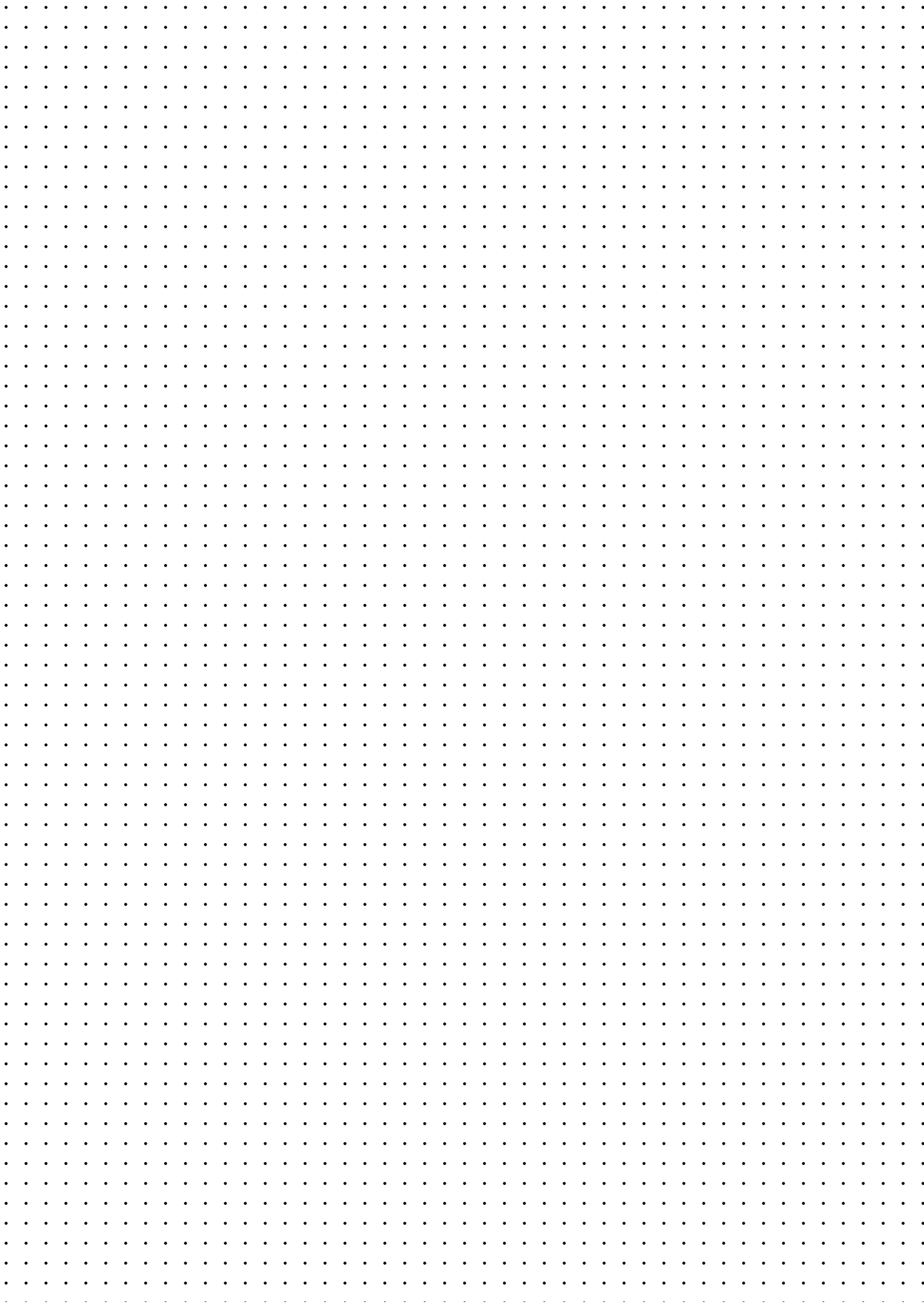
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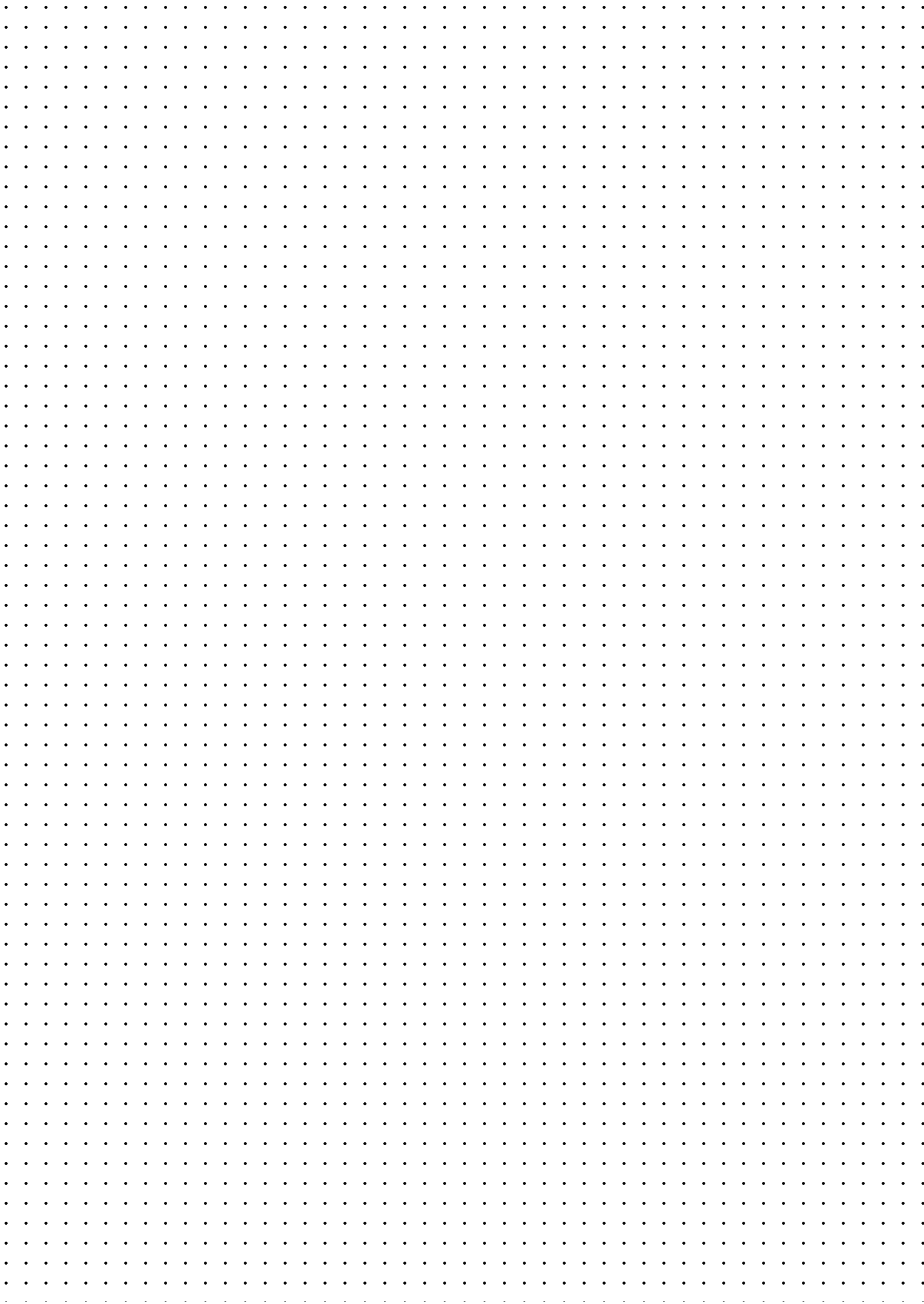
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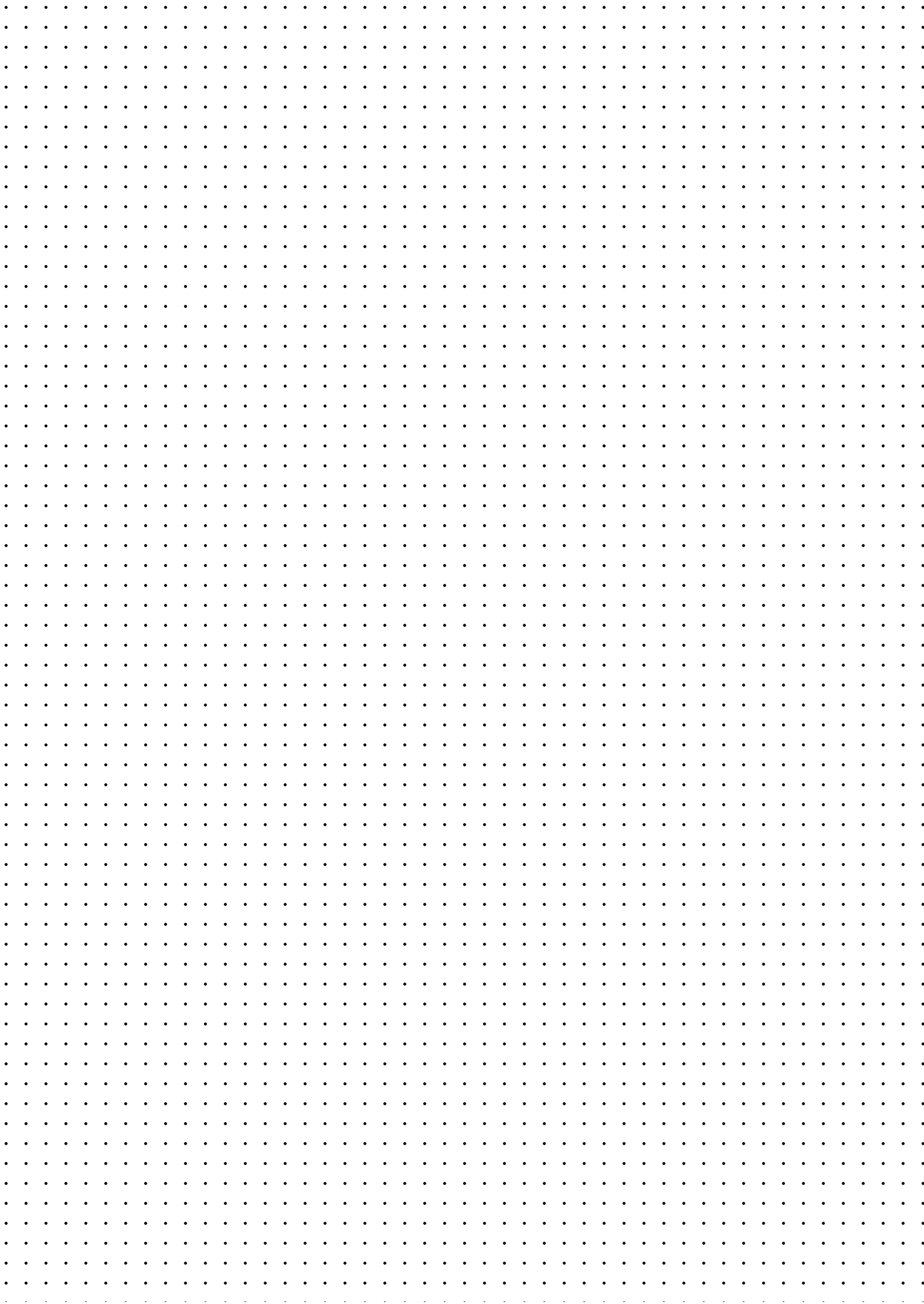
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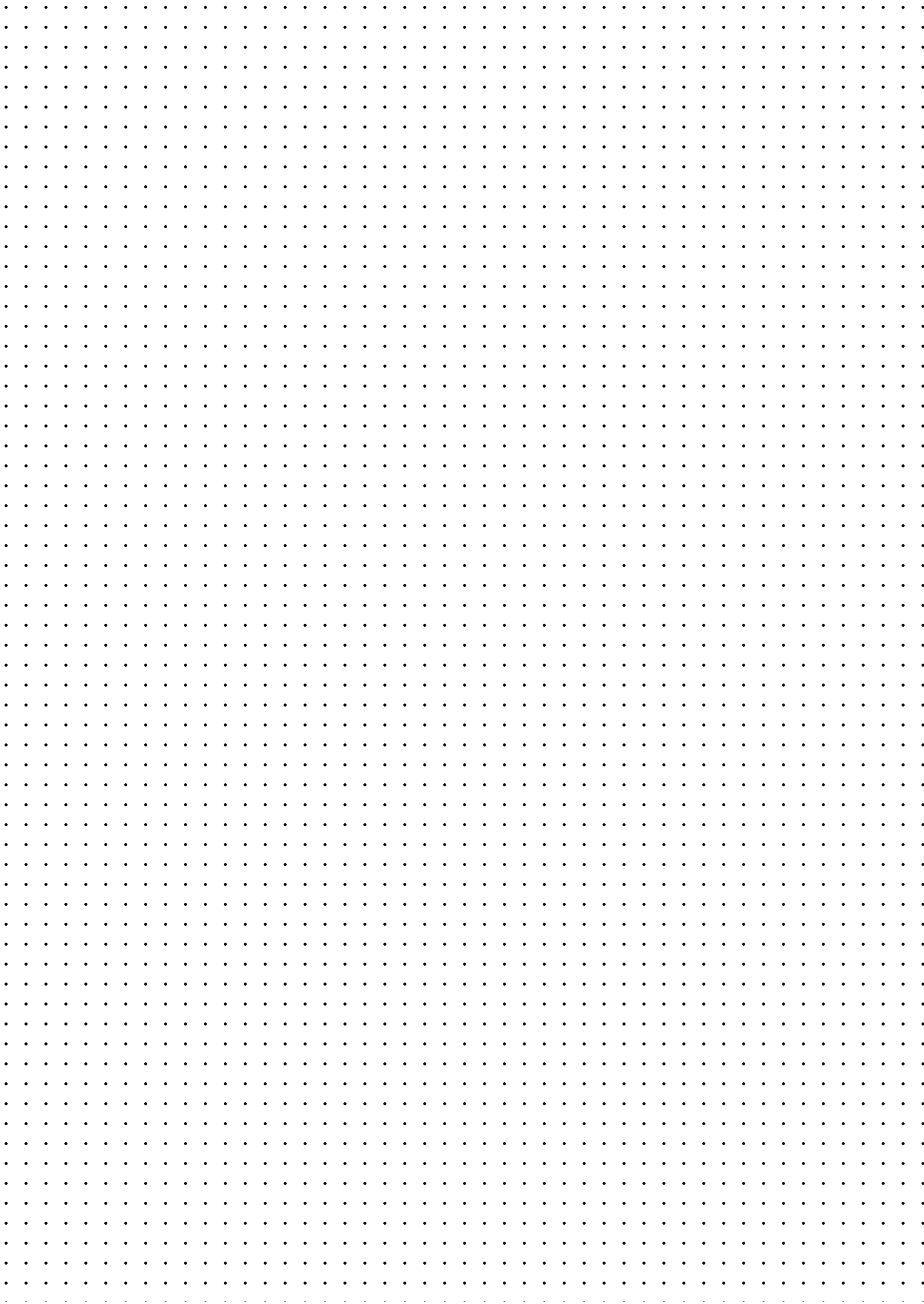
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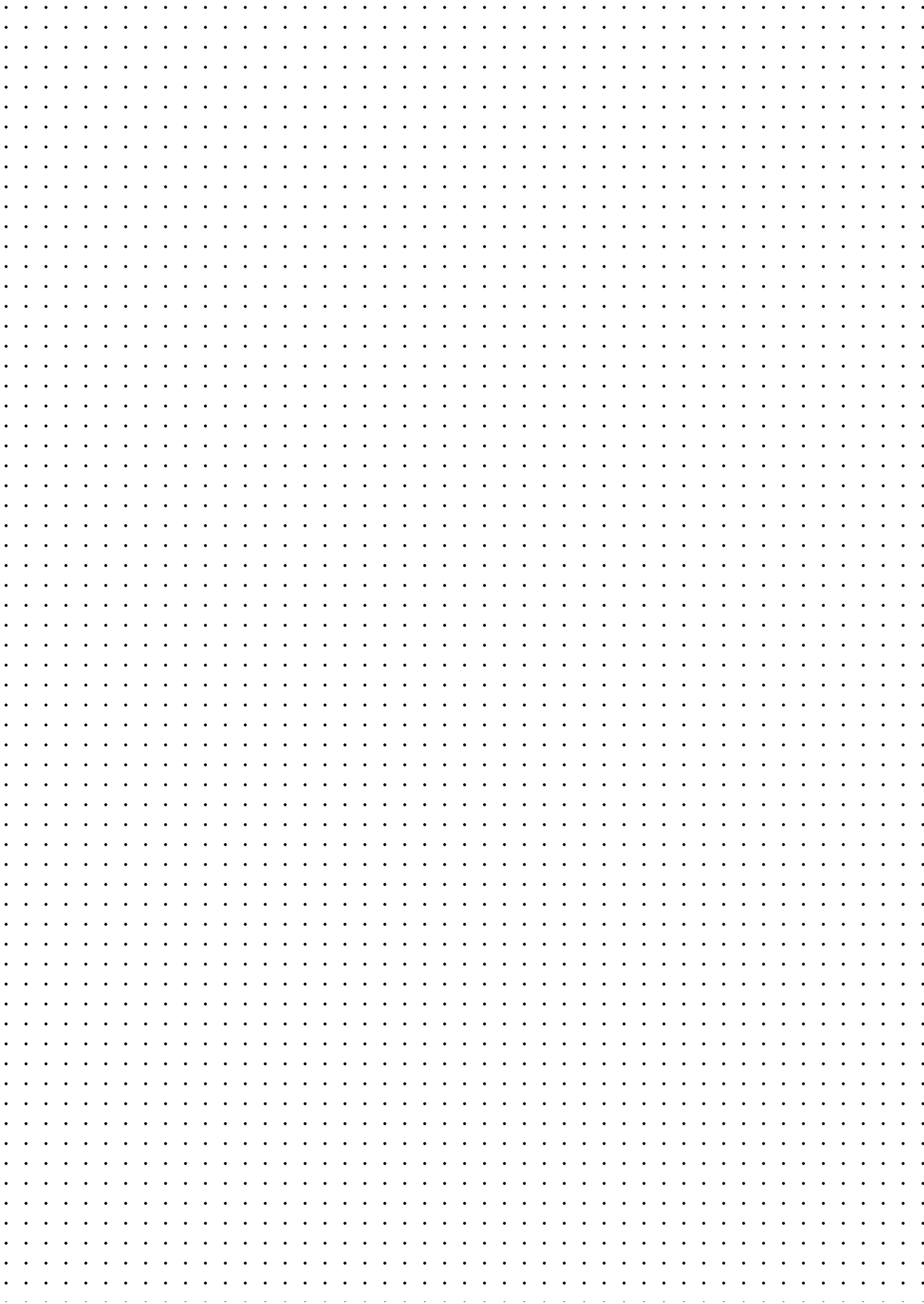
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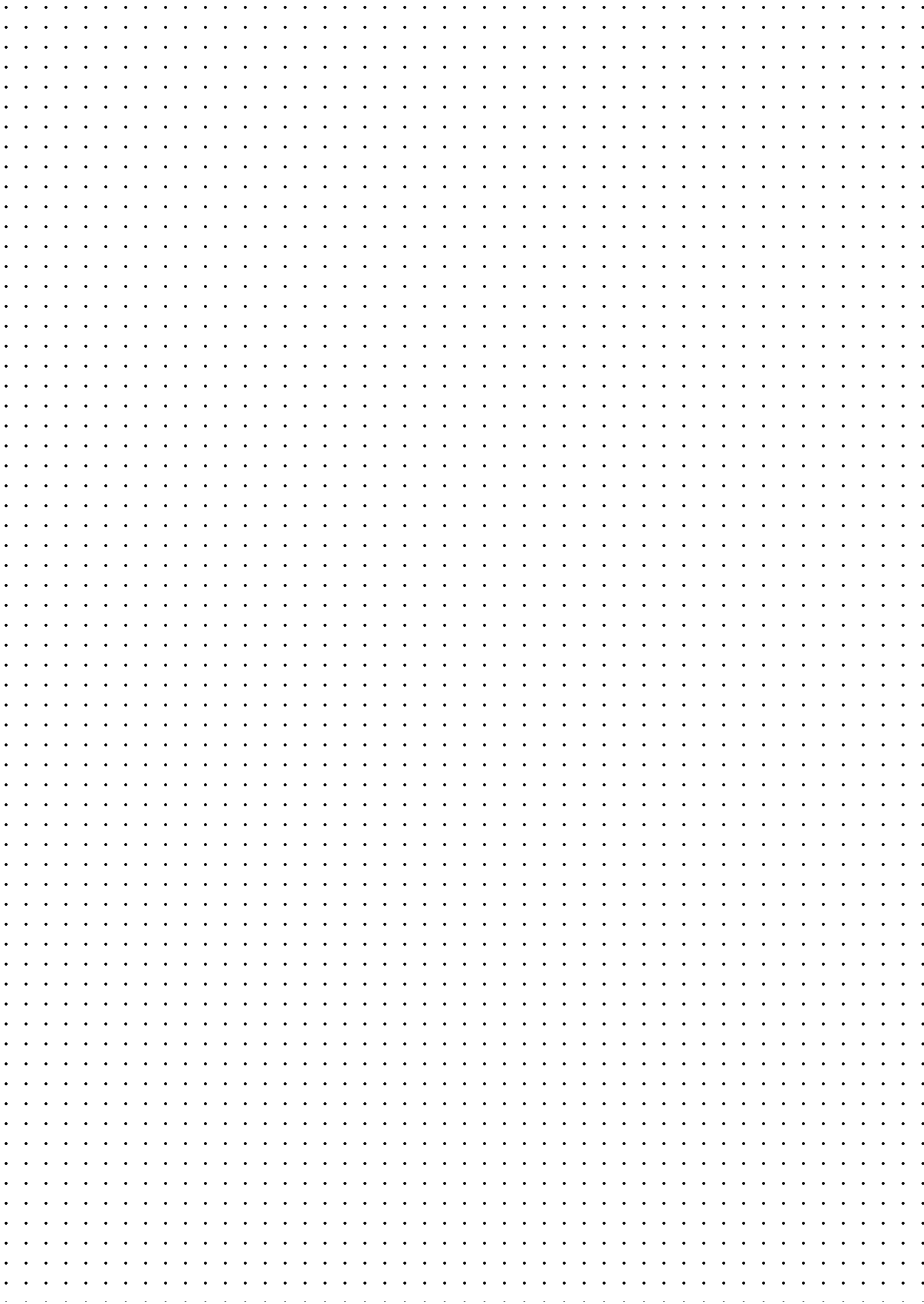
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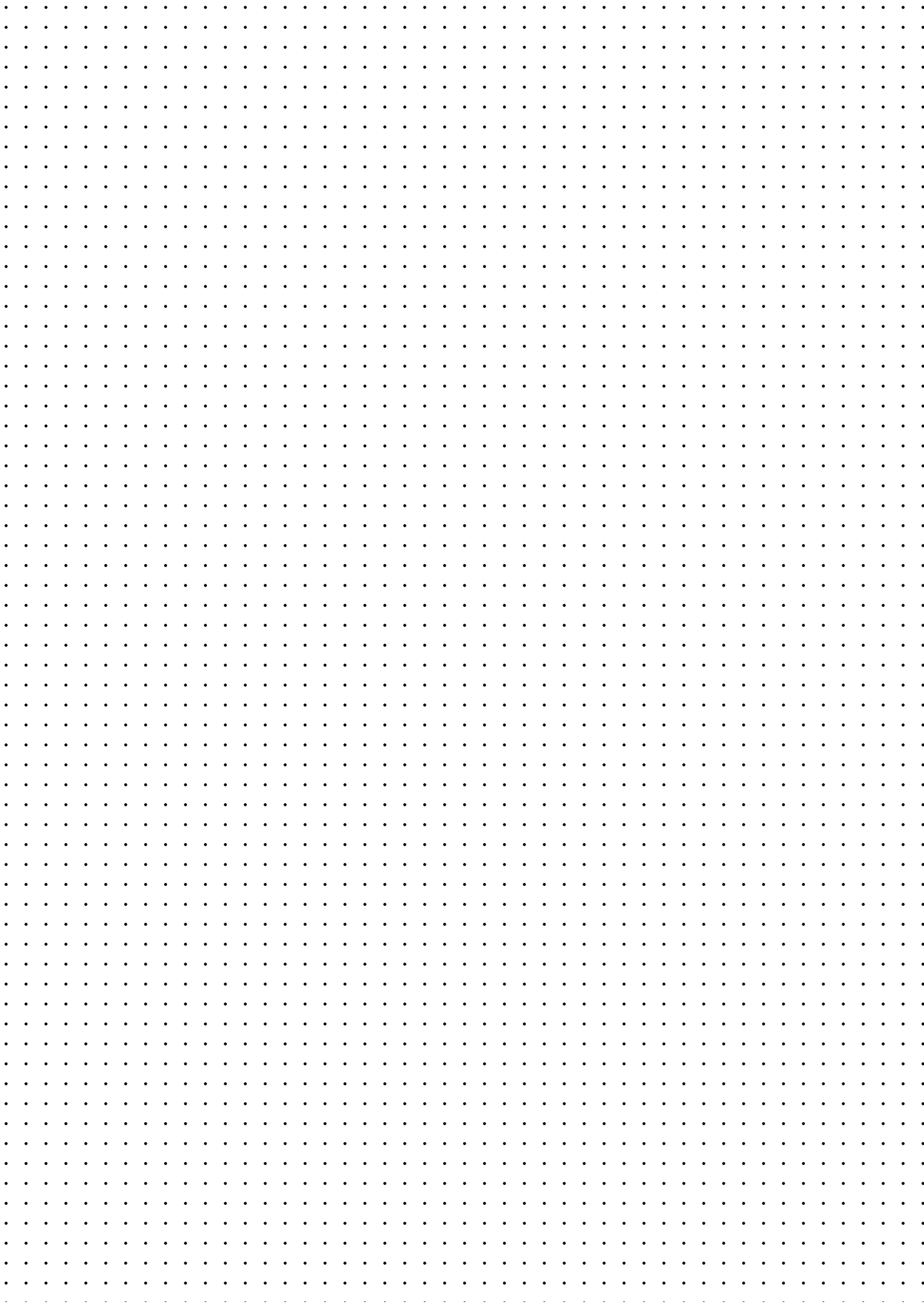
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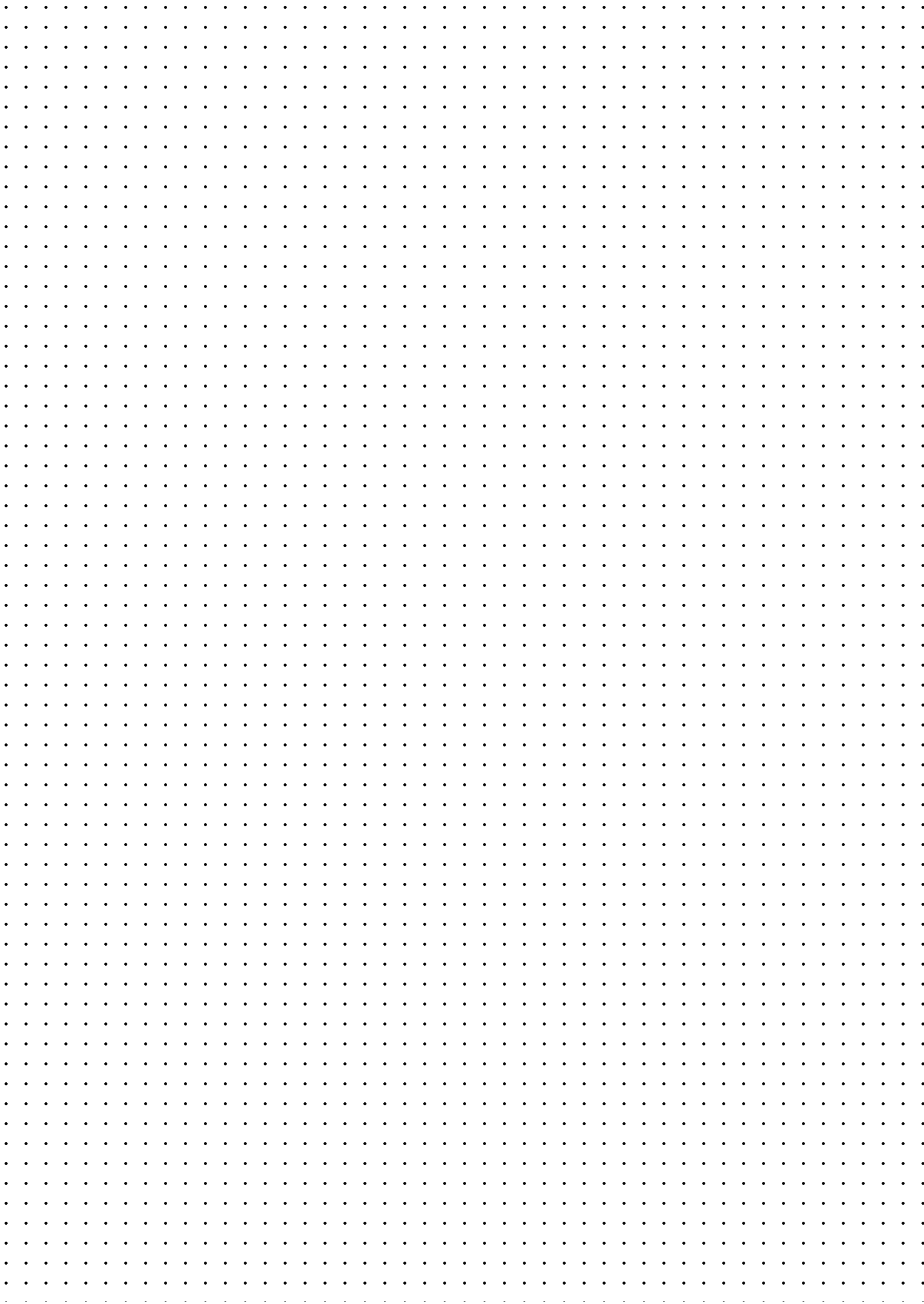
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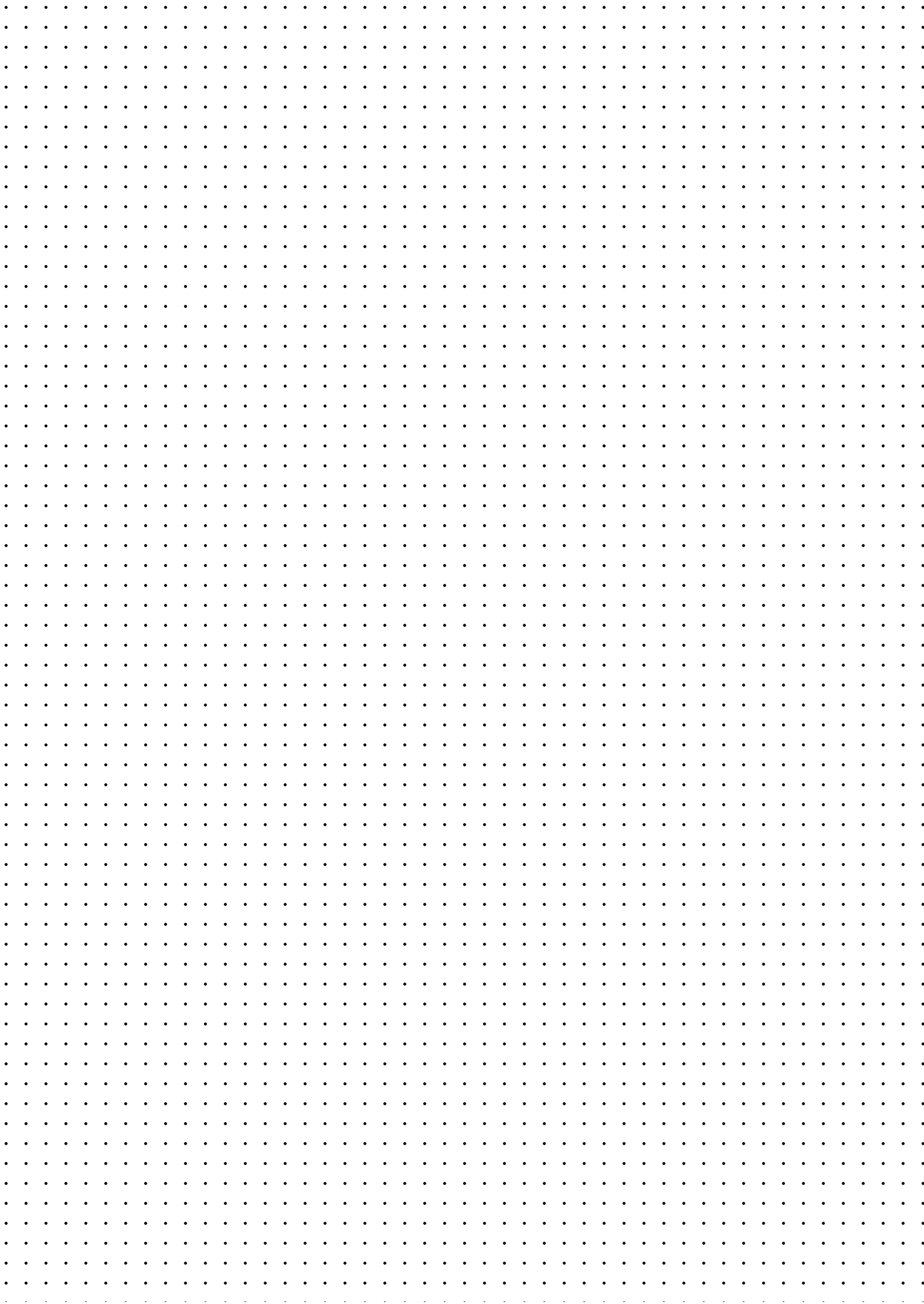
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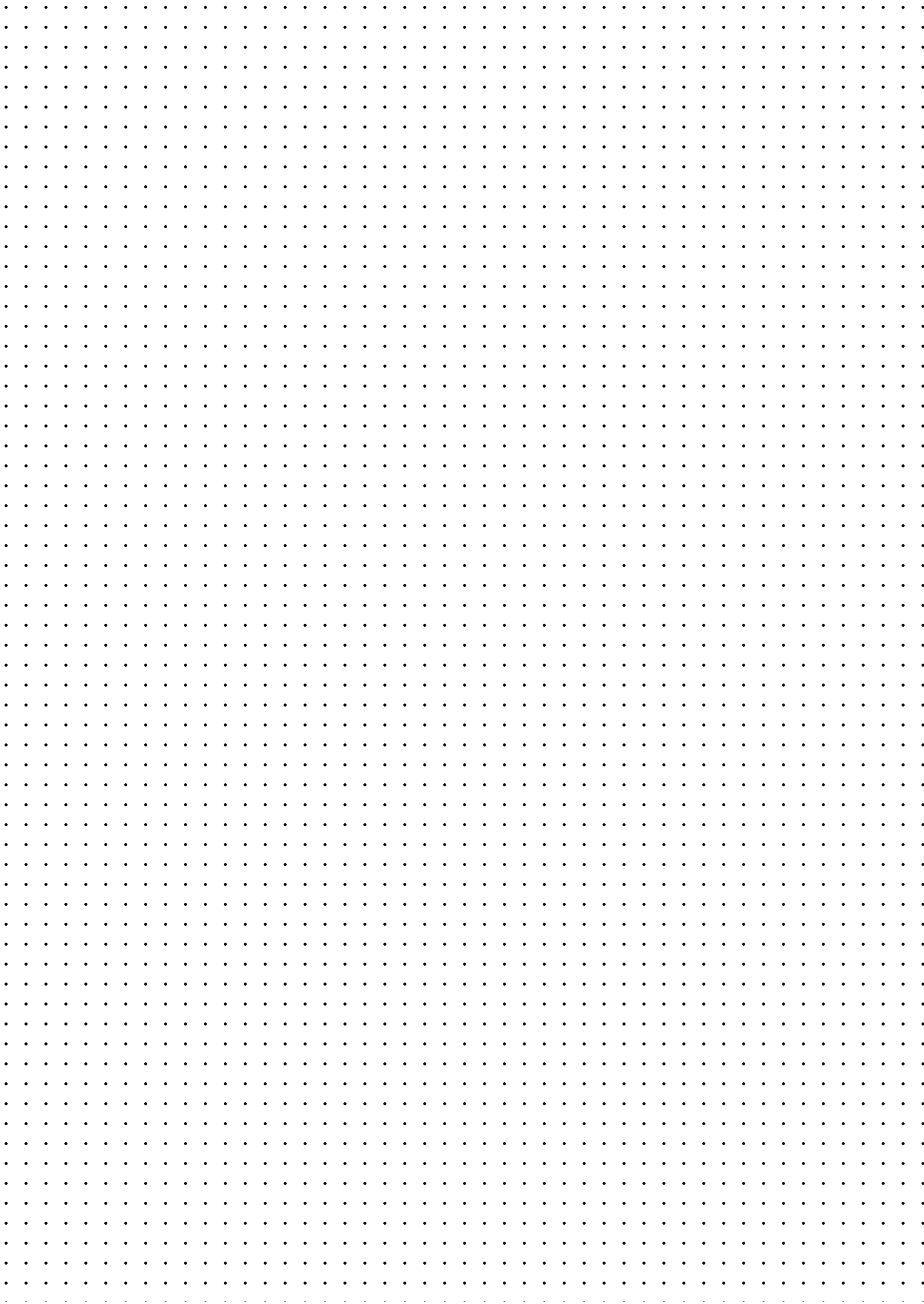
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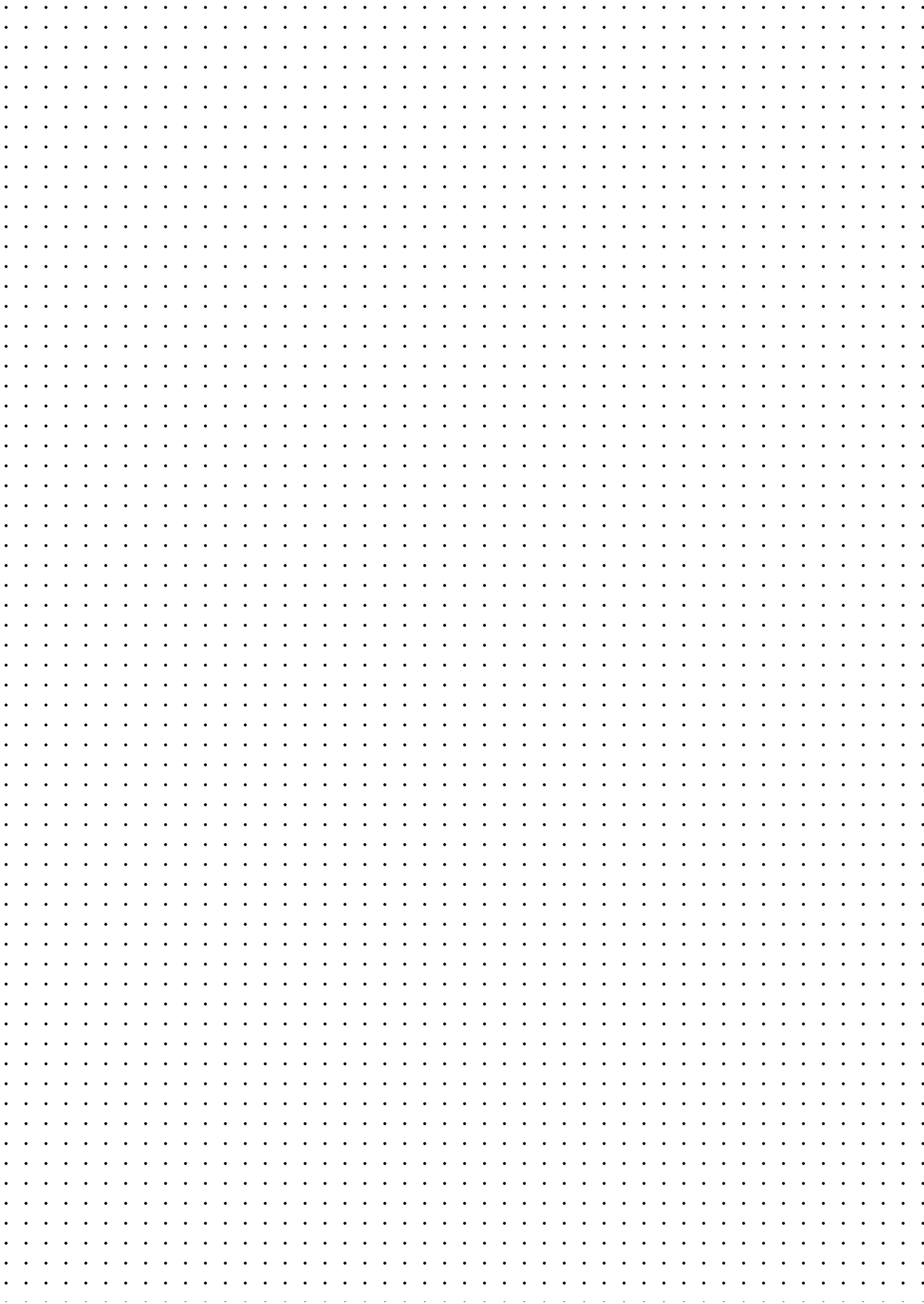
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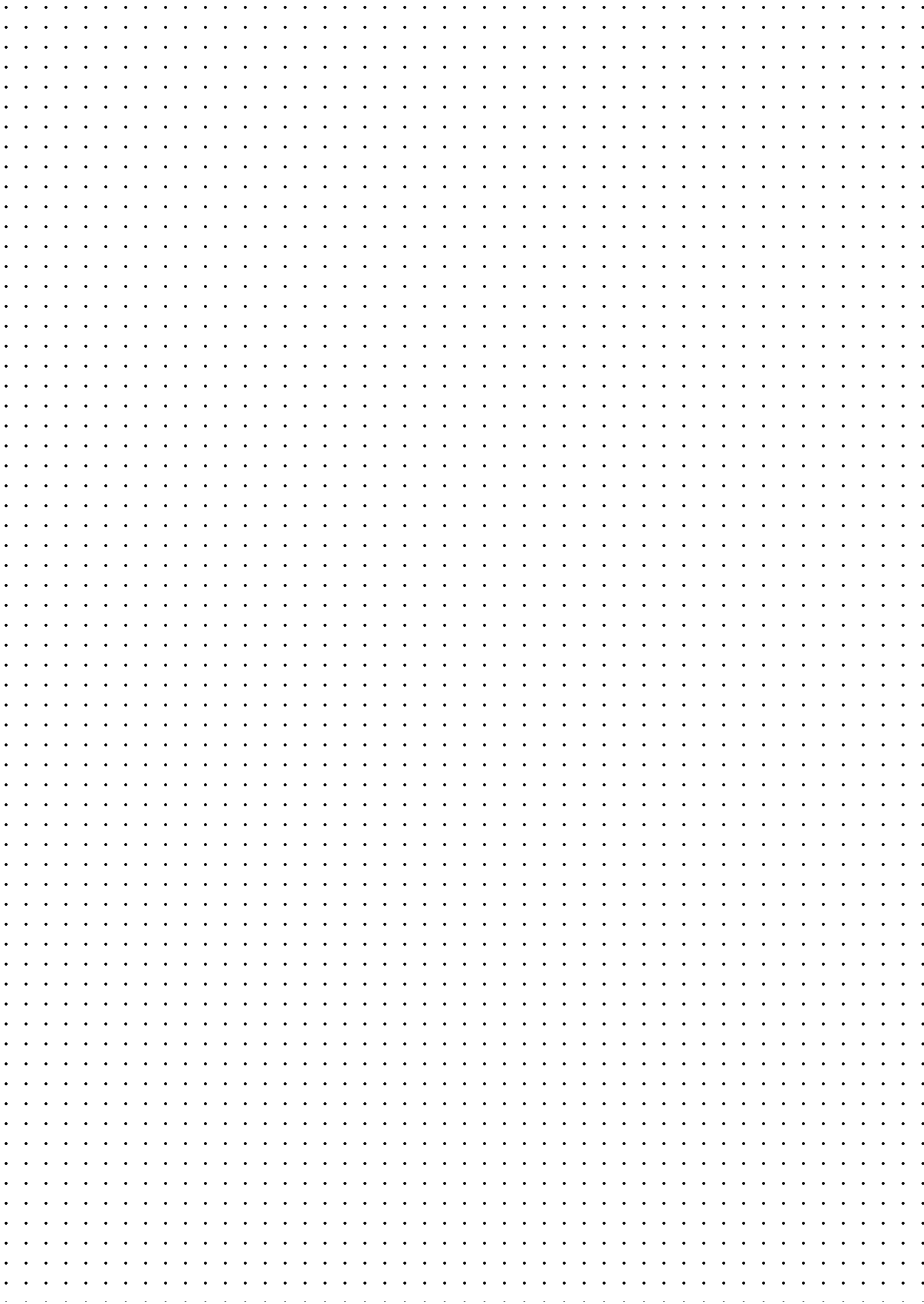
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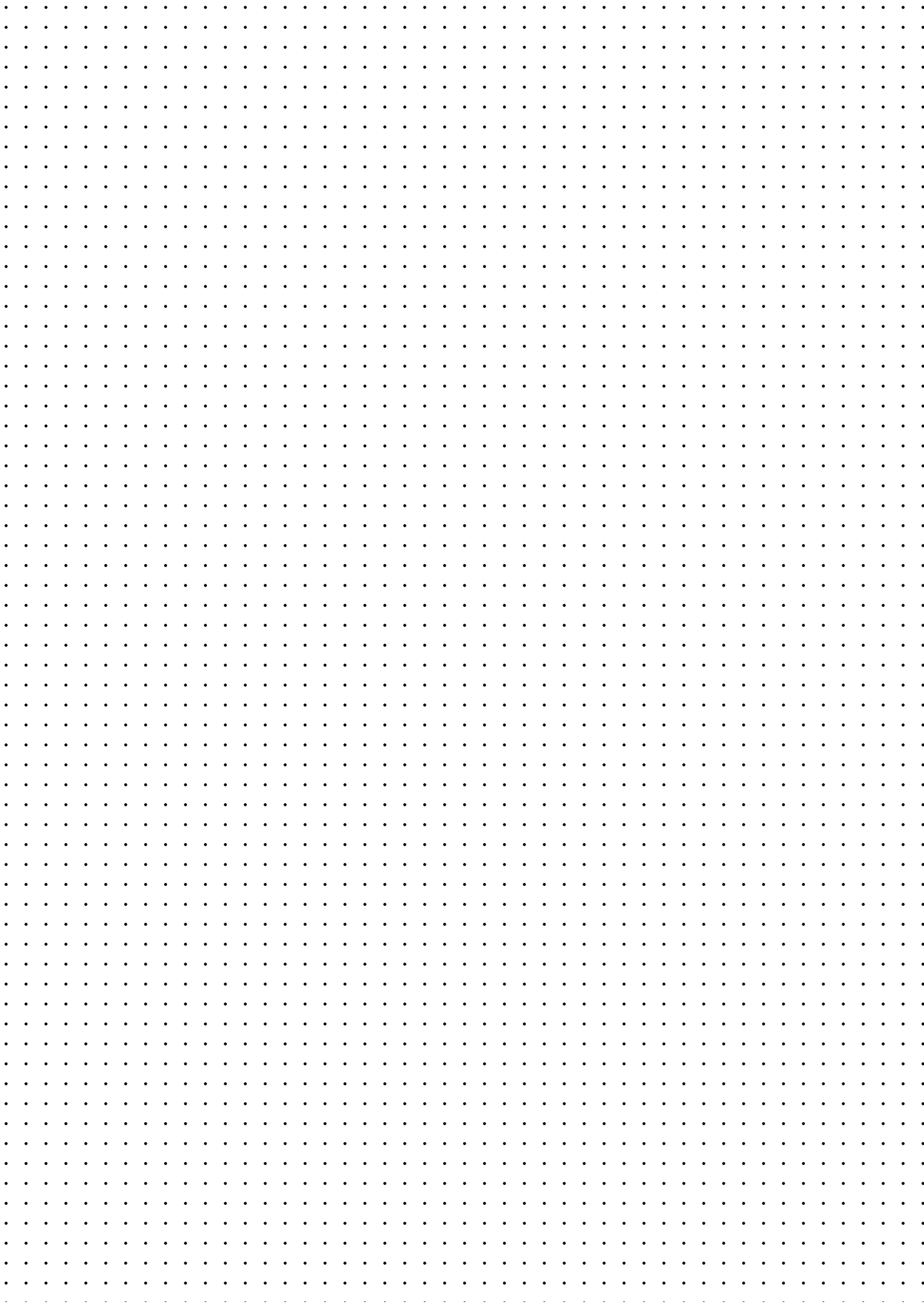
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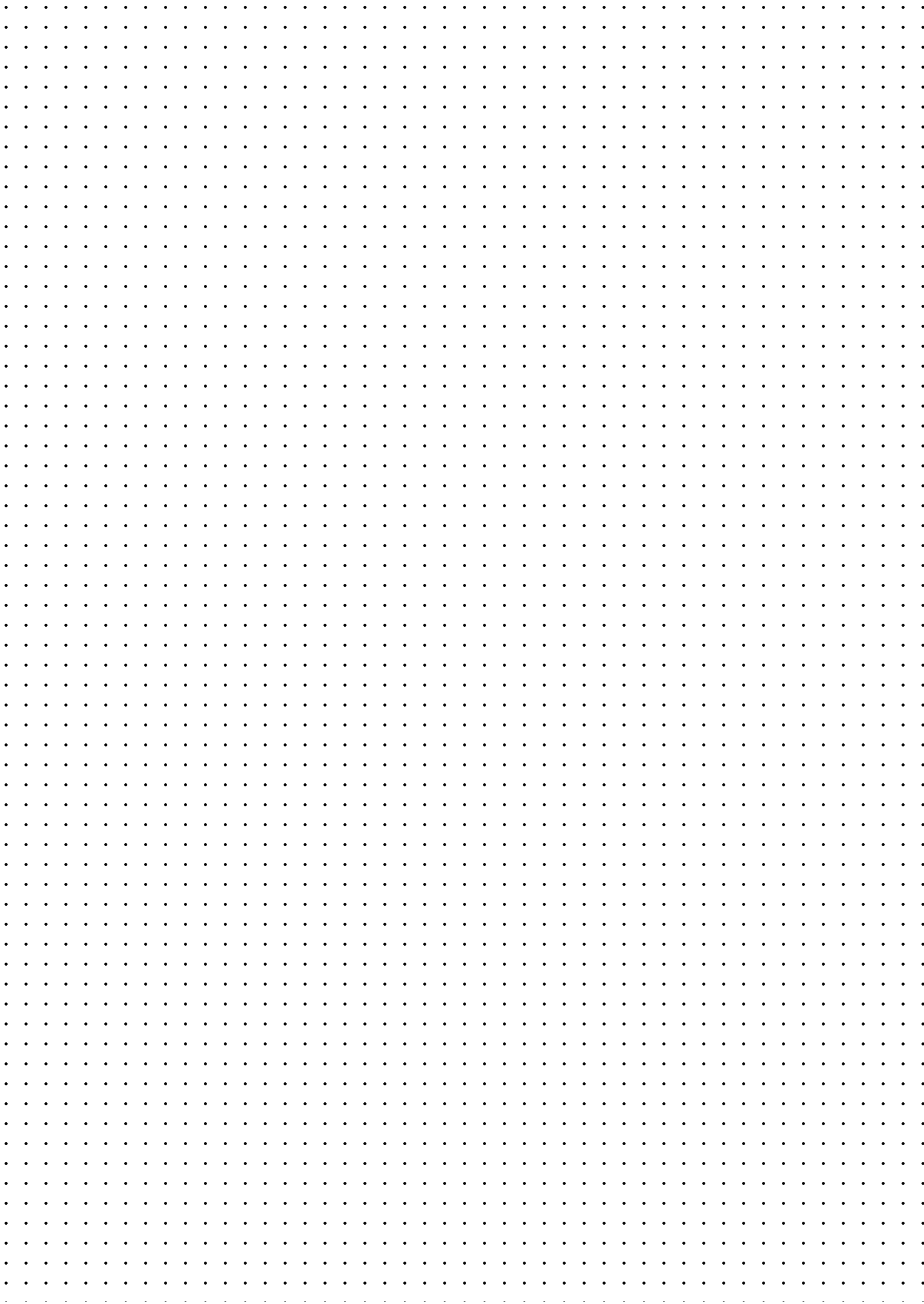
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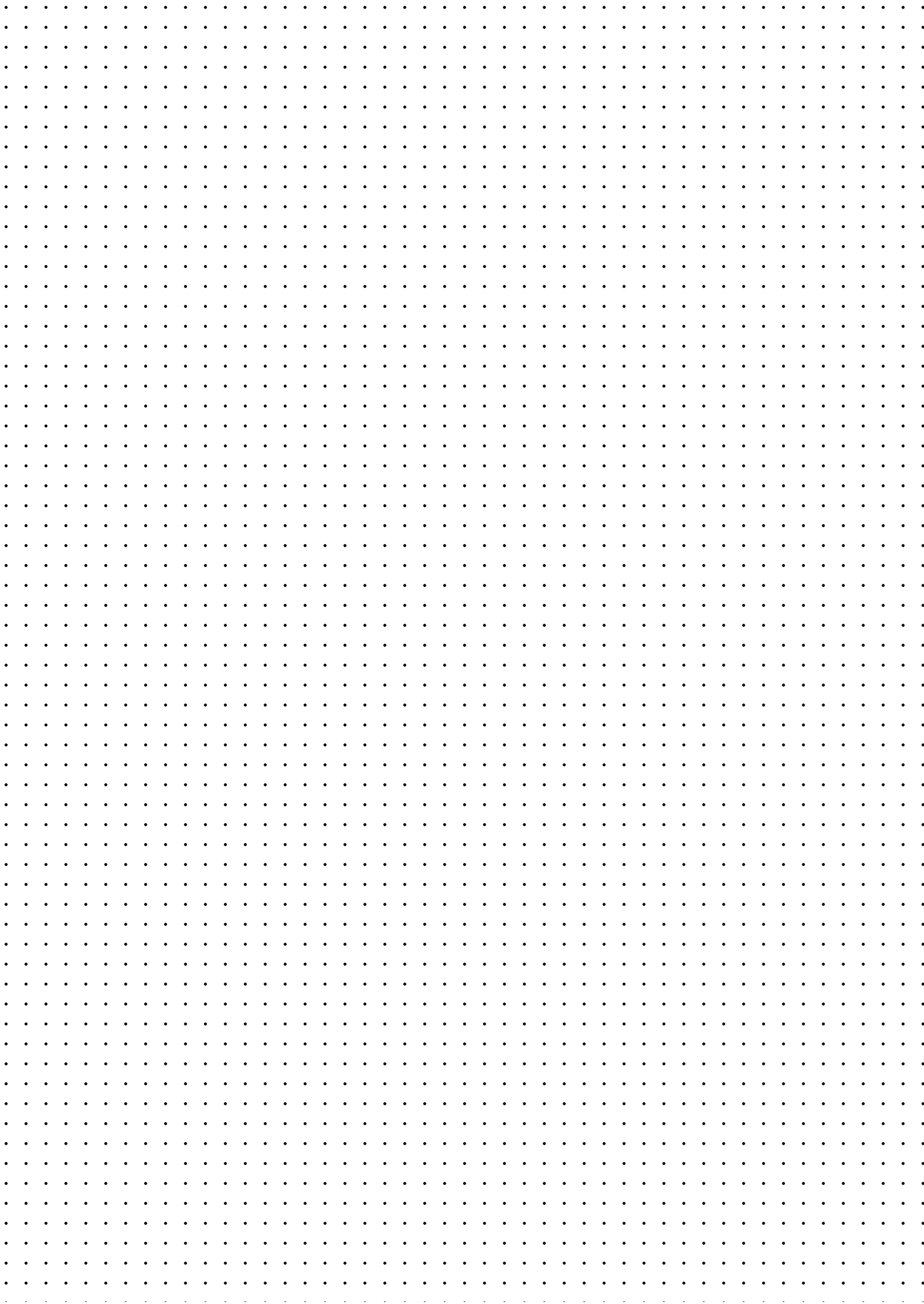
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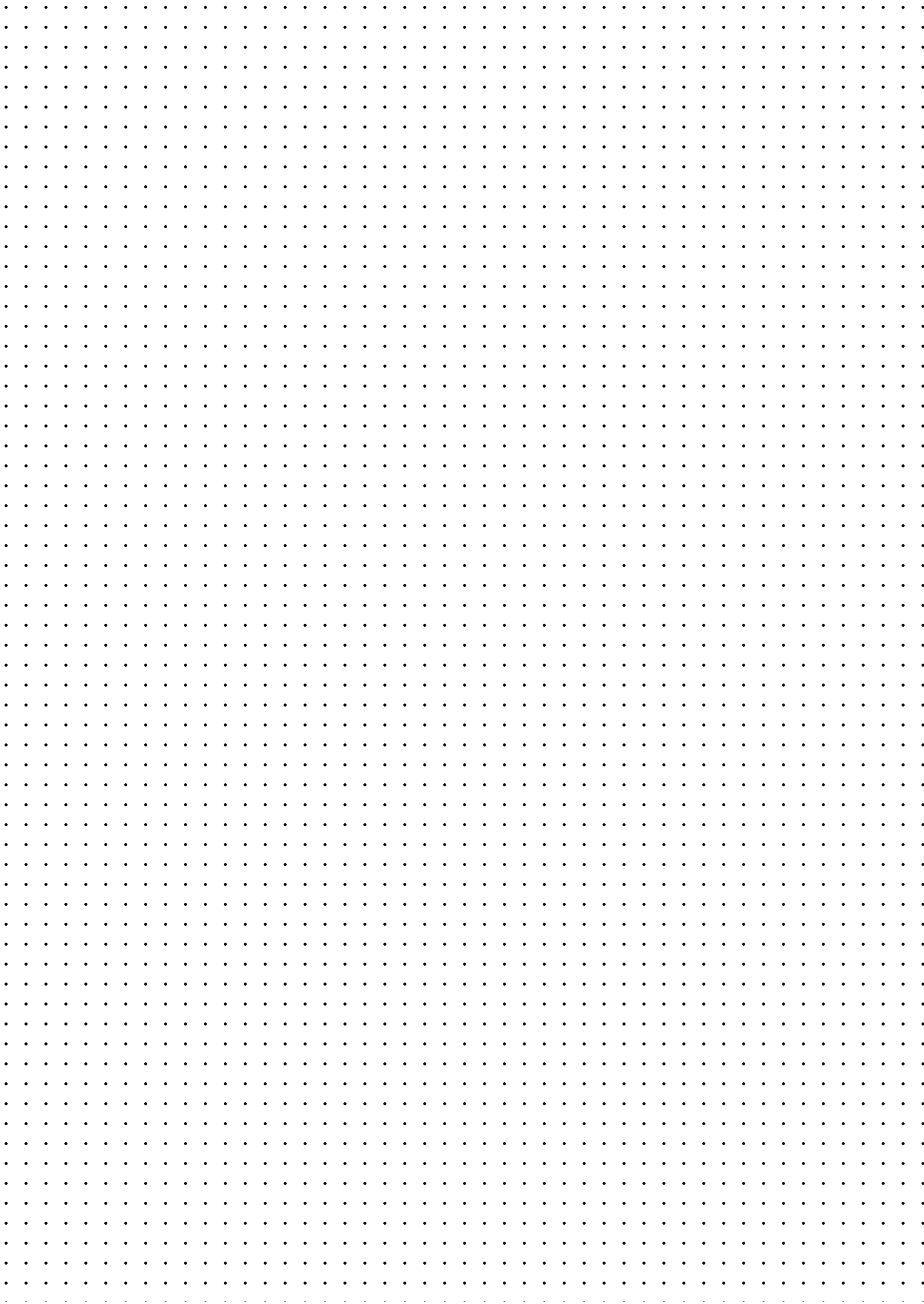
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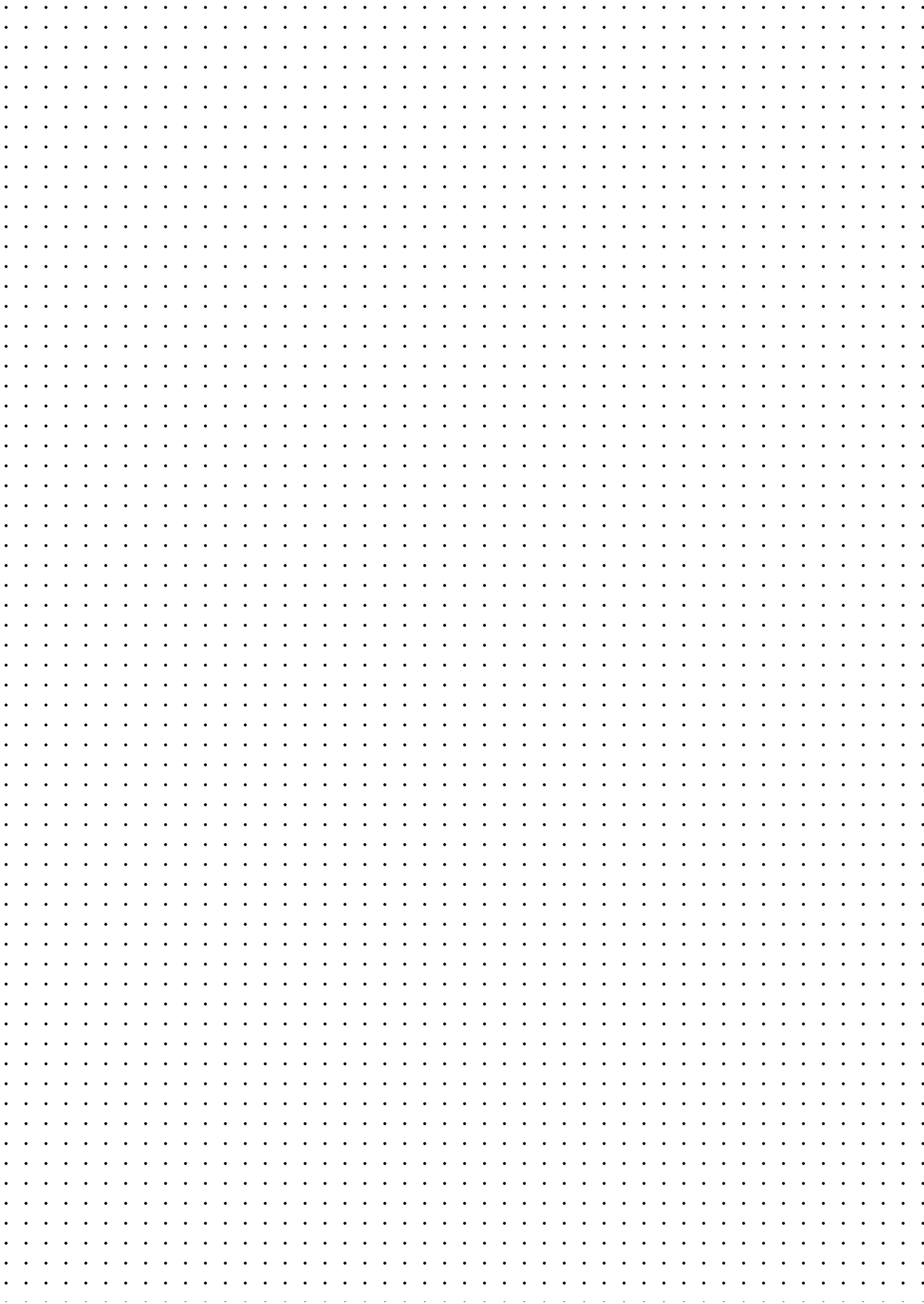
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